STROMEX

GROUP

The StromeX Editorial Bible

The Constitution, Operating Manual and Strategic Corpus

Ten Volumes · Edition II

RATIFIED AS THE GOVERNING CORPUS OF THE STROMEX GROUP

13 AUGUST 2026 · VERSION 2.0

The StromeX Editorial Bible

EDITION II

S T R O M E X   G R O U P

The StromeX
Editorial Bible

The Constitution, Operating Manual and Strategic Corpus of StromeX

Edition II · Supersedes Edition I

Lagos · London · New York

The StromeX Editorial Bible, Edition II.
Version 2.0. Published 13 August 2026.

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External data cited in this corpus is attributed at the point of use and listed in the Bibliography. Figures describing the group’s own delivered work are drawn from the sources recorded in Internal Sources.

Composed for A4. The text is set in a serif book face with a companion sans for tabular and navigational matter, following the typographic roles defined in Volume I, Chapter 10; the StromeX brand faces Archivo and Fraunces are shipped alongside this edition and are selected by the build when they are present in the rendering environment. The press-quality PDF edition is generated from this master and the two are content-identical.

DOCUMENT CONTROL

Document Control

TITLE	The StromeX Editorial Bible
EDITION	Edition II
VERSION	2.0
STATUS	Ratified — governing corpus
CLASSIFICATION	Confidential — internal and authorised parties
OWNER	Office of the Founder, StromeX Group Holdings
CUSTODIAN	Executive; Pricing Council for Volume III
SUPERSEDES	The StromeX Editorial Bible, Edition I (Version 1.0)
AUTHORITY	Supreme. Where any roadmap, contract, pricing sheet, presentation or line of code conflicts with this corpus, the corpus governs until formally amended.
AMENDMENT PROTOCOL	Volume I, Chapter 9. Four tiers: Entrenched (board supermajority, published rationale, 30 days' notice), Constitutional, Strategic, Operational.
REVIEW CYCLE	Volume III quarterly (Pricing Council). Volume II risk register quarterly. All other volumes annually.
MASTER SOURCE	This DOCX. The press-quality PDF edition is generated from it and is content-identical.
CORPUS LOCATION	docs/bible/ in the StromeX repository, under version control.
ISSUED	13 August 2026

REVISION HISTORY

Revision History

VERSION	DATE	AUTHOR / AUTHORITY	SUMMARY OF CHANGE
1.0	August 2026	Office of the Founder	Edition I ratified. Established the vision, product, user and intelligence philosophy, editorial standards, design philosophy, AI architecture philosophy, and the trust, safety and scalability constitutions for StromeX as an AI operating system for knowledge work.
2.0	13 August 2026	Office of the Founder	Edition II ratified; supersedes Edition I in full. Reframes StromeX from a product to a group. Adds the doctrine of ecosystems, the Free and Commercial Constitutions, the Ethics Constitution, the amendment protocol, the brand system and logo decision (Volume I); market positioning, the six competitor classes and the moat assessment (II); the seventeen-division catalogue with published module-level pricing (III); the Security Bible and credential architecture (IV); the Customer Success Playbook and partner economics (V); the craft standard and print coordination (VI); the industry ecosystems by wave (VII); the three-phase roadmap, phase gates and three financial scenarios (VIII); the innovation laboratory, research institute, talent system and 100-Year Plan (IX); and SpaceTalk (X).
2.0-p	13 August 2026	Office of the Founder	Flagship publication edition. Adds front matter, executive summaries, apparatus, appendices, glossary, bibliography and index. Content-identical to the corpus of record.

Superseded text is struck through and dated for one edition, then archived — never silently deleted. The record of what the group used to believe is part of the corpus (Volume I §9.2).

EXECUTIVE FOREWORD

Foreword

This corpus exists because of a specific failure that is easy to observe and difficult to correct.

Institutions across much of the world know exactly what they should become. A school knows its certificates ought to be verifiable, that parents should see results on a phone, that admissions should not consume six weeks of a registrar's life. A hospital, a ministry, a publisher and a bank each know the equivalent. What they lack is not ambition. It is a supplier they can trust, a price they can plan around, and a partner who will still exist in year three.

The world does not have a shortage of software. It has a shortage of finished transformation.

A company built to close that gap will face a particular temptation, repeatedly and persuasively. It will be offered revenue in exchange for opacity — for charging where value was not created, for making cancellation difficult, for holding data as leverage, for promising delivery it cannot staff. Each individual concession will be small, defensible, and argued for by intelligent people with a real business case.

This document exists to make those decisions in advance, in daylight, while nothing is at stake — so that the person who faces them later does not face them alone.

It is therefore written as a constitution rather than as a plan. A plan is a prediction and is usually wrong. A constitution is a commitment and can be kept. Where this corpus states a number it states the assumptions beneath it; where it states a principle it states the reasoning, so that a future reader may argue with the argument rather than guess at it; and where it entrenches a provision it does so precisely because that provision will one day be inconvenient.

Everything here is expected to change except the small number of things that must not.

FROM THE FOUNDER

A Message from the Founder

I did not set out to write a constitution. I set out to build a school's website.

What became clear in the building is that an institution's website, portal, admissions, results, certificates, identity and payments are not seven systems. They are one system wearing seven costumes, and the original error — the one nearly every institution in our markets has made — is buying them separately, from separate people, at separate times, and then employing someone whose real job becomes reconciling them by hand.

Once that is clear, it does not stay confined to schools. A hospital has the same shape. So does a ministry, a publisher, a law firm, a mosque. The vocabulary changes and the underlying structure does not.

I want StromeX to be the company institutions call when they decide to become modern. Not a vendor of components — a partner that takes responsibility for the whole outcome, charges a price it publishes openly, and is still there in a decade.

Two commitments matter more to me than the rest, and I have asked for both to be entrenched so that they cannot be traded away by a future version of this company under pressure.

The first is that verification is free, forever, for whoever is checking. A graduate's certificate belongs to the graduate. An employer, a university or a visa officer must never pay to confirm it is real, and it must never stop working because the school's subscription lapsed. That single commitment costs us a revenue line we could easily have taken, and it is the most valuable thing we will ever give away.

The second is that a customer's data is theirs, completely, and leaves with them whenever they wish. If institutions stay with us it must be because we are good, not because leaving is painful.

I am aware that the ambitions recorded in Volume VIII are large, and that most companies which set out with ambitions of that size do not reach them. This corpus is deliberately honest about that. It models three futures rather than one, and it says plainly which of them we should plan against and which we should merely keep possible.

What I am certain of is the standard. A child in Ikorodu should learn in an institution whose systems are as good as any in London, and should carry a certificate that is believed anywhere in the world without a phone call. Whether we become large is not fully within our control. Whether we are trustworthy is entirely within it.

If those two things ever conflict, Volume IX, Chapter 9 already contains the answer.

Ahmad Sulaimiy

Founder, StromeX Group Holdings

VISION AND MISSION

Vision & Mission

THE VISION — VOLUME IX §9.1

By 2127, that institutions serving ordinary people have extraordinary tools — and that where a person is born no longer determines the quality of the systems that shape their life.

THE MISSION — VOLUME I §1.2

To digitise, automate, modernise and intelligently transform organisations across every sector we can serve competently — and to make world-class execution available at a price the institution can actually plan around.

Three clauses of the mission are load-bearing. Every sector we can serve competently — breadth is the strategy, competence is the gate. Transform — the deliverable is the institution working better, measured in the institution's own units. A price the institution can plan around — pricing opacity is the primary mechanism by which institutions in emerging markets are overcharged, and removing it is both an ethical position and the group's sharpest competitive weapon.

ORIENTATION

How to Read This Corpus

This corpus is written to be consulted rather than read through. It is long by design: a document intended to govern for decades must answer questions its authors did not anticipate, and brevity is bought by leaving those answers out.

Read Volume I first regardless of your role. It is the root from which every other volume draws its authority, and a decision made without it will eventually contradict it.

Thereafter, read by need. Each volume opens with an executive summary sufficient for a reader who will not go further, and each chapter states its reasoning before its conclusion so that a reader who disagrees can locate precisely where.

Three conventions govern the whole corpus. Every recommendation states why it is included, what value it creates, what risk it introduces, what assumptions it rests on, and how success will be measured. Every estimate is labelled as an estimate and carries its assumptions inline — no figure here is a forecast of record or a guarantee. And nothing is fabricated: where the corpus reasons from established practice it says so, where it reasons from first principles it says so, and where it lacks a number it states the method by which the number should be obtained rather than inventing one.

NAVIGATION

Contents

A live Word field. In Microsoft Word press Ctrl+A then F9 (⌘A then fn+F9 on macOS) to repaginate it, or right-click and choose Update Field; every entry is an internal hyperlink. The lists of tables and figures and the index that follow are generated by the build and already carry final page numbers.

APPARATUS

List of Tables

Generated by the build from the corpus. Every table in the ten volumes is listed in order of appearance, grouped by volume.

VOLUME I — THE CONSTITUTION

VOLUME II — MARKET STRATEGY & COMPETITIVE POSITIONING

VOLUME III — THE CATALOGUE

APPARATUS

List of Figures

Generated by the build from the corpus. Architecture, stack and process figures in order of appearance.

VOLUME I — THE CONSTITUTION

VOLUME IV — ENGINEERING, AI ARCHITECTURE, CLOUD & SECURITY

VOLUME V — GO-TO-MARKET

VOLUME X — SPACETALK

VOLUME

I

The Constitution

Vision · Philosophy · Governance · Culture · Brand

AUTHORITY

Entrenched & Constitutional

EXECUTIVE SUMMARY

Volume I is the root of the corpus; every other volume derives its authority from it. It states why StromeX exists — that the gap between what an institution could be and what it is has little to do with ambition and everything to do with access to competent execution — and sets the doctrine of ecosystems, in which eight architectural layers are shared across every sector so that breadth becomes affordable. It establishes the eleven product principles, the value-before-revenue doctrine, the Free Constitution (including the permanent commitment that credential verification is free for the verifier), the Commercial Constitution and its regional price bands, the Ethics Constitution, the group structure and decision rights, the amendment protocol with its four tiers of authority, the brand system and the logo decision, the editorial voice, the design philosophy and the standing non-goals.

CHAPTER 1 — WHY STROMEX EXISTS

1.1 The observation the company is built on

Across most of the world, the gap between what an institution *could* be and what it *is* has almost nothing to do with ambition and almost everything to do with access to competent execution.

A school in Kano knows perfectly well that its certificates should be verifiable, that parents should see results on a phone, that admissions should not consume six weeks of a registrar's life, and that its textbooks should not be photocopies of a photocopy. It does not lack the desire. It lacks a vendor it can trust, a price it can plan around, and a supplier who will still be there in year three.

A hospital in Lagos, a ministry in Abuja, a law firm in Manchester, a publisher in Cairo, a factory in Ogun — the same shape of gap, different vocabulary. Each of them is quoted a number by a consultancy that is either so high it is unserious or so low it guarantees abandonment. Each of them ends up with four disconnected systems from four vendors, none of which speak to each other, and a staff member whose actual job has become manually reconciling them.

The world does not have a shortage of software. It has a shortage of **finished transformation**: someone who takes responsibility for the whole outcome, prices it honestly, builds it properly, and stays.

1.2 The mission

To digitise, automate, modernise and intelligently transform organisations across every sector we can serve competently — and to make world-class execution available at a price the institution can actually plan around.

Three clauses, each load-bearing:

— *Every sector we can serve competently* — breadth is the strategy, but competence is the gate. We do not enter a sector because it is large. We enter when we have, or can hire, a person who has actually done the work inside that sector.

— *Transform* — not "supply software to". The deliverable is the institution working better, measured in the institution's own units: days-to-admit, fee collection rate, certificate fraud incidents, hours of staff time returned.

— *A price the institution can plan around* — transparent, itemised, published. Pricing opacity is the primary mechanism by which institutions in emerging markets are overcharged. Removing it is both an ethical position and our sharpest competitive weapon.

1.3 The twenty-year ambition, stated honestly

We intend to build a globally significant technology group with diversified product lines, majority-recurring revenue, operations on at least three continents, and a reputation for trustworthiness that is the company's most valuable asset.

The founder's stated target is a group valued in the **$15–40 billion** range within ten to twenty years. This corpus takes that target seriously enough to model it properly in Volume VIII rather than repeat it as a slogan. The honest position is this:

> A target of that magnitude is achievable only under a specific and demanding conjunction of conditions — sustained ~55–70% compound revenue growth through the first decade, a durable gross margin above 65%, net revenue retention above 115%, successful entry into at least two

> developed markets, and access to growth capital at each stage. It is not a forecast. It is a design specification: *build the company so that, if execution is excellent and conditions are favourable, an outcome of that magnitude is structurally possible.* Volume VIII models the conservative, expected and optimistic paths side by side, states every assumption, and is explicit that the conservative path — a valuable, profitable, independent company an order of magnitude below the headline — is the most probable good outcome, and is not a failure.

Anyone in this company who repeats the headline number without the qualification is misrepresenting the company. Anyone who dismisses the ambition because it is hard is misunderstanding the job.

1.4 What SHRS is and is not

Sultan Hanafi Royal Schools is **Reference Implementation №1**. It is where the group learned, on a real institution with real consequences, that:

- an institution's website, portal, admissions, results, certificates, identity and payments are one system wearing five costumes, and building them separately is the original error;
- the binding constraint on delivery is almost never the code, it is the institution's own data hygiene and decision latency;
- verifiable credentials change the institution's relationship with its own alumni, which is a bigger deal than the feature list suggests;
- a beautifully executed deployment is a permanent sales asset, and one real reference outsells ten decks.

SHRS is not the ceiling, not the template every client receives, and not the reason a decision gets made. Any sentence in this company beginning "because that's how we did it at SHRS" must be followed by "and here is why it generalises" or it is not an argument.

CHAPTER 2 — THE DOCTRINE OF ECOSYSTEMS

2.1 The rule

We do not sell components. We build ecosystems and sell entry into them at whatever depth the customer can afford today.

The distinction is not marketing. It determines architecture, pricing, sales motion, and org design.

COMPONENT THINKING	ECOSYSTEM THINKING
"We build school websites."	"We build complete education ecosystems."
Deliverable is a site	Deliverable is an institution that runs on StromeX
Revenue ends at handover	Revenue compounds for a decade
Each sale starts from zero	Each sale starts from an installed base
Competitor can replace one part	Competitor must replace a nervous system
Customer relationship is transactional	Customer relationship is infrastructural

Table 1 — The rule

2.2 The ecosystem inventory

Each of the following is a designated ecosystem. Each is destined to become its own division with its own P&L, its own domain experts, and its own chapter in Volume VII. The list is open — additions follow the Volume I, Chapter 9 protocol.

Wave numbering is bound to the three-phase roadmap ratified in Volume VIII, Chapter 1 (**Phase I Foundation 2027–2030 · Phase II Race 2031–2035 · Phase III Global Scale 2036–2040**). No wave may open early because an opportunity looks attractive; the phase gates in Volume VIII, Chapter 2 govern.

Wave 1 — proven or adjacent · opens in Phase I (2027–2030) Education (K-12) · Higher Education · Publishing · Faith Institutions · Professional Services (law, accounting, consulting) · SME & Retail

Wave 2 — high fit, requires domain hire · opens in Phase II (2031–2035) Healthcare · Government & Public Sector · Financial Services & Fintech · NGO & Development · Hospitality · Real Estate & Construction

Wave 3 — large, slow, high-value · opens in Phase III (2036–2040) Manufacturing & Industry · Agriculture & Agritech · Logistics & Transport · Media & Broadcast · Energy & Utilities · Telecommunications · Research Institutions · Smart Cities

2.3 The anatomy of an ecosystem

Every ecosystem, in every sector, decomposes into the same eight layers. This is the group's central architectural insight and the reason breadth is affordable: **the layers are shared; only the top layer is sector-specific.**

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8 │ SECTOR SURFACE      Sector-specific apps, vocabulary, workflows, compliance
7 │ INTELLIGENCE        AI agents, analytics, forecasting, decision support
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6 | AUTOMATION         Workflow engine, approvals, document generation, integrations
5 | RECORDS            The system of record: people, money, assets, documents, events
4 | IDENTITY           Who is who: accounts, roles, cards, biometrics, credentials
3 | EXPERIENCE         Web, mobile, portals, kiosks, print — every surface
2 | PLATFORM           Auth, storage, search, messaging, payments, audit
1 | INFRASTRUCTURE     Compute, network, database, backup, security, observability
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Figure 1 — The anatomy of an ecosystem

A hospital and a school differ almost entirely at layer 8 and partially at layer 5. Layers 1–4 are identical. This is why StromeX can credibly serve twenty sectors with an engineering organisation sized for three — and it is why any proposal to fork the lower layers for a single customer requires executive approval (Volume IV, Chapter 3).

2.4 Depth before breadth, within a wave

Breadth across sectors is the twenty-year plan. Within any twelve-month window, the rule inverts: **go deeper in one ecosystem before opening another.** An ecosystem is “deep enough to leave” when it has ≥25 paying institutions, ≥110% net revenue retention, a reference customer willing to take calls, and a partner who can deliver without headquarters. Until then, opening a new sector is dilution wearing the costume of ambition.

CHAPTER 3 — PRODUCT PHILOSOPHY

3.1 What StromeX is

— An **institutional operating system** — the persistent substrate on which an organisation's people, money, documents, credentials and decisions live.
— A **transformation partner** — software plus the humans, design, consulting and hardware required to make the software actually land.
— A **compounding record** — the longer an institution runs on StromeX, the more it would cost them to leave, not because we lock them in, but because the accumulated record is genuinely valuable.

3.2 What StromeX is not

— Not a website agency. Websites are layer 3 of eight.
— Not a systems integrator that resells other people's software with a margin.
— Not an outsourcing shop that sells hours.
— Not an engagement-maximising, ad-funded attention product.
— Not a company that ships a demo and calls it a deployment.
— Not a company that will take a contract it cannot deliver in order to make a quarter.

3.3 The eleven product principles

P1 — Modularity is a right, not a feature. Every offering decomposes to a line item the customer can accept or decline. If a customer wants a website with no chatbot, no CRM and no app, that is a valid, respected, profitably-served configuration. Bundles exist to *simplify*, never to force.

P2 — Every capability is specified before it is priced. No line item enters Volume III without: purpose, target user, feature list, technical specification, delivery time, what is explicitly excluded, and the price. "Excluded" is mandatory. Most disputes in this industry originate in an unstated exclusion.

P3 — Progressive disclosure. A first-time visitor sees six choices. A procurement officer building a 60-module configuration sees everything. Same product, different depth. Complexity is available, never imposed.

P4 — The free tier must be genuinely, embarrassingly good. Governed by Chapter 5. A crippled demo is a broken promise with a countdown timer.

P5 — Nothing important is unexportable. Every customer can export every record they own, in an open format, at any time, without asking, without a fee. Lock-in earned by quality is legitimate; lock-in engineered through data hostage-taking is forbidden and will be treated as a P0 defect.

P6 — Boring where it matters, ambitious where it counts. Payments, auth, backups and audit logs are built conservatively with proven technology. Novelty is spent on the customer-visible frontier, not on the parts that must never fail.

P7 — Reversibility. Every destructive user action has an undo or a recovery window. Every deployment has a rollback. Every migration has a tested reverse path.

P8 — Offline and low-bandwidth are first-class. Large parts of our market operate at 2–5 Mbps with intermittent power. A product that assumes fibre and mains electricity is not a product for our market. Every surface has a defined degraded mode.

P9 — One institution, one record. No customer should ever be asked for data StromeX already holds. Re-keying is a defect.

P10 — Human-in-the-loop where stakes are high. Religious rulings, legal conclusions, medical diagnosis, disciplinary outcomes, admission rejections and financial write-offs are surfaced to a qualified human, never adjudicated silently by a model. Volume IV, Chapter 8 specifies the enforcement.

P11 — Speed is a feature and slowness is a bug. Perceived performance targets are contractual with ourselves: first meaningful paint under 1.8s on a 3G connection, interactive under 3.5s, API p95 under 400ms. A regression against these is a release blocker.

3.4 The tier ladder

Every offering that can be tiered uses this ladder, and only this ladder. No division invents its own tier names.

TIER	FOR	COMMERCIAL CHARACTER
Foundation	Anyone, forever	Free. Genuinely useful. No time limit, no credit card.
Standard	Small institutions, SMEs, individuals	Affordable, self-serve, transparent, no negotiation
Professional	Established institutions	The default serious choice; most customers belong here
Premium	Large or demanding institutions	Deeper modules, faster support, priority delivery
Elite	Flagship institutions	Bespoke design, named team, contractual SLAs, on-site
Enterprise	Multi-site groups, corporates	Custom-scoped, negotiated, multi-year
Government	Public sector	Procurement-track, tender-responsive, compliance-heavy
Global	Multinational deployments	Multi-jurisdiction, multi-currency, multi-residency

Table 2 — The tier ladder

The upgrade rule: upgrades happen because the customer outgrew the tier, never because we degraded the one below it. Deliberately withholding a trivially cheap capability from a lower tier purely to force an upgrade is prohibited. The legitimate reasons a capability sits in a higher tier are: it costs us materially more to deliver; it is only meaningful at that scale; or it carries risk requiring a higher-touch relationship.

CHAPTER 4 — THE VALUE-BEFORE-REVENUE DOCTRINE

4.1 The doctrine

Maximise trust. Maximise usefulness. Maximise delight. Revenue is the consequence, never the objective function.

This is not sentiment; it is the observed mechanism by which the companies we admire actually became large. Cloudflare gave away DDoS protection that competitors charged five figures for. GitHub gave away unlimited public repositories. Figma gave away the entire editor to viewers. Canva gave away a design tool good enough that most users never paid. Visual Studio Code was given away outright, and the ecosystem it created was worth more than any licence revenue would have been. In each case the free layer was not marketing spend — it was the distribution strategy, and it worked because the free thing was *genuinely excellent* rather than strategically hobbled.

4.2 The three-sentence test

Before any offering ships, it must pass:

1. **Would a stranger recommend this to a colleague without being asked?**
2. **Does the customer save more than we charge — and can they name the number?**
3. **If we doubled the price tomorrow, would we be ashamed, or merely nervous?** (Shame = the price is not earned. Nervous = the price is fine.)

4.3 Trust is the balance sheet

The group's most valuable asset is never in the accounts: it is the willingness of an institution to act on StromeX output without independently re-checking it. That asset takes years to build and one incident to destroy. Consequently:

- —We disclose incidents before we are asked, in public, with a timeline and a fix.
- —We never quietly change a price on an existing customer mid-term.
- —We never ship a "dark pattern": no manufactured urgency, no obstacle-course cancellation, no pre-ticked upsells, no auto-renewal without notice, no "contact sales to downgrade".
- —Cancelling must take fewer clicks than subscribing. This is measured and reported.
- —When we are wrong we say "we were wrong" in those words, without a passive voice.

4.4 The value ledger

Every customer relationship maintains a **value ledger** — a running, honest account of what the institution has gained: staff-hours returned, fraud incidents prevented, collection rate improvement, admissions cycle-time reduction, print cost avoided. It is shown to the customer at every renewal, including the periods where the number is unimpressive. A renewal conversation that cannot point to a value ledger is a renewal we have not earned.

CHAPTER 5 — THE FREE CONSTITUTION

5.1 The standard

The Foundation tier must provoke the reaction: **“I cannot believe this is free.”**

Not a trial. Not a demo. Not a watermarked tease. A permanently free set of tools that measurably improves the working life of a registrar, a bursar, a teacher, an imam, a small-business owner, a designer or a student who will never pay us a naira.

5.2 Why this is commercially rational, not charitable

Four mechanisms, each independently sufficient:

4. **Distribution.** Our market cannot be reached economically by paid acquisition. A registrar who found a free timetable optimiser at 11pm is a cheaper, warmer lead than any advertisement.
5. **Trust transfer.** An institution that has used a free StromeX tool for a year has already answered the hardest question in enterprise sales: “are these people real?”
6. **Data-free product intelligence.** Aggregate, anonymised usage of free tools tells us which sectors have which pain, months before a sales team could discover it.
7. **Standard-setting.** A free template used by 40,000 schools becomes the de-facto format. Whoever owns the format owns the workflow that surrounds it.

5.3 The Foundation catalogue

Permanently free, no account required unless marked †, no time limit, no watermark unless marked ‡.

Documents, templates and policy School admission forms · Examination and result templates · Report card templates · Transcript templates · Staff contract templates · Employee handbook skeleton · Student handbook skeleton · Parent handbook skeleton · Safeguarding policy generator · Data protection policy generator (NDPA/GDPR-aware) · Acceptable use policy generator · Anti-bullying policy generator · Fee policy templates · Procurement policy templates · Board charter templates · Governance calendar · HR letter library (offer, warning, promotion, exit) · Payroll spreadsheet templates · Accounting chart-of-accounts starters for schools, clinics, mosques and SMEs · Petty cash and requisition forms · Asset register template · Inventory templates · Meeting minute templates · Risk register template

Design and brand Logo preview generator ‡ · Colour system builder · Type-pairing tool · Icon pack (500+, open licence) · Certificate templates (25 designs) ‡ · ID card templates ‡ · Letterhead templates · Business card templates · Social media kit · Presentation templates · School branding starter kit · Signage templates · Prospectus template · Newsletter templates

Utilities QR code generator (unlimited, no expiry, no tracking) · Barcode generator · Website speed and SEO audit · Accessibility audit · Broken-link checker · Domain and SSL expiry checker · Image compressor · PDF merge/split/compress · Document format converter · Timetable conflict checker · GPA and grade calculators (multiple national systems) · Fee instalment calculator · Payroll tax estimators (NG, UK, US) · Currency and unit converters · Hijri/Gregorian calendar converter · Prayer time generator · Zakat calculator

Education Curriculum framework templates · Scheme of work templates · Lesson plan templates · Question bank starter (10,000 items, CC-licensed) · Marking rubric library · Reading list builder · Continuous assessment templates · Timetable optimiser (up to 30 classes) † · Attendance register (up to 200 students) † · Parent communication templates

Islamic and faith resources Qur'an text, translations and recitation index · Memorisation tracker † · Tajweed reference · Arabic learning starter pack · Hijri calendar tools · Mosque service schedule templates · Khutbah planning templates · Madrasah register templates · Islamic certificate templates ‡ · Waqf and endowment record templates

Developer and open source StromeX Design System (open source) · Component library · Arabic-first typography toolkit · Credential verification API (10,000 free verifications/month) · QR verification widget · Open-format exporters for every StromeX record type · Public status page and incident history · Documentation for every public API

Verification — permanently free, on principle Anyone, anywhere, may verify any StromeX-issued credential, forever, at no cost, without an account. The *issuer* pays. The *verifier* — an employer, a university admissions officer, a visa officer, a parent — never pays and never will. This is a constitutional commitment, not a pricing decision, and amending it requires the supermajority threshold in Chapter 9.

5.4 What Foundation deliberately excludes

Free stops where our marginal cost becomes material or where the capability only matters at institutional scale:

- Sustained compute (hosting, AI inference beyond the monthly allowance, video transcoding)
- Human labour (design services, consulting, editing, support beyond community and documentation)
- Physical goods (cards, hardware, printing)
- Institution-scale operation (unlimited students, multi-site, custom domains, SSO, SLAs)
- Compliance surface (audit exports, data residency, DPAs, penetration test reports)

Stating the boundary plainly is part of the gift. A free tier whose limits are hidden until you hit them is a trap, and traps are prohibited.

5.5 Open source policy

POSTURE	WHAT GOES HERE	WHY
Open source (permissive)	Design system, component library, Arabic typography toolkit, SDKs, exporters, verification widget, document format specs	Adoption is worth more than protection; these define standards and cost us little to give
Open source (copyleft)	Reference implementations of credential formats, integration adapters	We want derivatives to stay open so the format stays honest
Source-available	Sector modules where a customer needs to audit or self-host	Enterprise and government require inspectability; commercial terms still apply
Proprietary	Orchestration and agent layer, pricing/configuration engine, fraud and anomaly detection, operational tooling, the aggregated model of how institutions actually work	These are the accumulated advantage; giving them away funds a competitor without buying us adoption

Table 3 — Open source policy

The test: **open what spreads the standard; hold what compounds the advantage.**

CHAPTER 6 — THE COMMERCIAL CONSTITUTION

6.1 The monetisation principle, stated carefully

The founder's instruction is that everything of value should be monetisable, and that the cumulative total should be substantial without any individual price being frightening. That instruction is adopted, with one binding qualification that makes it durable rather than extractive:

> **If it creates measurable value for the customer, it may carry a price. If it does not create measurable value, it must not carry a price — no matter how easily it could.**

The failure mode we are guarding against is the airline model: a low headline price surrounded by fees for things that cost the seller nothing and that the customer reasonably expected to be included. That model produces revenue and destroys affection. We are building the opposite: a fair headline price, an itemised menu where each item is real, and a customer who feels they got more than they paid for.

Concretely, the following must never be separately charged: basic SSL, standard backups, security patches, uptime, the ability to export your own data, cancellation, or being spoken to by a human when something we built is broken.

6.2 The eleven revenue engines

#	ENGINE	CHARACTER	GROSS MARGIN BAND
1	**Subscription software**	Recurring, per-institution or per-seat	78–88%
2	**Consumption**	AI credits, compute, storage, verifications, messages	55–70%
3	**Implementation**	One-time setup, migration, configuration	40–55%
4	**Professional services**	Consulting, curriculum, policy, accreditation	45–60%
5	**Creative services**	Design, publishing, editing, translation, media	45–65%
6	**Print & fulfilment**	International print coordination, cards, signage	22–35%
7	**Hardware**	Gates, readers, kiosks, printers, IoT	18–30%
8	**Cloud infrastructure**	Hosting, VMs, storage, GPU, email, domains	45–65%
9	**Marketplace**	Third-party templates, agents, modules (rev-share)	70–85% on our share
10	**Licensing & white-**	Partners deploying	80–90%

#	ENGINE	CHARACTER	GROSS MARGIN BAND
	label	under their own brand	
11	**Financial services**	Payment facilitation, instalment plans, fee financing	35–55%

Table 4 — The eleven revenue engines

The portfolio rule: by 2031, engines 1, 2, 8 and 10 — the recurring, high-margin engines — must together exceed **65% of group revenue**. Services and hardware are strategically essential (they win deals nobody else can win and they create the relationship) but they do not compound, and a group whose majority revenue is human-hours cannot reach the scale described in Chapter 1.

6.3 Pricing philosophy

Six rules, binding on every division:

8. **Published by default.** Standard, Professional and Premium tier prices are public on the website, for everyone, in the customer's currency. Only Elite, Enterprise, Government and Global are quote-based, and only because their scope genuinely varies.
9. **Priced against value delivered, not cost incurred.** Cost sets the floor. Value sets the price. Competitors set the sanity check.
10. **Regionally fair.** Purchasing power differs by more than an order of magnitude across our markets. Charging Lagos and London the same number is not fairness, it is arithmetic. See §6.4.
11. **No surprise.** The configurator shows the running total, the annual total, and the three-year total *before* the customer gives us their name.
12. **The premium tiers must look premium.** An Elite price that looks cheap signals that the work is cheap. Where we do our finest work, the price must carry the same confidence as the work.
13. **Grandfathering.** A customer's price is fixed for their contract term. Increases apply at renewal, with 90 days' notice, and never exceed 12% annually without a documented, communicated increase in delivered scope.

6.4 Regional price bands

All prices in Volume III are quoted in **USD at Band C (list)**. Two multipliers derive the other bands.

BAND	MARKETS	MULTIPLIER	RATIONALE
Band A	Nigeria, West Africa, most of Sub-Saharan Africa, South Asia	**0.32×**	Serves the founding market at a price institutions can actually pay; still gross-margin positive because delivery labour is local
Band B	MENA, South-East Asia, Latin America, Eastern & Southern Europe, Turkey, South Africa	**0.65×**	Middle-income institutions with real budgets and real alternatives
Band C	United States, United Kingdom, EU, Canada, Australia, GCC premium,	**1.00×**	List. Priced against Western competitors, undercutting on

BAND	MARKETS	MULTIPLIER	RATIONALE
	Singapore, Japan		transparency rather than on rate

Table 5 — Regional price bands

Hardware, print and third-party pass-through costs are **band-invariant** — a card printer costs what it costs. Only labour- and software-derived prices band.

Anti-arbitrage: the band follows the *institution's operating jurisdiction*, verified at contract, not the billing address or the buyer's passport. A UK school group with a Lagos campus pays Band A for the Lagos campus and Band C for the UK ones.

6.5 The revenue-per-relationship ladder

The commercial objective for any institution is not the first invoice; it is the ten-year trajectory. A typical Professional-tier secondary school in Band A illustrates the intended shape:

YEAR	ENGAGED SURFACE	INDICATIVE ANNUAL VALUE (BAND A)
0	Free tools, then website + basic portal	$1,100 setup + $780/yr
1	+ Admissions, results, parent app, certificates	$2,400/yr
2	+ Payments, ID cards, AI assistants, SMS	$4,100/yr
3	+ Full ERP, HR/payroll, library, transport	$6,300/yr
5	+ Smart campus hardware, publishing, custom textbooks	$9,800/yr
7	+ Multi-campus, analytics, white-label parent brand	$14,500/yr
10	Institution fully resident on StromeX	$19,000–24,000/yr

Table 6 — The revenue-per-relationship ladder

Assumption stated: this ladder assumes no fee inflation, no student-roll growth, and successful cross-sell at the historical attach rates modelled in Volume VIII, Chapter 4. It is an illustration of the intended commercial shape, not a projection.

6.6 Prohibited commercial practices

Absolutely and permanently prohibited, in every market, regardless of local norm or competitive pressure:

- Bribery, facilitation payments, or "processing fees" to officials
- Bid-rigging, collusive tendering, or shell-company bidding
- Charging for security patches, uptime, or data export
- Auto-renewal without notice, or cancellation flows that exceed sign-up flows in length
- Contracts that claim ownership of customer data or student records
- Selling, brokering, or training third-party models on customer data (Volume IV, Chapter 9)

—Deliberately degrading a lower tier to manufacture an upgrade
—Quoting a price we know will need to be revised upward after signature
—Overstating what has been built. A demo is called a demo, in writing, always.

CHAPTER 7 — THE ETHICS CONSTITUTION

7.1 The automation question

We sell automation. Automation displaces tasks and sometimes people. The founder's framing — that with StromeX "not much more people will be required" — is commercially accurate and must be handled with precision rather than either denial or indifference.

Our position: we automate *tasks*, and we take responsibility for helping institutions redeploy *people*.

- —Every automation proposal above $10,000 in contract value includes a **workforce impact note**: which roles are affected, which tasks disappear, which new capacity is created.
- —We offer **redeployment training at no charge** for staff whose roles are materially changed by a StromeX deployment, capped at a defined number of seats per contract. It costs us little and it is the difference between being welcomed and being resisted — and resistance from staff is, practically, the single largest cause of failed deployments.
- —We do not market on headcount reduction. We market on capacity, accuracy and speed. A registrar who no longer spends six weeks on admissions is a registrar who can run alumni relations, which the institution never had.
- —Where a client's explicit goal is mass redundancy, we may still serve them — that is their lawful decision — but we will say plainly what we think, and we will not design the communication strategy for it.

7.2 The AI honesty rules

14. **We disclose that AI is involved.** The founder's instruction was that our blend of AI and human effort should be seamless, not that it should be concealed. Seamless is a quality standard. Concealment is a lie, and in publishing, education and credentialing it is a disqualifying one. We say "AI-assisted, human-reviewed" and we make it true.
15. **Human review is real or it is not claimed.** If a deliverable is marketed as human-reviewed, a named human reviewed it and is accountable for it.
16. **No fabricated authority.** No invented citations, no fictional expert quotes, no imaginary case studies, no synthetic testimonials, no fake reviews. Ever, anywhere, including in marketing.
17. **Refuse where stakes exceed competence.** Religious rulings, legal conclusions, medical diagnoses, and formal assessment decisions are escalated to qualified humans. Volume IV, Chapter 8.
18. **Attribution and licensing.** Training data, fonts, images, code and text used in deliverables are licensed. Volume VI, Chapter 9.

7.3 Data and privacy

- —Customer data belongs to the customer. Full stop.
- —We are a processor, not an owner. We do not sell, broker, or monetise customer data, and we do not train third-party models on it.
- —Children's data (a large share of our record, given education) receives the strictest handling in the group: minimisation, short retention, no behavioural profiling, no advertising use, parental access rights.
- —Data residency is offered where the law or the customer requires it, and honoured technically rather than contractually.
- —Breach notification: customers within 24 hours of confirmation, regulators per statute, public disclosure with a full timeline. No exceptions for embarrassment.

7.4 Sectors and work we decline

- Surveillance systems whose purpose is political monitoring, or biometric systems intended to track individuals outside a consented institutional context
- Weapons, munitions, and military targeting systems
- Gambling, predatory lending, and payday-loan style products
- Disinformation operations, astroturfing, or engineered political persuasion
- Credential systems designed to be forgeable, or “verification” that does not verify
- Any deployment where we have credible reason to believe the output will be used to persecute a group

A refusal is documented, communicated in one paragraph without moralising, and the nearest legitimate service we *can* provide is offered.

7.5 Faith and cultural fidelity

A significant part of our founding market is Muslim, and a significant part of our work touches Islamic education, publishing and institutions. The standard is exacting:

- Qur’anic text is never generated, paraphrased, or altered — only reproduced from verified sources with established provenance.
- Translation and tafsir are attributed to their named scholars and editions.
- No novel religious ruling is issued by a StromeX system. Questions of ruling are routed to qualified human scholars, with the routing visible to the user.
- Islamic design work respects the conventions of the tradition it serves, and where conventions differ between traditions we ask rather than assume.
- The same standard of fidelity applies to every faith and culture we serve. We do not have a “primary” tradition and a set of afterthoughts.

CHAPTER 8 — CORPORATE STRUCTURE & GOVERNANCE

8.1 Group structure

The intended structure, to be established as the group grows into it. Nothing here is a claim about the present state.

```
                        STROMEX GROUP HOLDINGS
                       (holding company, IP owner)
                                  │
      ┌───────────────┬──────────────┼──────────────┬───────────────┐
      │               │              │              │               │
STROMEX       STROMEX         STROMEX       STROMEX         STROMEX
TECHNOLOGY    CLOUD           CREATIVE      SERVICES        VENTURES
(products,    (infra,         (publishing,  (consulting,    (investments,
 platform,    hosting,        design,       training,       acquisitions,
 AI)          identity)       print, media) implementation) partner fund)
      │               │              │              │               │
      └───────────────┴──────────────┴──────────────┴───────────────┘
                                  │
                OPERATING ENTITIES BY JURISDICTION
        StromeX Nigeria · StromeX UK · StromeX US · StromeX MENA ...
```

Figure 2 — Group structure

Why a holding structure: it isolates liability by division (hardware and print carry different risk than SaaS), permits jurisdiction-specific compliance and data residency, allows divisions to be capitalised or partnered separately, and keeps IP ownership clean at the top — which matters enormously at any future financing or transaction.

8.2 Divisional P&L discipline

Each division owns a real P&L with its own gross margin target (Chapter 6.2), its own headcount plan, and its own quarterly review. Cross-division work is transfer-priced at documented internal rates. A division that persistently misses its margin band is restructured, repriced, or closed — breadth is not a licence to subsidise indefinitely.

Exception: **Foundation (free) tier costs are borne centrally**, not charged to divisions. Otherwise every division has an incentive to erode the free tier, which is the one thing this constitution most wants to protect.

8.3 Decision rights

DECISION	AUTHORITY	REQUIRES
Feature within an approved roadmap	Product lead	—
New module in Volume III	Division head + Pricing Council	Spec, price, margin model
Price change >10%	Pricing Council	Impact analysis, customer comms plan
New sector entry	Executive	Domain hire secured, Vol. VII chapter drafted

DECISION	AUTHORITY	REQUIRES
New country entry	Executive + Board	Vol. VIII entry checklist complete
Contract >$250k	Executive	Delivery capacity confirmed in writing
Anything touching Ch. 5 (Free) or Ch. 7 (Ethics)	Board	Supermajority per Ch. 9
Fork of platform layers 1–4 for one customer	CTO	Written justification, sunset date

Table 7 — Decision rights

8.4 The Pricing Council

A standing body — commercial, delivery, finance, and one engineer — meeting quarterly. It owns Volume III. Its duties: review every price against actual delivered margin, kill line items that nobody buys, add line items customers keep asking for, audit for creeping unfairness, and publish the changelog. Its single most important duty is to say no to prices that would work commercially and fail the Chapter 6.1 test.

8.5 The Ethics Review

Any of the following triggers a written ethics review before work starts: government surveillance adjacency, biometric deployment, work in a jurisdiction under active sanctions, a client whose stated goal is mass redundancy, religious content for a tradition we do not have a qualified reviewer for, or any deployment involving children's biometric data. The review is a memo, not a committee ritual, and it is filed and kept.

8.6 Board composition (target state)

Founder-led, with independent voices earned rather than decorative: at least one independent director with operating experience at scale, one with sector depth in education or public sector, and — once the group operates in three countries — one with international expansion experience. An audit function independent of the finance function from the point the group exceeds $25M revenue.

CHAPTER 9 — THE AMENDMENT PROTOCOL

9.1 Tiers of authority

TIER	CONTENT	TO AMEND
Entrenched	Ch. 5 §5.3 (free verification forever), Ch. 6.6 (prohibited practices), Ch. 7 (ethics), P5 (data export)	Board supermajority (≥75%), written rationale published internally, 30-day notice
Constitutional	The rest of Volume I	Executive consensus + board notification
Strategic	Volumes II, VII, VIII	Executive approval
Operational	Volumes III, IV, V, VI	Division head + relevant council; Vol. III additionally requires Pricing Council

Table 8 — Tiers of authority

9.2 Mechanics

19. Amendments are proposed as pull requests against `docs/bible/`.
20. Every amendment states: what changes, why, what it supersedes, and what becomes false if it is wrong.
21. Amendments to Volume III carry a mandatory impact note: affected customers, revenue delta, migration path for existing contracts.
22. Superseded text is struck through and dated for one edition, then archived — never silently deleted. The record of what we used to believe is part of the corpus.
23. Every volume carries a review date. Overdue review is a flag, not a failure — but an unreviewed chapter may not be cited as authority in a dispute.

9.3 The dissent record

Any employee may attach a dissent to any amendment. Dissents are preserved with the amendment permanently. A company that cannot remember who disagreed cannot learn from being wrong.

CHAPTER 10 — BRAND, IDENTITY & THE LOGO DECISION

10.1 The logo decision

Four identity candidates were presented. This chapter records the decision and the reasoning, because the reasoning will outlive the decision.

The evaluation criteria, in priority order, derived from what this company actually is:

24. **Reduction** — does it survive at 16px in a browser tab, embossed on a PVC card, and etched on a turnstile?
25. **Monochrome integrity** — does it work in one colour, on a fax, on a stamp, in a newspaper tender notice?
26. **Reproduction cost** — can a Lagos printer reproduce it correctly on the first attempt without a colour proof?
27. **Category signal** — does it read as *enterprise infrastructure* (Stripe, Microsoft, Oracle) rather than as luxury goods, crypto, or a gaming brand?
28. **Longevity** — will it look deliberate in 2046, or will it date to the rendering fashions of the mid-2020s?
29. **Ownability** — is the mark distinctive enough to be protected and recognised without the wordmark?

Assessment:

CANDIDATE	VERDICT
A — Chrome double-chevron X, blue gradient	Confident and legible, but the chevron pair is a heavily occupied space in tech identity (media players, logistics, fintech). Ownability is the weakness.
B — Blue/silver ribbon infinity-S	The most beautiful of the four as an image, and the weakest as a logo. Multi-stop gradients, self-intersecting ribbons and a dotted texture collapse below ~64px, cannot be embossed, and cannot be printed reliably in one colour. It is an illustration, not an identity.
C — Gold-and-silver hexagonal S with full system sheet	The most *complete* submission — it already thinks in app icon, monochrome, embossed, neon and gold variants, which is exactly the right instinct. But gold-on-black with a beveled isometric cube reads luxury/crypto rather than institutional infrastructure, and it is the hardest and most expensive of the four to reproduce faithfully at volume.
D — Interlocking ribbon S, silver with single blue accent	**Selected.**

Table 9 — The logo decision

The decision: Candidate D is the primary StromeX identity

Reasoning. Candidate D is the only one of the four that is *geometrically* rather than *pictorially* constructed. The mark is a single interlocking form with one accent element — which means it reduces cleanly to a favicon, survives single-colour reproduction, embosses on a card, cuts in vinyl, engraves on metal, and prints correctly on a Lagos press without a proof cycle. It carries the wordmark's blue X accent as a deliberate system rather than a coincidence, giving the identity an internal logic that scales into sub-brands. It reads as infrastructure — the category we are actually in — rather than as luxury, and its restraint is the visual argument for the Chapter 12 design philosophy. It is the mark that will still look correct when it is stamped on a government contract, a smart gate, a textbook spine and a 16px browser tab on the same day.

Candidate C's system sheet is adopted as the specification template for the brand kit: every StromeX identity asset ships with app icon, monochrome, embossed, reversed and digital variants, exactly as C proposed. C's *method* was right even though C's *mark* is not the one we ship.

Candidate C is retained as the approved treatment for the **StromeX Elite** service line and premium publishing imprints, where a heavier, more ornamental register is appropriate and reproduction is controlled. Candidates A and B are archived and not used.

10.2 The identity system

Primary mark: Candidate D, interlocking S. **Wordmark:** STROMEX, extended letterspacing, with the terminal X in the accent blue. **Lockups:** vertical (mark above wordmark), horizontal (mark left of wordmark), mark-only (≥24px), wordmark-only (for constrained horizontal spaces). **Clear space:** minimum of one mark-height on all sides. Non-negotiable. **Minimum sizes:** mark-only 16px digital / 8mm print. Full lockup 96px digital / 25mm print.

Palette

ROLE	NAME	VALUE	USE
Ground	Obsidian	`#05070A`	Primary dark ground, hero surfaces
Ground (light)	Paper	`#FBFBFD`	Light-mode ground
Primary	StromeX Blue	`#1B6EF3`	The accent. Actions, the X, focus states
Primary deep	Depth Blue	`#0B3C91`	Gradient anchor, pressed states
Metal	Platinum	`#D9DEE5`	Mark rendering, dividers on dark
Metal shadow	Graphite	`#6B7480`	Secondary text on dark
Ink	Ink	`#0D1117`	Body text on light
Success / Warning / Danger	—	`#12A150` / `#D98A00` / `#D62B34`	Semantic only, never decorative
Elite accent	Brass	`#C9A227`	**Elite tier and premium imprints only.** Prohibited in core product UI.

Table 10 — The identity system

Typography

ROLE	FAMILY	RATIONALE
Display / headings	Archivo (or Archivo Expanded at large sizes)	Geometric, wide, engineered — matches the mark

ROLE	FAMILY	RATIONALE
Body / UI	Inter	Optimised for screen at small sizes, huge language coverage
Arabic	Cairo (UI), Amiri (literary and Qur’anic)	Cairo for interface parity, Amiri for texts requiring traditional letterforms
Editorial / imprints	Fraunces	The publishing voice; not used in product UI
Code	JetBrains Mono	—

Table 11 — The identity system

Arabic is never rendered in a Latin font’s fallback. Bidirectional layout is a first-class requirement of every surface, not a localisation task (Volume IV, Chapter 11).

10.3 Sub-brand architecture

A **branded house**: everything is StromeX, with a descriptor. This maximises the transfer of trust from one division to another, which is the entire point of building a group rather than a portfolio.

`StromeX Cloud` · `StromeX AI` · `StromeX Identity` · `StromeX Pay` · `StromeX Learn` · `StromeX Health` · `StromeX Gov` · `StromeX Press` (publishing) · `StromeX Studio` (design) · `StromeX Intelligence` (analytics) · `StromeX Marketplace` · `StromeX Academy` (training & certification) · `StromeX Labs` (R&D) · `StromeX Ventures`

Sub-brands share the mark, the palette and the type system. They differ only in the descriptor and, where useful, a single accent variation. No sub-brand gets its own logo. Ever.

10.4 The naming rules

- —Descriptive over clever. `StromeX Admissions`, not `Gateway`.
- —No invented words for products. The company name is the only coined word we ask people to learn.
- —No version numbers in product names.
- —No internal codenames in customer-facing material.
- —Sector products take the sector’s own vocabulary: schools have a *registrar*, hospitals have *records*, ministries have *case files*. We adopt their word.

CHAPTER 11 — VOICE & EDITORIAL STANDARDS

11.1 The voice

Precise, warm, confident, unpretentious. We sound like a very good engineer who respects your intelligence and your time.

WE ARE	WE ARE NOT
Clear	Simplistic
Confident	Boastful
Warm	Chummy
Direct	Blunt
Serious about the work	Serious about ourselves

Table 12 — The voice

11.2 Rules of writing

30. Lead with what the reader gets, not with what we built.
31. One idea per sentence. Short sentences carry authority; long sentences hide in it.
32. Name the number. “Admissions in four days” beats “dramatically faster admissions”.
33. Active voice, especially when reporting failure. “We lost the data” — not “data loss occurred”.
34. No superlatives we cannot defend. Banned unless independently substantiated: *world-class*, *cutting-edge*, *revolutionary*, *best-in-class*, *unparalleled*, *game-changing*, *seamless* (as a claim rather than a standard), *AI-powered* used as a synonym for “good”.
35. No fear-selling. We do not sell by frightening a registrar about certificate fraud; we show them what verification does and let the comparison work.
36. Plain-language first. Every technical document opens with a paragraph a non-technical trustee could read aloud in a board meeting.
37. Localise, don’t translate. Nigerian English, British English and American English are all correct in their own market. Arabic is written by an Arabic writer, never machine-translated for publication.
38. Say “we don’t know” when we don’t know.
39. Never write a sentence about a customer that we would not read to them.

11.3 Document standards

Every customer-facing document carries: a title, a version, a date, an owner, and a statement of what it supersedes. Every proposal carries a scope, an explicit exclusion list, a delivery timeline with dependencies on the customer named, and a total that does not move after signature unless the customer changes the scope in writing.

CHAPTER 12 — DESIGN PHILOSOPHY

12.1 The eleven qualities

Every StromeX surface — product UI, website, certificate, ID card, book, proposal, invoice, signage, kiosk — must be: **premium, elegant, minimalist, fast, reliable, beautiful, professional, accessible, human, modern, timeless.**

“Timeless” is the demanding one. It rules out the styling that is currently fashionable and will read as dated in five years: heavy glassmorphism, animated gradient meshes as decoration, oversized rounded corners, illustration styles tied to a single era. It rules in typography, spacing, hierarchy, restraint and craft — which have never dated.

12.2 The design laws

40. **Typography is the interface.** Get the type scale and rhythm right and most layout problems disappear.
41. **Space is a component.** Whitespace is designed, budgeted and reviewed like any element.
42. **Motion explains, never entertains.** Animation communicates state, causality or continuity. Under 200ms for state, under 400ms for transition. `prefers-reduced-motion` is honoured everywhere.
43. **Colour carries meaning.** Semantic colours mean one thing. Decorative colour is rationed.
44. **Contrast is non-negotiable.** WCAG 2.2 AA minimum, AAA for body text where achievable. Verified in CI, not by eye.
45. **One-handed on a phone.** Primary actions inside the thumb arc.
46. **Empty states teach.** A blank screen is the highest-attention moment in the product and must be designed harder than the full one.
47. **Errors are apologies with instructions.** State what happened, what it means, what to do, and how to reach a human.
48. **Print is a design surface.** Certificates, cards, reports and books carry the same standard as the screen. Bleed, trim, colour profile, paper stock and binding are design decisions, not production details.
49. **Test on the real device.** A 5-year-old Android on 3G in a room with bad light is the reference environment, not a MacBook on fibre.
50. **Every product is bidirectional.** RTL is not a mode we add. It is a state the design is verified in before ship.

12.3 The design system

`StromeX Design System` (open source, per Chapter 5.5) is the single source of truth: tokens, primitives, components, patterns, and per-sector theme layers. Divisions may extend it; they may not fork it. A component that exists in three products belongs in the system, and the third team to need it is responsible for promoting it there.

CHAPTER 13 — CULTURE & THE OPERATING RHYTHM

13.1 The nine cultural principles

1. Craft is the differentiator. In a market where most vendors ship "good enough", finished work is a competitive moat. The last 10% of polish is 50% of the perceived value and nearly all of the referral.

2. Own the outcome, not the ticket. "It's not my module" is not an answer available to anyone here.

3. Write it down. Decisions of consequence are written, with the reasoning, in the repository. A decision that exists only in someone's head does not survive their annual leave, let alone their departure.

4. Bad news travels fastest. Escalating a problem early is rewarded, always, including when it turns out to be nothing. Concealing a problem is the one performance issue that is not coached — it is a trust breach.

5. Show the work. Demos over status updates. Working software over slides. Real data over lorem ipsum.

6. Argue hard, then commit. Disagreement is expected before a decision and disloyal after it. Dissent goes in the record (Chapter 9.3), and then everyone rows.

7. The customer's constraint is the design constraint. Bad connectivity, unreliable power, low device specs, staff with limited digital confidence, a bursar who trusts paper — these are the requirements, not the excuses.

8. Hire for judgement. Skills are teachable at pace; judgement is not. In interviews we weight "tell me about a decision you got wrong and what you changed" above credentials.

9. Sustainable pace. Heroics indicate a planning failure. We build for twenty years, and people who are exhausted in year three do not reach year twenty.

13.2 The operating rhythm

CADENCE	RITUAL	OUTPUT
Daily	Team standup, ≤15 min	Blockers named and owned
Weekly	Delivery review per active engagement	RAG status, escalations
Bi-weekly	Ship review	What actually shipped, demoed live
Monthly	Division P&L	Margin vs. band, pipeline, capacity
Monthly	Customer value ledger review	Renewal risk flagged 90 days out
Quarterly	**Pricing Council**	Volume III amendments + changelog
Quarterly	Corpus review	Which chapters are stale
Quarterly	Ethics review docket	Cases reviewed and decided
Half-yearly	Sector readiness review	Which ecosystem is deep enough to leave
Annually	Strategy offsite	Volume VIII roadmap amendment

Table 13 — The operating rhythm

13.3 What gets someone promoted here

Judgement demonstrated under uncertainty. Work that other people build on. Customers who ask for them by name. Documents that outlive the project. Mistakes surfaced early and learned from publicly. Colleagues who are better because they work alongside them.

Not: hours worked, tickets closed, lines written, meetings attended, or proximity to the founder.

CHAPTER 14 — THE STANDING NON-GOALS

Stated plainly so that no one has to guess, and so that a future employee can point at this page and decline.

51. **We will not become an outsourcing body shop.** Selling undifferentiated hours has no margin, no compounding and no ceiling worth climbing.
52. **We will not take a contract we cannot deliver.** Ever, for any amount, in any quarter, for any reason.
53. **We will not build an advertising business.** Our record contains children's education data. It will never be monetised through attention.
54. **We will not compete on being cheapest.** We compete on being clearest, most complete, and most reliable. There will always be someone cheaper and they will always be worse.
55. **We will not grow headcount as a signal.** Revenue per employee is a tracked metric; headcount is a cost.
56. **We will not enter a sector without a domain expert.** Confidence is not competence.
57. **We will not ship a demo as a product.** A demo is labelled a demo, in writing, in the same font size.
58. **We will not hold customer data hostage.** Export is free, complete, open-format and immediate.
59. **We will not chase a valuation at the expense of the constitution.** If a term sheet requires amending Chapter 5 or Chapter 7, we decline the term sheet.

VOLUME

II

Market Strategy & Competitive Positioning

Who we serve · Who we compete with · Why we win

AUTHORITY

Strategic

EXECUTIVE SUMMARY

Volume II locates the company in its market. It identifies four converging structural forces — the collapsed cost of building software against the unchanged cost of finishing it, the decade-long lag in institutional digitisation, the growing enforceability of credential fraud, and the shift from projects to subscriptions — and sets out the founding market thesis with its risks stated alongside. It supplies a method for market sizing rather than a headline figure, anatomises the buyer and the four other people in the room, applies established findings in behavioural economics while naming the techniques the group refuses to use, classifies the competitive landscape into six classes each requiring a different response, sets out the seven wedges, and closes with an honest assessment of the moat in which every advantage is stated together with the condition under which it expires.

CHAPTER 1 — THE STRUCTURAL OPPORTUNITY

1.1 Four forces, converging

StromeX is not betting on a trend. It is positioned at the intersection of four independent structural forces, each of which would justify a company on its own, and whose convergence is the actual opportunity.

Force 1 — The cost of building software collapsed, but the cost of finishing _it did not. AI-assisted development has compressed the labour required to produce working software by a large factor. It has compressed the labour required to_ specify, integrate, migrate, train, support and maintain software by very little. The result is a widening gap: an explosion of software that mostly does not get adopted, and a scarcity premium on organisations that can actually land it. StromeX is architected to arbitrage exactly this gap — heavy AI leverage on production, deliberately human investment on landing.

Force 2 — Institutional digitisation in emerging markets is a decade behind and moving fast. Most African, South Asian and Middle Eastern institutions have already digitised communications (WhatsApp reached them before any ERP did) but have not digitised *records* — admissions, finance, credentials, HR, assets. The gap between "has a smartphone" and "has a system of record" is the single largest addressable inefficiency in these economies, and mobile penetration means the last-mile problem is already solved for us.

Force 3 — Credential fraud is becoming enforceable. Certificate and transcript forgery has always been endemic; what changed is that employers, universities and immigration authorities increasingly *check*, and increasingly check digitally. An institution whose credentials cannot be verified online is beginning to suffer real consequences for its graduates. This converts verification from a nice-to-have into a competitive necessity for the institution — which is the strongest possible position for a vendor.

Force 4 — Procurement is shifting from projects to subscriptions. Institutions that once bought a system every eight years now expect to rent capability continuously. This is unambiguously good for a company designed around recurring revenue and continuous delivery, and unambiguously bad for the incumbent consultancies whose model depends on large, infrequent, capitalised projects.

1.2 The founding market thesis

Begin in Nigerian education, for six specific reasons, none of which is sentiment:

60. **Volume.** Nigeria has tens of thousands of private secondary and primary schools and hundreds of tertiary institutions, overwhelmingly under-served by software.
61. **Willingness to pay.** Private schools charge fees and compete on prestige. Prestige is purchasable, and modern digital infrastructure is visible prestige to fee-paying parents.
62. **Proof exists.** Reference Implementation №1 (SHRS) demonstrates end-to-end delivery on a real institution.
63. **The buying committee is small.** A proprietor or principal can sign. Compare a hospital or a ministry, where the same decision takes eleven months.
64. **Referral density.** School proprietors know other school proprietors. Word of mouth in this market is unusually efficient, which matters enormously when paid acquisition is uneconomic.
65. **Everything generalises.** A school is a small institution containing every problem a large one has: identity, records, money, credentials, communications, compliance, assets, people. Solving education properly builds layers 1–7 of the Chapter 2 stack for every other sector.

The corresponding risk, stated honestly: the Nigerian private education market is price-sensitive, FX-exposed, and has a meaningful share of institutions whose ability to pay a recurring subscription is genuinely fragile. Volume VIII, Chapter 6 models churn and collection assumptions accordingly, and Chapter 10 of this volume carries the risk.

CHAPTER 2 — MARKET SIZING METHOD

2.1 Why this chapter contains a method rather than a number

This corpus does not print market-size figures it cannot source. Published TAM figures for “African EdTech” or “global digital transformation” vary by an order of magnitude between analysts, are frequently circular, and are almost never useful for deciding anything. What follows is the method by which the group produces its own sizing, so that any figure quoted internally can be traced to its inputs and challenged.

2.2 The bottom-up formula

For any market:

```
Serviceable Revenue  =  Σ (institutions in segment
                         × share reachable by our channel
                         × realistic win rate
                         × annual contract value at that segment's band)
```

Each of the four inputs must be independently sourced and dated:

INPUT	SOURCE OF TRUTH	REFRESH
Institution count	National registries, ministry data, accreditation bodies, licensing lists	Annually
Reachability	Our own channel data: partner coverage, sales capacity, marketing reach	Quarterly
Win rate	Our own pipeline history by segment — never an assumption	Quarterly
ACV	Volume III at the applicable band, times observed attach rate	Quarterly

Table 14 — The bottom-up formula

Rule: any market-size claim in a StromeX document must cite the four inputs and the date. A figure without inputs may not appear in a board pack, an investor document, or a proposal.

2.3 Segmentation model

Institutions are segmented by **decision complexity**, not by size or revenue — because decision complexity is what actually determines sales cost, cycle length and win rate.

SEGMENT	SIGNATURE	CYCLE	MOTION	WHERE VALUE COMES FROM
Self-serve	One person decides, pays by card	Minutes–days	Product-led, zero touch	Volume, expansion revenue
Owner-led	Proprietor/ principal/founder decides	2–8 weeks	Low-touch, one salesperson	The core of Phase I
Committee	3–8 stakeholders, budget cycle	3–9 months	Full sales cycle, proposal, pilot	Higher ACV, stickier

SEGMENT	SIGNATURE	CYCLE	MOTION	WHERE VALUE COMES FROM
Procurement	Formal tender, evaluation matrix	6–24 months	Bid team, compliance, references	Largest ACV, slowest
Programme	Multi-institution, donor or ministry funded	12–36 months	Consortium, partner-led	Transformational, high risk

Table 15 — Segmentation model

Phase I deliberately concentrates on self-serve and owner-led. Committee and procurement open in Phase II, and only once the reference base makes them winnable — bidding for a ministry contract without twenty referenceable schools is not ambition, it is a donation of proposal costs.

CHAPTER 3 — THE BUYER ANATOMY

3.1 The economic buyer

Who actually signs, by sector:

SECTOR	SIGNS	CARES MOST ABOUT	KILLS DEALS OVER
Private K-12	Proprietor / Director	Prestige, parent perception, fee collection	Price, and fear of disruption during term
Public K-12	Ministry / Board	Compliance, reporting, cost per student	Procurement irregularity
University	Registrar / VC / Bursar	Accreditation, integrity of records	Data migration risk
Hospital	MD / CMD / Admin	Patient throughput, regulatory compliance	Clinical risk, downtime
Government	Permanent Secretary / DG	Delivery visibility, audit defensibility	Anything resembling scandal
Bank / Fintech	COO / CTO	Regulatory posture, uptime	Security review failure
Publisher	Publisher / MD	Speed to market, unit economics	Quality of finish
SME	Owner	Getting customers, getting paid	Complexity, monthly cost
NGO	Country Director	Donor reporting, beneficiary data	Cost per beneficiary
Mosque / Church	Imam / Board / Trustee	Community trust, propriety	Anything culturally careless

Table 16 — The economic buyer

3.2 The other four people in the room

The economic buyer signs, but four other roles decide:

- —**The champion** — usually an ICT coordinator, a young registrar, or an ambitious head of department. They want the institution modernised and their own standing raised by having brought it. *We arm them with materials they can present as their own.*
- —**The blocker** — typically the person whose manual process is being automated, or an incumbent vendor's relationship holder. Their objection is real and is never actually about the feature list. *We address it by naming it: this changes your role, here is the training, here is what you get to do instead.*
- —**The technical evaluator** — asks about hosting, backups, security, integration, and exit. *We answer with documentation, not reassurance.* The published security posture and the free export guarantee close more technical evaluations than any demo.
- —**The finance gate** — the bursar. Wants predictability above all, and hates variable bills. *We win here on published, itemised, plannable pricing — this is where transparency converts directly into revenue.*

3.3 The five objections and their answers

OBJECTION	WHAT'S UNDERNEATH	OUR ANSWER
"It's too expensive"	*I can't predict the total*	The configurator: itemised, running total, 3-year total, before we take their name
"We'll do it next term"	*I'm afraid of disruption mid-year*	Phased go-live; parallel running; nothing is switched off until the replacement is proven
"Our staff won't use it"	*Correct, if you don't train them*	Included training, in-language, on their devices, plus adoption reporting we're accountable for
"What if you disappear?"	*We've been abandoned before*	Free complete export, open formats, source-available options, published status history
"Can't we just use WhatsApp/Excel?"	*Genuinely, sometimes yes*	Then do — and here's the free tier. We win them when they outgrow it, and they will

Table 17 — The five objections and their answers

That last answer is doctrine, not politeness. Telling a 40-student school that they do not need our ERP builds more revenue over ten years than selling it to them today.

CHAPTER 4 — BUYER PSYCHOLOGY

Synthesis of well-established findings in behavioural economics, consumer psychology and pricing research, applied to our specific market. No fabricated studies; where a principle is contested, it is flagged.

4.1 Principles we deliberately apply

Anchoring. The first number a buyer sees frames every subsequent number. We anchor with the *complete ecosystem* price and then show how little the entry configuration costs — the entry price feels like relief rather than expense. This is legitimate because the anchor is a real, purchasable configuration, not an inflated decoy.

Loss aversion, applied to their loss, not manufactured fear. People weigh losses more heavily than equivalent gains. We frame around losses the institution is *already suffering* and can verify — staff hours, uncollected fees, forged certificates in circulation — never around invented threats. The distinction between “here is what you are already losing” and “here is what might happen to you” is the line between honest framing and fear-selling, and Volume I §11.2.6 makes the latter prohibited.

Choice architecture. Excessive choice reduces conversion. Our configurator’s default view offers a small number of curated configurations, with full modularity one click away. Both are real; neither is hidden.

The endowment effect, earned through the free tier. People value what they already possess. An institution that has used free StromeX tools for a year already feels ownership. This is the primary conversion mechanism of Chapter 5 of Volume I and is worth more than any discount.

Transparency as a trust shortcut. In low-trust markets, published pricing is not merely convenient — it is *evidence of character*. It signals we are not deciding what to charge based on how wealthy you look. In our founding market this is the single most disarming thing we do.

Social proof, correctly scoped. A proprietor is moved by the school down the road, not by a Fortune 500 logo. Reference customers are recruited and celebrated *locally*.

Reciprocity. Substantial free value given before any ask creates genuine obligation. This works only when the gift is real; a “gift” with strings attached inverts into resentment.

4.2 Techniques we deliberately refuse

Each of these reliably increases short-term conversion and reliably degrades trust. All are prohibited:

- Manufactured scarcity (“3 slots left this month”)
- Fake countdown timers
- Decoy pricing designed purely to make the middle tier look good
- Drip pricing — a low headline that grows during checkout
- Pre-ticked add-ons
- Confirmshaming (“No thanks, I prefer wasting money”)
- Cancellation obstacle courses
- Roach-motel onboarding — easy in, hard out
- Sales pressure on institutions that clearly cannot afford the commitment

4.3 The presentation of price

Derived from the founder’s instruction that prices should never look shocking, and that the elite tiers should look genuinely premium:

66. **Show unit economics, not just totals.** “$0.12 per certificate” lands very differently from “$1,200 for 10,000 certificates” — and both are true, so we show both.
67. **Show cost per beneficiary.** “$1.40 per student per year” is the number a proprietor can defend to a board.
68. **Show the comparison the buyer would make anyway.** Manual cost, incumbent cost, cost of doing nothing.
69. **Never hide the total.** The running total is always on screen.
70. **Let premium look premium.** Elite pricing is presented without apology, hedging, or discount language. A discounted Elite tier is a contradiction — the price is part of the assurance.
71. **Annual framing for recurring, one-time framing for capital.** Institutions budget in these two shapes; matching their mental accounting reduces friction honestly.

CHAPTER 5 — THE COMPETITIVE LANDSCAPE

5.1 The six competitor classes

We do not have one competitor. We have six kinds, and they must be beaten differently.

Class 1 — Global horizontal SaaS

Examples: Microsoft 365, Google Workspace, Salesforce, Zoho, HubSpot **Strength:** enormous, trusted, cheap at the entry point, already installed. **Weakness:** generic. They supply capability, not outcome. Nobody at Microsoft will migrate your student records or design your certificate. **Our posture: integrate, never fight.** We are the sector layer on top. A school on Google Workspace is a *better* prospect, not a worse one — the plumbing is already there. Explicit interoperability with these platforms is a permanent product requirement, not a roadmap item.

Class 2 — Global vertical SaaS

Examples: PowerSchool, Blackbaud, Ellucian, Instructure, Epic, Tyler Technologies **Strength:** deep domain functionality, decades of accumulated edge cases, entrenched. **Weakness:** priced for North America and Western Europe; implementation costs that exceed our entire contract value; localisation that stops at currency; and near-total absence from our founding markets. **Our posture:** avoid direct collision in their home markets during Phases I–II. Win our markets so completely that when we enter theirs in Phase III we arrive with scale, references and a structural cost advantage rather than as a challenger.

Class 3 — Regional and local vendors

Examples: local school-management systems, regional ERP shops, national portal builders **Strength:** local presence, local language, local price, existing relationships. **Weakness:** thin engineering, weak design, no security posture, no roadmap, high abandonment risk, single-product. Most cannot survive their founder leaving. **Our posture: this is the real competitive fight of Phase I.** We win on completeness, craft, reliability and published pricing. We must be careful not to win on arrogance — many of these vendors are the incumbent relationship, and the graceful path is often to acquire them, partner with them, or absorb their customers when they fail rather than to attack them.

Class 4 — Consultancies and systems integrators

Examples: Big Four advisory arms, regional SI firms, agency networks **Strength:** boardroom access, procurement fluency, credibility with ministries and donors. **Weakness:** they sell hours. Their incentive is duration, not outcome. They own no IP, so every engagement restarts from zero and the client pays for it. **Our posture: partner where they have access we lack; compete where the client has been burned.** Our decisive advantage is that we own the product, so our incentive is a fast, correct, permanent implementation. Say this plainly in competitive situations, without disparagement.

Class 5 — Freelancers and small agencies

Strength: cheap, fast, personally accountable, everywhere. *Weakness:* no continuity, no security, no support, no scale, no exit path. The single largest cause of abandoned institutional software in our market. **Our posture:** we do not compete at the bottom of this market — we *recruit* it. The best of these become StromeX Partners (Volume V, Chapter 6), which converts the most fragmented part of the competitive landscape into our distribution network.

Class 6 — AI-native newcomers

Strength: fast, cheap, technically current, no legacy. *Weakness:* almost all are a thin layer over a foundation model with no system of record, no implementation capability, no hardware, no compliance posture, and no reason for an institution to trust them with student data. **Our posture:** this is the class that could genuinely disrupt us, and it deserves respect rather than dismissal. Our defence is not our model — models commoditise. Our defence is layers 1–5 of the Volume I stack: the record, the identity, the credentials, the money, the physical estate. **A competitor can replicate our intelligence layer in a quarter. Replicating an institution's ten-year system of record takes ten years.**

5.2 The competitive rules

72. Never disparage a competitor by name to a customer. Compare on specifics, in writing, verifiably.
73. Never claim a capability we do not have. If a competitor genuinely does something better, say so and explain the trade.
74. Never win a deal by promising a delivery date we have not capacity-checked.
75. Losing to a better fit is acceptable. Losing to a worse product because we communicated badly is a defect and is reviewed like one.
76. Every loss is logged with a real reason. "Price" is almost never the real reason and is not accepted as one without evidence.

CHAPTER 6 — WHY WE WIN — THE SEVEN WEDGES

Seven advantages. None is individually decisive; the combination is very hard to assemble, which is precisely why it is the strategy.

Wedge 1 — Radical price transparency. Published, itemised, module-level pricing in a market where opacity is the norm. It disarms the finance gate, shortens the cycle, removes the negotiation tax, and signals character. Competitors can copy this — but Class 3 and 4 competitors mostly *cannot*, because their margins depend on the opacity.

Wedge 2 — Completeness. Website, portal, ERP, identity, credentials, payments, hardware, design, print, publishing and consulting from one accountable supplier. The institution stops being the integrator of last resort. No competitor class covers more than a third of this span.

Wedge 3 — The free layer. A permanently free tier substantial enough to be the market's default toolkit, and free verification forever. It builds distribution where paid acquisition does not work and converts the trust problem into a trust asset.

Wedge 4 — AI leverage with human accountability. Heavy AI leverage on production (design, drafting, code, translation, layout, data migration), deliberate human investment on judgement and landing. This produces developed-market quality at emerging-market cost — the structural margin advantage the whole plan rests on. Disclosed, never concealed (Volume I §7.2).

Wedge 5 — Craft. In a market where "good enough" is the norm, finished work is a moat. It is the reason for referral, the reason for premium pricing, and the reason an institution shows its portal to a rival proprietor.

Wedge 6 — The physical bridge. Almost no software company will also print your ID cards, install your gate, coordinate your UK book printing and design your prospectus. It is operationally messy and lower-margin — which is exactly why it is defensible. It also anchors the relationship in the institution's physical reality, where switching costs are real.

Wedge 7 — Emerging-market-native architecture. Offline-first, low-bandwidth, low-spec-device, intermittent-power, multi-currency, bidirectional-language, cash-adjacent. Retrofitting this into a product designed for fibre and mains power is close to impossible, which is why Class 1 and 2 competitors will not do it.

CHAPTER 7 — POSITIONING STATEMENTS

7.1 The group positioning

> **For institutions that need to become modern and cannot afford to get it wrong, StromeX is the digital transformation company that builds the whole ecosystem — software, design, identity, credentials, payments, publishing and physical infrastructure — at a published price, with the craft of a developed-market firm and the economics of an emerging-market one.**

7.2 The one-liner

> **StromeX builds the institution's operating system.**

7.3 By audience

AUDIENCE	STATEMENT
School proprietor	"Everything your school needs to run and to be seen as serious — website, portal, admissions, results, fees, verifiable certificates, ID cards — from one company, at a price on the website."
University registrar	"Records you can defend at accreditation, credentials the world can verify, and a migration we take responsibility for."
Ministry	"Delivery you can see, records you can audit, and a supplier whose pricing is public."
Hospital administrator	"Patient records, staff, billing and compliance in one system that works when the power doesn't."
SME owner	"Get found, get paid, get organised. Start free."
Publisher / author	"From manuscript to a book printed in the UK or US, designed to a standard your readers will notice."
Developer	"Open design system, open formats, documented APIs, free verification. Build on us."
Investor	"Recurring institutional infrastructure revenue in the fastest-growing markets on earth, with a structural cost advantage and a free layer that owns distribution."
Prospective employee	"Build things that matter for institutions that have never had them, to a standard that would pass in London, without moving there."

Table 18 — By audience

7.4 The five proof points every asset must be able to substantiate

77. Published, itemised pricing — verifiable on the website, right now.
78. Reference Implementation №1 delivered end-to-end.
79. Free tier and free credential verification, permanent, no card.
80. Complete data export, open format, no fee — contractual.

81. Public status page and published incident history.

Marketing may not claim a sixth thing until a sixth thing is true.

CHAPTER 8 — WHAT WE REFUSE TO COMPETE ON

Price as the primary axis. There is always someone cheaper. Competing on price against an operator with no engineering costs, no security posture and no intention of existing in three years is a race to a place we do not want to arrive at. We compete on transparency and completeness, and we let the cheap option be cheap.

Feature count. Feature-list warfare produces bloated products and is won by whoever has the most engineers. We compete on whether the institution's actual problem is solved.

Model benchmarks. We do not market on which foundation model we use or how it scores. Models commoditise; the record does not.

Speed of announcement. We do not pre-announce. A thing is announced when a customer can buy it. This costs us press and saves us the credibility that press would eventually consume.

Being everywhere at once. Geographic and sector breadth is the twenty-year plan, and depth is the twelve-month rule (Volume I §2.4).

Prestige logos we did not earn. No customer appears in our marketing without written permission and an accurate description of what we actually did for them.

CHAPTER 9 — THE MOAT, HONESTLY ASSESSED

A moat claim is worthless without an assessment of how it fails. Each of ours is stated with its expiry condition.

MOAT	STRENGTH	TIME TO BUILD	HOW A COMPETITOR BEATS IT
System of record	Very strong	3–10 years per customer	Only by waiting for us to fail a customer badly enough to justify migration. This is the real moat.
Switching cost from physical estate	Strong	1–3 years	Rip-and-replace of gates, cards and readers — expensive and disruptive, rarely worth it
Free-tier distribution	Strong	2–4 years	A better-funded rival giving away more. Real risk in Phase III. Defence: quality and breadth, not volume of giveaway
Craft and reputation	Moderate–strong	3–7 years	Hiring better designers than us. Entirely possible; requires them to also want our markets
Price transparency	Moderate	Immediate to copy	Trivially copyable in principle; blocked in practice by competitors whose margins require opacity. Assume erosion by Phase III
Partner network	Moderate	2–5 years	Outbidding us on partner economics. Defence: partners who are genuinely better off with us
Cost structure	Moderate	1–3 years	Any competitor adopting the same AI-leverage model. **Assume this advantage is temporary and largely gone by 2032.** It funds the durable moats; it is not itself durable
Regulatory / data residency posture	Moderate	2–4 years per market	Investment. Slow but not hard

Table 19 — The Moat, Honestly Assessed

The honest conclusion: only the system of record and the physical estate are genuinely durable. Everything else is a head start. The strategic implication is direct and governs Phase I: **get institutions onto the record as fast as possible, at whatever depth they will accept, and make leaving unattractive by being good rather than by being sticky.** Every other advantage exists to buy time for that one.

CHAPTER 10 — RISK REGISTER

Reviewed quarterly by the Executive. Each risk carries its owner and its live mitigation.

#	RISK	LIKELIHOOD	IMPACT	MITIGATION
R1	**FX volatility** erodes Band A revenue in USD terms	High	High	Price locally, cost locally; USD-denominate international contracts; maintain hard-currency reserve; band multipliers reviewed, not fixed
R2	**Collection failure** — institutions that sign and do not pay	High	Medium	Term-aligned billing; direct debit and mandate where possible; automatic feature degradation before disconnection; credit assessment above a threshold
R3	**Key-person dependency** on the founder	High	Very high	Write everything down (this corpus); build a genuine executive layer in Phase I; no undocumented decisions
R4	**Over-extension** — too many sectors before depth	High	High	Volume I §2.4 depth gate; Volume VIII phase gates; sector entry requires a domain hire
R5	**Delivery quality collapse under growth**	Medium	Very high	Partner certification; templated delivery; capacity confirmed in writing before signature; hard rule: never sign what we cannot deliver
R6	**Security incident involving children's data**	Medium	Catastrophic	Volume IV Chapters 6–9; independent audit cadence; minimisation by default; breach

#	RISK	LIKELIHOOD	IMPACT	MITIGATION
				protocol rehearsed, not merely written
R7	**Foundation model dependency** — pricing, availability, terms	Medium	High	Multi-provider routing (already in the MVP); no capability may depend on a single provider; self-hosted fallback for critical paths
R8	**Well-funded AI-native entrant** targeting the same wedge	Medium	High	Depth of record; physical estate; partner network; move faster in Phase I than a newcomer can
R9	**Political / regulatory shock** in a concentrated market	Medium	High	Geographic diversification is a risk control, not only a growth strategy — this is a core reason for the UK/US entries
R10	**Free tier costs outrun the funding capacity**	Medium	Medium	Central budget with a hard ceiling; per-tool unit-cost monitoring; degrade generosity before degrading quality
R11	**Reputational damage from an automation-driven redundancy story**	Medium	Medium	Volume I §7.1 workforce policy applied and visible; never market on headcount reduction
R12	**Hardware supply chain and import volatility**	Medium	Medium	Multi-supplier; local assembly where viable; hardware treated as an enabler, never a margin engine
R13	**Talent loss to overseas remote employers**	High	Medium	Pay competitively against remote-work benchmarks, not local ones; equity; the work itself is the

#	RISK	LIKELIHOOD	IMPACT	MITIGATION
				retention argument
R14	**Founder ambition outrunning capital**	Medium	High	Volume VIII capital plan; phase gates that cannot be skipped by enthusiasm
R15	**AI regulation** restricting deployment in education or health	Medium	Medium	Human-in-the-loop is already architectural (Volume I P10); disclosure already policy; compliance is a small delta rather than a rebuild

Table 20 — Risk Register

VOLUME

III

The Catalogue

Products · Services · Modules · Transparent Pricing

AUTHORITY

Operational — Pricing Council

EXECUTIVE SUMMARY

Volume III is the commercial heart of the corpus and the sole authority on StromeX prices. Seventeen divisions carry itemised, published, module-level pricing — web and digital presence, the institutional platform, identity and credentials, cloud, AI, payments, automation, mobile, intelligence, security, hardware and smart campus, design, publishing and print, consulting, the academy, the marketplace, and support — each line stating its purpose, specification, price, exclusions and upgrade path. It defines the regional band multipliers, discount authority, credits and verification tokens, and the anti-bill-shock rules. Every line is marked according to whether it is built, partial, or specified but not yet built. Four worked configurations show what an institution actually pays, including a state ministry deployment at approximately $1.48 per student per year.

> **A price quoted from memory, from a deck, from an email, or from a previous deal is not a StromeX price.** This volume is the sole authority. Where a signed contract differs, the contract is honoured for its term and this volume governs at renewal.

CHAPTER 0 — HOW TO USE THIS PRICE BOOK

0.1 Reading a price

- **All prices are USD at Band C (list).** Apply the multiplier from Volume I §6.4 for other markets: **Band A ×0.32**, **Band B ×0.65**, **Band C ×1.00**.
- `/mo` = per month, billed monthly or annually (annual = 10× monthly — two months free).
- `/yr` = per year only.
- **one-time** = single charge at delivery.
- **Hardware, third-party licences, print, freight, duties and payment-processor fees are band-invariant pass-through** and are quoted at cost plus the stated coordination margin.
- Every line item shows **what is excluded**. Exclusions are contractual.

0.2 The tier ladder

Foundation (free) · Standard · Professional · Premium · Elite · Enterprise · Government · Global. Defined in Volume I §3.4. No division invents its own tier names.

0.3 The configurator principle

The customer assembles their own solution. Bundles (Chapter 19) exist to simplify a common shape, never to force one. Every bundle is decomposable to its line items, and the configurator shows the running total, annual total and three-year total before we ask for a name.

0.4 Discount authority

DISCOUNT	WHO MAY APPROVE	CONDITION
0–10%	Sales lead	Multi-year or prepayment
10–20%	Division head	Strategic reference, documented
20–35%	Executive	Lighthouse account, written business case
>35%	Executive + Pricing Council	Exceptional; must not become precedent
Non-profit / low-fee school	Standing policy	**Up to 40% off software lines**, applied by policy not negotiation, verified once
Public school / government primary	Standing policy	Band A pricing regardless of geography for software lines

Table 21 — Discount authority

Discounting hardware, print or pass-through below cost is prohibited. Discount is never applied to make an Elite tier “affordable” — the correct answer is a lower tier.

0.5 Margin governance

Every line carries a target gross margin band (Volume I §6.2). The Pricing Council reviews delivered margin against target quarterly. A line item persistently below band is repriced, re-scoped, or retired.

0.6 Evidence base

The catalogue is grounded in delivered work, not aspiration. Reference implementations in the group's repositories as of this edition include: a four-language (English, Arabic RTL, Yorùbá, French) institutional site with ~35 unique page templates per locale, four distinct verification surfaces (certificate, identity, receipt, graduation document), a certificate generation system with a documented identifier architecture, security model and cryptographic signing-key deployment, a digital identity and card system, a finance platform, parent/student/teacher/registrar/executive portals, a personalisation centre, push notifications, a marketplace surface, and a 73-table credential-and-commerce schema covering enrolments, assessments, competencies, evidence, awards, credential signatures, issued documents, verifying institutions, payments, refunds, instalment plans, promo codes, multi-currency routing, corporate accounts and alumni chapters.

Line items marked ✅ are built and delivered somewhere in the group. Items marked ◐ are partially built. Items marked ○ are specified but not yet built — they may be sold only with a written delivery date and executive sign-off (Volume I §14.7: a demo is called a demo).

DIVISION 1 — WEB & DIGITAL PRESENCE

Target gross margin: 55–70% (build), 80%+ (recurring). Layer 3 of the ecosystem stack.

1.1 Website tiers

TIER	PAGES	BUILD (ONE-TIME)	CARE PLAN	DELIVERY	STATUS
Foundation	1 (hosted subdomain)	**Free**	Free	Instant	✅
Landing	1 bespoke	$340	$19/mo	5 days	✅
Portfolio	3–5	$780	$29/mo	10 days	✅
Standard	6–10	$1,650	$45/mo	3 weeks	✅
Professional	11–25	$4,200	$89/mo	5 weeks	✅
Premium	26–50	$9,500	$179/mo	8 weeks	✅
Elite	51–120, bespoke art direction	$24,000	$390/mo	12–16 weeks	✅
Enterprise / Government	Unlimited, multi-site	From $60,000	From $850/mo	Scoped	◐

Table 22 — Website tiers

Every tier includes, at no extra charge: responsive design to WCAG 2.2 AA, DV SSL, CDN, daily backups, security patching, uptime monitoring, a sitemap, structured data, an accessibility statement, analytics, a contact form, and complete source/content export. *These are never separately charged (Volume I §6.1).*

Excluded from all tiers: content writing (§1.3), photography (Division 12), translation (§1.4), third-party licences, domain registration, and any module below.

The care plan, specified

Not a vague retainer. Monthly: security patching, dependency updates, uptime monitoring with alerting, daily off-site backup with a tested restore, SSL renewal, broken-link scan, performance budget check, and a defined allowance of content edits (Landing 1 / Portfolio 2 / Standard 4 / Professional 8 / Premium 16 / Elite unlimited within reason, per month). Unused allowance does not roll over. Additional edits $28 each.

1.2 Page-level and structural add-ons

ITEM	PRICE	NOTES
Additional page (template reuse)	$95	
Additional page (bespoke layout)	$260	
Landing page for a campaign	$180	

ITEM	PRICE	NOTES
Microsite (3 pages, own domain)	$650	
Blog / news system	$340 one-time + $12/mo	✅
Events & calendar system	$290 + $12/mo	✅
Gallery / media library	$240 + $9/mo	✅
Downloads & document centre	$210 + $9/mo	✅
Announcements ribbon & notice system	$180 + $8/mo	✅
Policy / governance document library	$320 + $12/mo	✅
Staff & faculty directory	$290 + $12/mo	✅
Careers / vacancies module	$260 + $12/mo	
FAQ / knowledge base	$220 + $9/mo	
Testimonials & reviews	$160 + $6/mo	
On-site search (indexed, per-language)	$380 + $18/mo	✅
Personalisation centre (audience-tailored surfaces)	$1,400 + $45/mo	✅
Progressive Web App (installable, offline)	$680 + $22/mo	✅
Service-worker offline mode	$420 + $15/mo	✅
Dark mode	$180	
Print stylesheet	$120	

Table 23 — Page-level and structural add-ons

1.3 Content

ITEM	PRICE
Content audit of an existing site	Free
Page copywriting (standard, ≤600 words)	$130/page
Page copywriting (premium, researched)	$290/page
Institutional prospectus copy	From $1,900
Blog article (900–1,200 words,	$140

ITEM	PRICE
SEO-structured)	
Long-form article (2,000+ words, researched)	$340
Editorial style guide for the institution	$850
Content migration from an existing site	$6/page, min $180

Table 24 — Content

1.4 Multilingual

Pricing follows the founder's question directly: **charge per language, per page, with a maintenance component** — because that is where the cost actually sits.

ITEM	PRICE
Additional language — setup (locale registry, fonts, bidi/RTL layout, hreflang, switcher)	**$420 per language**
Human translation, standard content	**$11/page** (to 600 words)
Human translation, technical/legal/religious content	**$24/page**
AI-assisted translation, human-reviewed	**$4.50/page**
Right-to-left (RTL) layout engineering — Arabic, Hebrew, Urdu, Farsi	**$540 one-time per site**
Non-Latin typography engineering (Arabic, Yorùbá diacritics, CJK, Devanagari)	**$260 per script**
Multilingual maintenance	**$0.90/page/language/mo**
Full-site language audit (parity, missing strings, fallbacks)	$290

Table 25 — Multilingual

Standing practice (from the reference implementations): locale strings live in one flat dot-keyed dictionary per language; chrome is a single language-neutral template carrying translation tokens rather than a hand-maintained copy per language. This is what makes the fourth language cost a fraction of the second, and it is why we price setup separately from per-page translation. **We do not ship machine translation as finished work** — AI-assisted output is human-reviewed before publication, and Volume I §7.2 governs the disclosure.

1.5 SEO, performance & accessibility

ITEM	PRICE
SEO audit	**Free**
Performance audit (Core Web Vitals)	**Free**
Accessibility audit (WCAG 2.2)	**Free**
Technical SEO implementation	$480 one-time
Structured data / rich results	$220
Local SEO & business listings	$340 + $29/mo
SEO retainer — Starter	$290/mo
SEO retainer — Growth	$780/mo
SEO retainer — Enterprise	$2,200/mo
Accessibility remediation to AA	From $640
Accessibility remediation to AAA	From $1,500
Performance optimisation engagement	$560
Multi-region CDN & edge tuning	$240 + $35/mo

Table 26 — SEO, performance & accessibility

1.6 Domains & DNS

ITEM	PRICE
`.com` / `.net` / `.org` registration	$16/yr
`.ng` / `.com.ng`	$28/yr
`.edu.ng` / `.sch.ng` (with eligibility handling)	$65/yr
`.co.uk`	$14/yr
`.ai`	$95/yr
Premium & aftermarket domains	Market + 12%
Domain transfer & consolidation	Free
Managed DNS	$6/mo
DNS with DDoS-resilient anycast	$22/mo
Domain privacy	$8/yr
Multi-domain portfolio management (10+)	$45/mo

Table 27 — Domains & DNS

1.7 Certificates (TLS/SSL)

ITEM	PRICE	NOTE
DV SSL	**Free, always**	Auto-renewing. Never charged, on principle
OV SSL (organisation validated)	$89/yr	Where a tender requires it
EV SSL (extended validation)	$279/yr	Rarely necessary; we say so
Wildcard SSL	$169/yr	
Multi-domain (SAN, to 5)	$210/yr	
Code-signing certificate	$340/yr	For app distribution
Document-signing certificate	$290/yr	For sealed PDFs
Certificate lifecycle management	$18/mo	Expiry monitoring, auto-renewal, alerting

Table 28 — Certificates (TLS/SSL)

1.8 Email

ITEM	PRICE
Business mailbox — Lite (10 GB)	**$2.20/mailbox/mo**
Business mailbox — Standard (50 GB, calendar, contacts)	$4.60/mailbox/mo
Business mailbox — Pro (200 GB, archiving, retention)	$9.40/mailbox/mo
Role aliases (`info@`, `admissions@`, `bursary@`, `ict@`)	**Free, unlimited**
Shared / departmental mailbox	$3.20/mo each
Email migration from another provider	$3.50/mailbox, min $60
SPF / DKIM / DMARC configuration & monitoring	$140 one-time + $12/mo
Transactional email (API)	$0.60 per 1,000, first 5,000/mo free
Bulk / newsletter email	$0.90 per 1,000
Email signature management	$1.20/user/mo
Email archiving & legal hold	$2.80/mailbox/mo
Anti-phishing & advanced threat protection	$2.40/mailbox/mo

Table 29 — Email

DIVISION 2 — THE INSTITUTIONAL PLATFORM

The system of record. Layers 5–6. Target gross margin: 80–88%. Priced per institution per month unless stated. Student/staff counts are the primary scaling dimension.

2.1 Core platform

TIER	RECORDS INCLUDED	PRICE
Foundation	50 students / 10 staff	**Free**
Standard	300 students / 40 staff	$89/mo
Professional	1,000 students / 120 staff	$215/mo
Premium	3,000 students / 400 staff	$490/mo
Elite	8,000 students / 1,000 staff	$1,100/mo
Enterprise / Multi-campus	Unlimited	From $2,400/mo

Table 30 — Core platform

Additional students beyond tier: $0.14/student/mo. Additional staff: $0.70/staff/mo.

Core includes: people records, roles and permissions, org structure, document store, audit log, notifications, search, data export, and one administrative portal.

2.2 Portals & role surfaces ✅

MODULE	ONE-TIME	RECURRING
Student portal	$520	$34/mo
Parent / guardian portal	$520	$34/mo
Staff / teacher portal	$480	$30/mo
Registrar's office	$640	$42/mo
Bursary / finance portal	$640	$42/mo
Executive / board reporting portal	$780	$58/mo
Founder / proprietor dashboard	$560	$38/mo
Alumni portal & chapters	$580	$36/mo
Applicant portal	$420	$26/mo
Corporate / sponsor portal (seat management)	$690	$48/mo
Governor / trustee portal	$520	$32/mo
Instructor review & moderation console	$610	$40/mo

MODULE	ONE-TIME	RECURRING
Role & permission matrix configuration	$340	included

Table 31 — Portals & role surfaces ✅

2.3 Academic modules ✅

MODULE	ONE-TIME	RECURRING
Admissions & applications pipeline	$740	$52/mo
Enrolment & enrolment-integrity events	$520	$34/mo
Class, section & cohort management	$310	$20/mo
Timetable & scheduling	$460	$30/mo
AI timetable optimiser	$290	$24/mo
Attendance (manual + kiosk + biometric-ready)	$380	$26/mo
Gradebook & continuous assessment	$520	$36/mo
Examinations & question banks	$680	$46/mo
Online quizzes & assessment engine	$540	$38/mo
Competency & skills framework	$610	$42/mo
Learning outcomes & evidence mapping	$580	$40/mo
Evidence centre (portfolio of proof)	$720	$48/mo
Report cards & result publication	$440	$30/mo
Transcripts	$520	$36/mo
Curriculum & scheme-of-work management	$490	$32/mo
Programme definition & level architecture	$560	$36/mo
Course & unit progression tracking	$470	$32/mo
Time-on-task analytics	$380	$28/mo
Learning management	$860	$62/mo

MODULE	ONE-TIME	RECURRING
(LMS) & course player		
Live session / virtual classroom integration	$420	$34/mo
Assignment submission & marking	$460	$32/mo
Plagiarism & AI-authorship screening	$340	$38/mo
Spaced-repetition memorisation engine	$520	$34/mo
Qur’an memorisation & tajweed tracking	$640	$42/mo
Recitation recording & pronunciation feedback	$890	$68/mo
Listening lab / audio comprehension	$620	$44/mo
Vocabulary & exercise sets	$410	$28/mo
Self-check & formative assessment	$330	$22/mo
Awards, distinctions & honours register	$480	$32/mo
Graduate profile & graduate register	$690	$46/mo
CPD records (staff development)	$380	$26/mo
Academic bodies & external relations register	$340	$22/mo
Scholarships & bursaries administration	$520	$34/mo

Table 32 — Academic modules ✅

2.4 Administrative & operational modules

MODULE	ONE-TIME	RECURRING
Fee management & invoicing	$620	$44/mo
Instalment plans & payment scheduling	$410	$30/mo
Receipts & receipt verification	$340	$24/mo
HR & staff records	$560	$38/mo

MODULE	ONE-TIME	RECURRING
Payroll (per jurisdiction)	$740	$52/mo + $0.60/payslip
Leave & absence management	$310	$20/mo
Recruitment & applicant tracking	$480	$32/mo
Performance review & appraisal	$420	$28/mo
Procurement & requisitions	$520	$36/mo
Approval workflow engine	$580	$40/mo
Inventory & stores	$440	$30/mo
Asset register & maintenance	$460	$32/mo
Library management	$520	$34/mo
Digital library & e-resources	$640	$46/mo
Hostel / boarding management	$580	$40/mo
Transport & route management	$620	$44/mo
Vehicle tracking integration	$380	$32/mo + device
Catering & meal management	$420	$28/mo
Health / clinic records	$560	$40/mo
Discipline & pastoral records	$440	$30/mo
Safeguarding case management	$680	$48/mo
Visitor management	$360	$24/mo
Facilities & room booking	$390	$26/mo
Event management & ticketing	$470	$32/mo
Governance decisions & resolution register	$580	$38/mo
Board pack generation	$460	$34/mo
Statutory & regulatory reporting	$720	$54/mo
Data lifecycle & retention	$490	$32/mo

MODULE	ONE-TIME	RECURRING
register		
Data ownership register	$380	$24/mo

Table 33 — Administrative & operational modules

2.5 Communications

MODULE	ONE-TIME	RECURRING
In-app messaging & inbox	$380	$26/mo
Push notifications	$340	$22/mo
Email broadcast	$260	$18/mo
SMS gateway	$220	$16/mo + per-message
WhatsApp Business integration	$520	$44/mo + per-conversation
Voice / IVR broadcast	$480	$38/mo + per-minute
Emergency alert (multi-channel)	$560	$42/mo
Newsletter & mailing lists	$240	$16/mo
Parent–teacher appointment booking	$390	$26/mo

Table 34 — Communications

Per-message pass-through (band-invariant, at cost + 15%): SMS from $0.021 (NG) / $0.032 (UK) / $0.009 (US); WhatsApp conversation from $0.014; voice from $0.031/min. Published live in the configurator; never marked up beyond the stated coordination margin.

2.6 Implementation & migration

ITEM	PRICE
Standard onboarding & configuration	$480
Professional implementation	$1,900
Premium implementation	$4,800
Elite implementation (named team, on-site)	$12,000
Enterprise implementation	From $30,000
Data migration — clean spreadsheet	$0.35/record, min $180
Data migration — messy or mixed sources	$0.85/record, min $420

ITEM	PRICE
Data migration — legacy system extraction	From $1,600
Historical records digitisation (scan + key + verify)	$0.55/page
Parallel-running support (per month, during transition)	$340/mo
Data-readiness assessment	**Free**

Table 35 — Implementation & migration

Note on the data-readiness assessment being free: it is the single highest-value thing we do before a contract, it prevents most failed implementations, and charging for it would mean fewer institutions do it. It also qualifies the deal better than any sales call.

DIVISION 3 — IDENTITY, CREDENTIALS & VERIFICATION

The strategic heart of the group. Layer 4. Target gross margin: 78–88% (software), 25–35% (physical cards).

> **Constitutional commitment (Volume I §5.3): verification is free for the verifier, forever.** An employer, university, immigration officer or parent checking a StromeX-issued credential never pays and never needs an account. The issuer pays. This is not amendable at division level.

3.1 Credential issuance ✅

ITEM	PRICE
Credential platform — base	$640 one-time + $46/mo
Certificate template design — Standard	$180/template
Certificate template design — Premium (guilloche, engraved seal, foil-ready)	$520/template
Certificate template design — Elite (bespoke security artwork, custom cartouche, press specification)	$1,400/template
Identifier architecture design (institution-wide numbering scheme)	$680 one-time
Cryptographic signing-key generation & deployment	$420 one-time
Key rotation & custody procedure	$260/yr
Certificate generation (bulk)	**$0.11 each**, min 100
Certificate generation (on-demand, single)	$0.45 each
Transcript generation	$0.90 each
Graduation document generation (multi-page, sealed)	$1.60 each
Receipt generation & sealing	$0.06 each
Issued-document register	$340 + $24/mo
Revocation register & revocation workflow	$280 + $20/mo
Batch issuance workflow & approval chain	$460 + $30/mo
Certificate reissue / replacement handling	$0.80 each

Table 36 — Credential issuance ✅

3.2 Verification ✅

ITEM	PRICE
Public verification portal	$420 one-time + $32/mo
Verification by the public	**Free, unlimited, forever**
QR verification on every credential	Included
Verification API	Free to 10,000 checks/mo, then $0.004/check
Verifying-institution accounts (universities, employers, embassies)	Free
Bulk verification for an employer / registry	Free to 50,000/mo, then $0.002/check
Verification analytics for the issuer	$18/mo
Blockchain / public-ledger anchoring (optional)	$0.28/credential
Tamper-evidence report for a disputed credential	$40/report
Legacy credential back-loading (pre-StromeX certificates)	$0.22/record

Table 37 — Verification ✅

3.3 Digital identity ✅

ITEM	PRICE
Digital identity system — base	$580 one-time + $42/mo
Institutional identity number architecture	$520 one-time
Digital ID card (in-app, verifiable)	$0.22/holder/yr
Identity card design — Standard	$150
Identity card design — Premium (guilloche, holographic-ready, portrait system)	$460
Identity card design — Elite (bespoke, security-printed specification)	$1,200
Staff identity platform	$480 + $32/mo
Executive / dignitary identity design	$640
Photo capture & normalisation workflow	$0.18/portrait
Identity migration from a legacy scheme	$0.30/record

ITEM	PRICE
Single sign-on (SSO / SAML / OIDC)	$520 + $44/mo
Multi-factor authentication	$180 + $0.30/user/mo
Passkey / WebAuthn support	$340 + $18/mo
Biometric enrolment (fingerprint / face template)	$620 + $0.35/enrolee
Profile sharing & public graduate profile	$380 + $26/mo

Table 38 — Digital identity ✅

3.4 Physical credential production

Band-invariant. Card stock, printing and freight at cost + 22% coordination.

ITEM	UNIT PRICE
PVC card, plain, edge-to-edge print, both sides	$2.10
PVC card, premium (matte, spot UV, signature panel)	$3.40
Card with barcode / QR	$2.35
Card with magnetic stripe	$2.90
RFID / MIFARE proximity card	$3.60
NFC card (NTAG-class)	$4.10
Smart card with secure element	$7.80
Composite / polycarbonate card (high durability, 5–10 yr)	$9.40
Holographic overlay / hot-stamp foil	+$1.20
Laser-engraved personalisation	+$2.60
Lanyard, printed	$1.40
Card holder / sleeve	$0.55
Certificate printing — 250gsm premium	$1.90/sheet
Certificate printing — 300gsm cotton, deckled	$4.20/sheet
Certificate printing — security paper (watermark, UV fibres, microtext)	$6.80/sheet
Foil-blocked seal	+$1.60
Embossed / blind-debossed seal	+$1.10

ITEM	UNIT PRICE
Certificate folder / presentation case	$3.90
Personalisation & issuance run (labour)	$0.40/item, min $45

Table 39 — Physical credential production

3.5 The verification tokens

The founder's token concept, formalised. These are prepaid consumption units for verification-adjacent services that carry a real marginal cost. Verification by the public remains free.

TOKEN	PRICE	BUYS
Check	$0.50	One deep verification with a signed, timestamped, downloadable attestation PDF (for visa, employment or registry submission)
Attest	$1.00	One institution-to-institution attestation with counter-signature and audit trail
Seal	$2.00	One notarial-grade sealed document pack (multi-credential, sealed, ledger-anchored)
Legalise	$5.00	One credential prepared for consular/apostille submission, formatted to destination requirements

Table 40 — The verification tokens

Sold in packs (100 / 500 / 2,000 / 10,000) with 8% / 14% / 20% / 26% volume discounts. Institutions may resell these to their own alumni at their own margin — a revenue line for the institution and a reason to stay.

DIVISION 4 — STROMEX CLOUD

Target gross margin: 45–65%. Priced to be plannable, not to be the cheapest.

4.1 Managed hosting

TIER	SPEC	PRICE
Foundation	Static, 1 GB, subdomain	**Free**
Starter	1 vCPU, 1 GB RAM, 20 GB SSD	$7/mo
Business	2 vCPU, 4 GB RAM, 60 GB SSD	$22/mo
Professional	4 vCPU, 8 GB RAM, 160 GB SSD	$64/mo
Premium	8 vCPU, 16 GB, 320 GB SSD	$148/mo
Elite	16 vCPU, 32 GB, 640 GB SSD	$290/mo
Enterprise	Dedicated, HA, multi-AZ	From $780/mo

Table 41 — Managed hosting

4.2 Compute, storage & network (metered)

RESOURCE	PRICE
vCPU	$0.0115/hr
RAM	$0.0058/GB/hr
Block storage (SSD)	$0.085/GB/mo
Block storage (NVMe)	$0.16/GB/mo
Object storage	$0.018/GB/mo
Archive / cold storage	$0.0038/GB/mo
Snapshot storage	$0.045/GB/mo
Egress	First 100 GB/mo **free**, then $0.042/GB
Ingress	**Free**
Static IP	$2.40/mo
Load balancer	$14/mo
Private network / VPC	$6/mo
VPN gateway	$18/mo
Managed firewall	$12/mo
DDoS protection	**Free, included**

Table 42 — Compute, storage & network (metered)

4.3 Managed data services

SERVICE	PRICE
PostgreSQL — Starter (1 vCPU, 10 GB)	$16/mo
PostgreSQL — Business (2 vCPU, 50 GB)	$54/mo
PostgreSQL — Professional (4 vCPU, 200 GB, replica)	$165/mo
PostgreSQL — Enterprise (HA, PITR, multi-AZ)	From $420/mo
Redis / cache	$12–$180/mo by size
Vector database (embeddings)	$24–$340/mo by size
Search cluster	$28–$390/mo by size
Managed backups (daily, 30-day retention)	**Free, included**
Extended retention (1 yr)	$0.012/GB/mo
Point-in-time recovery	$18/mo
Cross-region replication	$34/mo + egress
Disaster-recovery site (warm standby)	45% of primary cost
Disaster-recovery drill (tested, documented)	$420/drill

Table 43 — Managed data services

4.4 GPU & AI compute

RESOURCE	PRICE
Inference GPU (entry class)	$0.68/hr
Inference GPU (standard class)	$1.75/hr
Training GPU (high-memory class)	$3.90/hr
Reserved GPU (monthly commitment)	32% discount
Model hosting (managed endpoint)	$46/mo + compute
Fine-tuning job	$18/hr + compute
Embedding generation	$0.018 per 1M tokens

Table 44 — GPU & AI compute

4.5 Platform services

SERVICE	PRICE
Authentication service	Free to 5,000 MAU, then $0.014/MAU
File storage & CDN delivery	$0.018/GB/mo + $0.042/GB egress
Queue / job runner	$0.40 per million operations
Cron / scheduled jobs	$4/mo
Webhooks & event bus	$0.30 per million events
Logging & retention	7 days free, then $0.09/GB/mo
Metrics & monitoring	**Free**
Uptime monitoring & alerting	**Free**
Error tracking	Free to 10,000 events/mo, then $0.0004/event
Status page (public)	**Free**
API gateway & rate limiting	$12/mo
Secrets management	$8/mo
Container registry	$0.09/GB/mo
CI/CD minutes	2,000/mo free, then $0.006/min

Table 45 — Platform services

4.6 Data residency & sovereignty

REGION	AVAILABILITY	PREMIUM
Nigeria	Phase I	Base
European Union	Phase I	+12%
United Kingdom	Phase II	+12%
United States	Phase II	+8%
GCC	Phase II	+18%
Customer-specified jurisdiction	Phase III	Quoted
On-premise / air-gapped deployment	Phase II	From $24,000 + $640/mo

Table 46 — Data residency & sovereignty

DIVISION 5 — STROMEX AI

Target gross margin: 55–70% after inference cost. Every agent is disclosed as AI (Volume I §7.2) and every high-stakes decision is human-reviewed (Volume I P10).

5.1 AI agents — subscription per agent, per institution

Administrative agents

AGENT	WHAT IT DOES	PRICE
AI Receptionist	Answers enquiries across web, WhatsApp and phone; routes and books	$44/mo
AI Admissions Officer	Guides applicants, screens completeness, schedules, chases documents	$86/mo
AI Registrar	Answers record queries, prepares transcripts, flags anomalies	$98/mo
AI Bursar / Finance Officer	Fee reminders, reconciliation prompts, arrears triage, plain-language statements	$92/mo
AI HR Assistant	Drafts letters, screens applications, answers policy questions	$74/mo
AI Secretary	Minutes, action tracking, correspondence drafting, diary triage	$58/mo
AI Procurement Officer	Requisition triage, quote comparison, supplier correspondence	$68/mo
AI Compliance Officer	Regulatory calendar, gap flags, evidence assembly	$110/mo
AI Executive Analyst	Board packs, executive summaries, KPI narrative	$124/mo

Table 47 — AI agents — subscription per agent, per institution

Academic agents

AGENT	WHAT IT DOES	PRICE
AI Teacher (per subject)	Explains, sets work, adapts to level, tracks misconceptions	$118/mo
AI Tutor (per student)	1:1 adaptive tutoring across subjects	$5.40/student/mo
AI Homework Marker	Marks and gives feedback; teacher approves before release	$0.035/script page
AI Curriculum Designer	Schemes of work, lesson plans, differentiation, mapping to standards	$96/mo
AI Examination Officer	Question generation, blueprint balance, difficulty calibration	$104/mo

AGENT	WHAT IT DOES	PRICE
AI Librarian	Catalogue Q&A, reading recommendation, research guidance	$62/mo
AI Research Assistant	Source retrieval, synthesis with citations, literature mapping	$88/mo
AI Translator	Document and page translation, human-review workflow built in	$54/mo
AI Arabic Tutor	Grammar, morphology, vocabulary, graded reading	$78/mo
AI Qur'an Memorisation Coach	Scheduling, revision, error tracking; recitation review is human-confirmed	$84/mo
AI Timetable Optimiser	Constraint-solved timetables, conflict resolution	$24/mo
AI Inspection Readiness Assistant	Evidence assembly against an inspection framework	$92/mo
AI Parent Assistant	Answers parent questions from the institution's own record	$58/mo
AI Student Companion	Study planning, wellbeing check-ins, escalation to a human	$6.20/student/mo

Table 48 — AI agents — subscription per agent, per institution

Sector agents (Volume VII)

AGENT	PRICE
AI Patient Coordinator (health)	$128/mo
AI Clinical Documentation Assistant	$164/mo
AI Case Officer (government)	$148/mo
AI Legal Research Assistant	$156/mo
AI Contract Analyst	$142/mo
AI Underwriting Assistant (insurance)	$168/mo
AI Loan Officer Assistant (finance)	$158/mo
AI Customer Support Agent (any sector)	$68/mo
AI Sales Development Agent	$88/mo
AI Marketing Assistant	$72/mo
AI Editor (publishing)	$94/mo
AI Fact-Checker	$86/mo

Table 49 — AI agents — subscription per agent, per institution

Every agent includes: institution-scoped knowledge grounding, citation of the institution's own source documents, conversation logs, an escalation path to a named human, configurable refusal boundaries, and a monthly credit allowance (Chapter 18). Beyond allowance, credits are consumed at published rates.

5.2 AI platform

ITEM	PRICE
Knowledge base & retrieval (institution-scoped)	$340 + $38/mo
Document ingestion & indexing	$0.008/page
Enterprise search across all institutional data	$520 + $64/mo
Custom agent builder	$180 + $34/mo
Agent marketplace access	Free (Division 16)
Multi-model routing (provider-agnostic)	Included
Bring-your-own-model / key	Included, no surcharge
Self-hosted model deployment	$1,900 setup + compute
Fine-tuning on institutional data	From $1,400
Voice agent (speech in / speech out)	$86/mo + $0.024/min
Real-time translation (live sessions)	$0.09/min
Transcription	$0.014/min
OCR & handwriting recognition	$0.006/page
Document understanding & extraction	$0.011/page
Image generation (brand-constrained)	$0.03/image
AI content detection & authorship screening	$0.004/document

Table 50 — AI platform

5.3 Guardrails — included at no charge, mandatory, non-removable

Refusal boundaries per domain · human escalation routing · citation enforcement in factual modes · hallucination monitoring · prompt-injection defence · PII redaction in logs · full conversation audit trail · per-agent kill switch · the standing prohibition on training third-party models on customer data.

These are not a premium tier. Selling safety as an upgrade would mean selling an unsafe default, which Volume I forbids.

DIVISION 6 — STROMEX PAY

Target gross margin: 35–55%. Processor fees are pass-through and disclosed.

ITEM	PRICE
Payment module	$290 one-time + $24/mo
StromeX platform fee	**0.4%** on top of the processor's rate
Processor fees	Pass-through at cost, published
Multi-currency support	$180 + $18/mo
Country payment routing	$240 + $22/mo
Payment links	Free
Invoicing	$16/mo
Recurring billing & subscriptions	$34/mo
Instalment plans (school fees in tranches)	$30/mo
Promo codes & scholarships applied at checkout	$14/mo
Wallets & stored balance	$420 + $38/mo
Split payments / multi-beneficiary	$28/mo
Escrow (milestone release)	$46/mo + 0.6%
Donations & fundraising	$18/mo
Zakat & sadaqah collection with designation tracking	$22/mo
Refunds & reversal workflow	Included
Reconciliation & settlement reporting	$26/mo
Chargeback handling	$18/mo + $12/case
Cash & bank-transfer capture (for cash-heavy markets)	$20/mo
POS terminal integration	$32/mo + hardware
Fee financing referral (third-party lender)	Rev-share, disclosed
Payout to local bank accounts	$0.30/payout

Table 51 — Stromex Pay

Rule: we never take a hidden spread on FX. The rate applied and the margin taken are both shown on the transaction record.

DIVISION 7 — AUTOMATION & INTEGRATION

Target gross margin: 60–75%. The founder's brief: anything repetitive, anything a person does ten times a week, anything involving forms, approvals, signatures, documents or reconciliation.

7.1 Automation platform

ITEM	PRICE
Automation engine — base	$290 + $28/mo
Automation runs	5,000/mo included, then $0.0009/run
Visual workflow builder	Included
Approval chains (multi-stage, delegated)	$180 + $18/mo
Digital signatures	$240 + $22/mo + $0.14/signature
Qualified / advanced e-signature	$0.85/signature
Document generation from templates	$160 + $16/mo + $0.02/document
Mail merge & bulk personalisation	$120 + $10/mo
Form builder (unlimited forms)	**Free**
Advanced forms (logic, payments, file upload)	$22/mo
PDF assembly, sealing & watermarking	$140 + $12/mo
Scheduled reports	$14/mo
Data sync between systems	$34/mo per connection
Robotic process automation (legacy screen automation)	From $1,400
Custom automation build	$95/hr or fixed quote

Table 52 — Automation platform

7.2 Prebuilt automation packs

PACK	WHAT IT AUTOMATES	PRICE
Admissions pack	Application → screening → interview → offer → acceptance → enrolment	$340
Fee pack	Invoice → reminder → instalment → receipt → arrears escalation	$340
Certificate pack	Result finalisation → eligibility → generation → seal → issue → verify	$420
Onboarding pack (staff)	Offer → documents → accounts → induction → probation	$290

PACK	WHAT IT AUTOMATES	PRICE
Offboarding pack	Exit → access revocation → asset return → final pay → reference	$260
Procurement pack	Requisition → quotes → approval → PO → receipt → payment	$360
Attendance pack	Capture → exception → parent notification → escalation	$220
Result-publication pack	Entry → moderation → approval → publication → parent alert	$300
Inspection pack	Evidence request → collection → mapping → dossier	$380
Alumni pack	Graduation → profile → verification → chapter → engagement	$280
Safeguarding pack	Concern → triage → case → escalation → record	$400
Board pack	Data collection → pack assembly → distribution → minutes → actions	$340

Table 53 — Prebuilt automation packs

7.3 Integration

ITEM	PRICE
Public REST API	**Free**
Webhooks	**Free**
SDKs (JS, Python, PHP)	**Free, open source**
Standard connector (Google Workspace, Microsoft 365, Zoom, Teams)	$120 each + $9/mo
Accounting connector (QuickBooks, Sage, Xero, Zoho Books)	$180 each + $14/mo
Payment gateway connector	$140 each + $11/mo
SMS / WhatsApp gateway connector	$120 each + $9/mo
Government / regulator reporting connector	From $680 + $46/mo
Examination board connector (WAEC, NECO, Cambridge, Edexcel)	From $520 + $38/mo
Bank statement / open banking connector	$340 + $28/mo
Custom connector build	From $890

ITEM	PRICE
Legacy system bridge	From $1,600
Data warehouse export (scheduled)	$46/mo

Table 54 — Integration

DIVISION 8 — MOBILE & APPLICATIONS

ITEM	PRICE
Progressive Web App (installable, offline)	$680 + $22/mo
Android app (branded wrapper, store-ready)	$2,400 + $58/mo
iOS app (branded wrapper, store-ready)	$2,900 + $58/mo
Android + iOS bundle	$4,600 + $88/mo
Native app (bespoke, single platform)	From $14,000
Native app (bespoke, both platforms)	From $24,000
App store submission & review management	$340/store
Store listing assets (icon, screenshots, copy)	$290
Push notification service	$340 + $22/mo
In-app purchases / subscriptions	$420 + $34/mo
Offline-first data sync	$560 + $38/mo
Deep linking & universal links	$180
App analytics & crash reporting	$24/mo
Over-the-air update channel	$28/mo
Annual store compliance & SDK maintenance	$420/yr per platform
White-label app for a partner’s client base	Division 16
Desktop application (Windows / macOS / Linux)	From $9,000
Kiosk application	$1,900 + $34/mo
Smart TV / digital signage app	$2,200 + $38/mo
Wearable companion	From $4,200
USSD service (feature-phone access)	$1,400 + $180/mo + telco fees

Table 55 — Mobile & Applications

DIVISION 9 — INTELLIGENCE & ANALYTICS

ITEM	PRICE
Standard dashboards	Included in platform
Custom dashboard	$290 each
Executive reporting system	$780 + $58/mo
Founder / proprietor dashboard	$560 + $38/mo
Business intelligence workspace	$640 + $86/mo
Predictive analytics — enrolment forecasting	$520 + $64/mo
Predictive analytics — attrition / dropout risk	$560 + $68/mo
Predictive analytics — fee default risk	$540 + $66/mo
Predictive analytics — demand & capacity planning	$580 + $70/mo
Cohort & longitudinal analysis	$420 + $46/mo
Benchmarking against anonymised peer set	$38/mo
Learning analytics & comprehension gap detection	$490 + $54/mo
Financial modelling & scenario planning	$610 + $72/mo
Data warehouse	$840 + $124/mo
ETL pipelines	$340 + $44/mo per pipeline
Report builder (self-service)	$28/mo
Scheduled report distribution	$14/mo
Anomaly detection & alerting	$380 + $42/mo
Fraud detection (fees, credentials, attendance)	$620 + $78/mo

Table 56 — Intelligence & Analytics

Benchmarking is opt-in, aggregated, anonymised and never resold. An institution can see how it compares without any institution being identifiable, and no third party ever receives the data (Volume I §7.3).

DIVISION 10 — SECURITY & COMPLIANCE

ITEM	PRICE
Security patching & dependency updates	**Free, always**
TLS, WAF, DDoS protection	**Free, included**
Backups with tested restore	**Free, included**
Audit log	**Free, included**
Security posture documentation	**Free**
Vulnerability scanning (continuous)	$34/mo
Penetration test — application	$2,400
Penetration test — infrastructure	$2,900
Penetration test — full scope with retest	$6,800
Red team engagement	From $14,000
Security monitoring (SIEM)	$86/mo
24/7 security operations	$420/mo
Incident response retainer	$340/mo
Incident response (ad hoc)	$180/hr
Data protection impact assessment	$980
NDPA compliance readiness (Nigeria)	$1,400
GDPR compliance readiness	$2,200
ISO 27001 readiness assessment	$3,400
ISO 27001 implementation support	From $11,000
SOC 2 readiness	From $9,000
Data processing agreement	**Free**
Data residency configuration	Division 4.6
Records retention policy design	$620
Business continuity plan	$1,200
Disaster recovery plan + drill	$1,600
Security awareness training	$22/staff
Phishing simulation programme	$180/mo
Access review & least-privilege	$740

ITEM	PRICE
audit	
Child data protection audit (education-specific)	$1,100

Table 57 — Security & Compliance

DIVISION 11 — HARDWARE & SMART CAMPUS

Band-invariant. Hardware at cost + 22%; installation and commissioning at local labour rates. ***Hardware is an enabler, not a margin engine*** *(Volume II R12) — we do not push hardware to make a quarter.*

11.1 Access & security

ITEM	PRICE
Smart gate controller (per lane)	$840
Automatic sliding gate motor + controller	$1,650
Boom barrier	$1,280
Full-height turnstile	$2,350
Waist-height tripod turnstile	$1,180
Speed gate (optical)	$4,600
Door access controller	$320
Electromagnetic lock	$95
Electric strike	$62
RFID / NFC reader	$88
Long-range UHF reader (vehicle)	$410
Fingerprint reader	$260
Face recognition terminal	$480
Palm-vein reader	$690
Number-plate recognition camera	$740
IP camera (indoor)	$110
IP camera (outdoor, IR)	$175
NVR (16-channel)	$520
Intercom / video door station	$290
Panic button (wired / wireless)	$65
Intrusion alarm panel	$340

Table 58 — Access & security

11.2 Identity & issuance hardware

ITEM	PRICE
Card printer — single-sided	$1,380
Card printer — dual-sided	$2,150

ITEM	PRICE
Card printer — retransfer (premium finish)	$4,200
Card printer ribbon (250 prints)	$145
Lamination module	$980
Photo capture station (camera, backdrop, lighting)	$620
Signature pad	$130
Document scanner (ADF)	$380
Passport / ID scanner	$690
Card encoder (MIFARE / DESFire)	$240

Table 59 — Identity & issuance hardware

11.3 Campus & operations

ITEM	PRICE
Attendance kiosk (tablet, mount, reader)	$1,080
Self-service kiosk (touchscreen, printer)	$2,400
Digital signage display (43”)	$610
Digital signage display (65”)	$1,240
Signage media player	$180
Interactive whiteboard (75”)	$2,900
Classroom projector + screen	$740
PA / bell system (per zone)	$480
Two-way radio (per handset)	$120
Thermal receipt printer	$165
Label / badge printer	$280
Barcode scanner	$85
POS terminal	$340
Cash drawer	$110
Laboratory sensor kit (IoT, STEM)	$520
Environmental sensor (temp/humidity/air)	$145
Water tank level sensor	$190
Energy meter (smart, per phase)	$230

ITEM	PRICE
Solar monitoring gateway	$410
Vehicle GPS tracker	$95
School bus student-tracking reader	$340
Network switch (24-port PoE)	$390
Wireless access point	$175
UPS (1 kVA)	$290
Inverter + battery (2 kVA)	$980
Server cabinet (12U)	$340

Table 60 — Campus & operations

11.4 Smart campus packages

PACKAGE	CONTENTS	PRICE
Smart Entry	1 gate controller, 2 readers, software, install	$2,900
Smart Attendance	4 kiosks, readers, software, install	$5,400
Smart Identity	Card printer, capture station, encoder, 1,000 cards, software, training	$6,800
Smart Security	8 cameras, NVR, 2 access doors, monitoring	$4,900
Smart Classroom (per room)	Interactive board, projector, AP, sensors	$4,300
Smart Campus Foundation	Entry + Attendance + Identity	$13,900
Smart Campus Complete	All above + security + 10 classrooms + signage	From $62,000
Smart Mosque	PA, signage, attendance, donation kiosk, prayer display	$7,400
Smart Clinic	Queue display, kiosk, ID reader, sensors	$6,200
Smart Office	Access, attendance, signage, meeting-room booking	$8,900

Table 61 — Smart campus packages

Included with every hardware package: site survey, installation, commissioning, staff training, 12-month warranty, and integration into the StromeX platform. **Excluded:** civil works, cabling beyond 30m, power provisioning, and structural mounting.

11.5 Hardware services

ITEM	PRICE
Site survey	$180 (credited against order)
Installation & commissioning	18% of hardware value, min $220
Extended warranty (year 2–3)	9% of hardware value/yr
Maintenance contract	12% of hardware value/yr
Emergency callout	$140 + parts
Spare parts holding	6% of hardware value/yr
Decommissioning & recycling	$95/site

Table 62 — Hardware services

DIVISION 12 — DESIGN & CREATIVE

Detailed method in Volume VI. Prices authoritative here. Target gross margin: 45–65%.

12.1 Brand identity

ITEM	PRICE
Logo preview / concept explorer	**Free**
Logo design — Standard (3 concepts, 2 revisions)	$420
Logo design — Premium (5 concepts, full exploration)	$1,150
Logo design — Elite (research-led, bespoke lettering)	$3,400
Institutional crest / coat of arms	$1,900
Brand identity system — Standard	$1,400
Brand identity system — Premium	$4,200
Brand identity system — Elite	$11,000
Brand guidelines document	$890
Brand kit (all variants, all formats, per Volume I §10.1)	$340
Brand refresh / modernisation	From $2,400
Naming & verbal identity	$1,600
Tagline development	$480
Sonic logo / audio identity	$1,200

Table 63 — Brand identity

12.2 Print & collateral

ITEM	PRICE
Letterhead	$140
Business card	$120
Complete stationery suite	$460
Compliment slip / memo	$95
Envelope	$85
Folder / presentation wallet	$210
Brochure (4pp)	$340
Brochure (8–12pp)	$680
Prospectus (24pp)	$2,200

ITEM	PRICE
Prospectus (48pp, art-directed)	$5,400
Prospectus (premium, bespoke spreads, photography direction)	$12,000
Annual report	$2,800
Poster	$180
Flyer	$130
Banner / roll-up	$160
Billboard artwork	$420
Signage system (wayfinding)	From $1,800
Vehicle livery	$640
Uniform & apparel design	$780
Merchandise design	$340
Exhibition stand design	$2,400

Table 64 — Print & collateral

12.3 Digital & product design

ITEM	PRICE
UI design (per screen)	$180
UX research & journey mapping	From $1,600
Wireframing (per flow)	$340
Design system creation	From $4,800
Prototype (interactive)	From $1,200
Usability testing (5 participants)	$1,400
Accessibility design review	$560
Email template design	$260
Social media template kit	$420
Presentation template	$480
Presentation design (per deck, ≤20 slides)	$740
Pitch deck (investor-grade)	$2,400
Infographic	$290
Data visualisation system	$1,100
Icon set (custom, 30 icons)	$840
Illustration (spot)	$140

ITEM	PRICE
Illustration (full page)	$420
Character design	$680

Table 65 — Digital & product design

12.4 Motion, 3D & media

ITEM	PRICE
Logo animation	$420
Motion graphics (per finished minute)	$1,200
Explainer video (60–90s)	$2,400
Corporate film (3–5 min)	From $6,500
Documentary (institutional)	From $18,000
2D animation (per finished minute)	$1,800
3D animation (per finished minute)	$3,600
3D product visualisation	$890
Architectural visualisation (still)	$740
Architectural visualisation (walkthrough)	$3,200
Virtual campus tour (360°)	$2,900
Drone photography & video	$680/day
Photography — portraits (per session, ≤20 people)	$520
Photography — institutional / editorial (per day)	$840
Photo retouching	$18/image
Podcast production (per episode)	$340
Audio branding & jingle	$980
Voice-over (per finished minute)	$95
AI voice-over (per finished minute)	$22
Subtitling & captions (per finished minute)	$14

Table 66 — Motion, 3D & media

DIVISION 13 — PUBLISHING & PRINT

Method in Volume VI. Target gross margin: 45–65% (services), 22–35% (print coordination).

13.1 Editorial

ITEM	PRICE
Manuscript assessment	$290
Developmental editing	$0.048/word
Structural editing	$0.036/word
Line editing	$0.030/word
Copy-editing	$0.017/word
Proofreading	$0.011/word
Fact-checking	$0.022/word
Indexing	$4.20/page
Ghostwriting	$0.34/word
AI-assisted drafting with human editorial control	$0.09/word
Translation — general	$0.075/word
Translation — literary	$0.135/word
Translation — religious / classical Arabic	$0.165/word
Transcription of dictation	$1.40/audio minute
Permissions & rights clearance	$85/item

Table 67 — Editorial

13.2 Design & production

ITEM	PRICE
Cover design — Standard	$290
Cover design — Premium	$780
Cover design — Elite (illustrated, bespoke lettering)	$1,900
Full jacket (with flaps, spine, case)	+$340
Interior layout — text-only	$3.10/page
Interior layout — illustrated	$6.40/page
Interior layout — children's / picture book	$12.50/page
Interior layout — **premium**	**$14.00/page**

ITEM	PRICE
Islamic typesetting (Arabic + translation, harakat, footnote apparatus)	
Qur'anic text setting (verified source, no generation)	$18.00/page
Complex academic layout (tables, equations, apparatus)	$9.80/page
Typesetting corrections round	$0.90/page
eBook conversion (EPUB3, reflowable)	$340
Fixed-layout eBook	$680
Accessible eBook (screen-reader optimised)	$520
Print-ready PDF preparation & preflight	$180
ISBN registration (service fee)	$85
Barcode generation	Free
Copyright registration support	$220
Legal deposit handling	$140
CIP data preparation	$95

Table 68 — Design & production

13.3 International print coordination

The founder's brief: give people access to UK and US printing standards they cannot reach directly. Printing cost is pass-through; StromeX charges a coordination margin and takes responsibility for the outcome.

ITEM	PRICE
Coordination margin	**18% of print cost, minimum $240**
Print quotation & specification	Free
Paper stock selection & dummy	$180
Colour proofing (digital)	$95
Colour proofing (wet proof, press-accurate)	$340
Press check attendance (UK/US)	$640/day
Print-run management	Included in coordination
Freight & customs coordination	Cost + 12%
Warehousing (per pallet/month)	$28

ITEM	PRICE
Fulfilment & distribution setup	$420
Print-on-demand setup (global)	$290
Rush handling	+25% of coordination

Table 69 — International print coordination

Print tiers offered:

TIER	WHERE	TYPICAL USE
Local	Nigeria / regional	Workbooks, exam papers, newsletters, volume text
Regional Premium	UAE / Turkey / India	Quality colour work at moderate cost
UK Premium	United Kingdom	Prestige titles, case binding, foil, specialist finishing
US Premium	United States	Trade distribution, large runs, POD network
Fine Press	UK / EU specialist	Limited editions, hand-finishing, archival stock

Table 70 — International print coordination

13.4 Audio & interactive

ITEM	PRICE
Audiobook — AI narration, human-QC'd	$170/finished hour
Audiobook — human narration (standard)	$620/finished hour
Audiobook — human narration (premium, studio)	$1,100/finished hour
Audiobook — Qur'an / religious recitation (qualified reciter)	Quoted
Audiobook mastering & distribution setup	$290
Interactive eBook (embedded quizzes, audio, video)	From $2,800
Animated children's book	From $4,600
App-based book	From $9,000
Read-along / karaoke text sync	$240/hour of audio

Table 71 — Audio & interactive

13.5 Institutional & educational publishing

ITEM	PRICE
Bespoke school textbook (single subject, single level)	From $4,800
Textbook series (6 levels, one subject)	From $24,000
Workbook / activity book	From $1,900
Teacher’s guide	From $2,200
Scheme of work & syllabus framework document	$1,400
Examination paper bank (per subject, per level)	$780
Institutional yearbook	From $3,200
Graduation programme (designed, printed, verifiable)	From $1,600
Institutional history / commemorative volume	From $14,000
Board & governance handbook	$1,900
Policy compendium	$1,400
Journal / periodical setup	$2,400 + $680/issue
Research publication support	$890
Custom edition of an existing title for one institution	From $1,200

Table 72 — Institutional & educational publishing

13.6 Author & publisher services

ITEM	PRICE
Publishing consultation	Free (45 min)
Publishing strategy & business plan	$980
Author website	Division 1
Author brand kit	$640
Book launch campaign	From $1,800
Amazon / retail listing optimisation	$340
Metadata & keyword optimisation	$190
Review & endorsement outreach	$620
Rights & licensing brokerage	15% of deal value

ITEM	PRICE
Royalty administration	$46/mo + 2% of royalties
Distribution setup (global retail)	$480

Table 73 — Author & publisher services

DIVISION 14 — CONSULTING & PROFESSIONAL SERVICES

Target gross margin: 45–60%. Day rates band: Band A $780 · Band B $1,650 · Band C $3,200 (senior consultant, per day).

14.1 Strategy & transformation

ITEM	PRICE
Discovery consultation	**Free (60 min, physical or remote)**
Digital readiness assessment	**Free**
Digital transformation roadmap	$3,400
Institutional audit (full)	$5,800
Technology strategy & architecture review	$2,900
Business process re-engineering	From $4,200
Organisational structure review	$2,600
Strategic plan (3–5 year)	$6,400
Feasibility study	$3,800
Market analysis	$2,400
Competitor benchmarking	$1,600
Change management programme	From $5,200
Workforce impact & redeployment plan	$1,900 (**free** where StromeX automation caused the change — Volume I §7.1)
Board advisory retainer	$1,400/mo
Interim CTO / CIO	$6,800/mo
Founder / executive coaching	$340/session

Table 74 — Strategy & transformation

14.2 Education services

ITEM	PRICE
Curriculum design (per subject, per level)	$680
Full curriculum framework (whole school)	From $8,400
Scheme of work (per subject, per term)	$190
Lesson plan library (per subject, per level)	$420

ITEM	PRICE
Assessment framework design	$1,600
Marking & moderation policy	$740
School establishment / start-up consultancy	From $6,800
School improvement plan	$2,900
Accreditation readiness — national	$4,600
Accreditation readiness — international (Cambridge, IB, Edexcel, COGNIA)	From $18,000
Inspection preparation	$2,400
Teacher observation & development programme	$3,200
Pedagogy audit	$1,900
Timetable design (manual, complex)	$890
Special educational needs framework	$2,200
Boarding / pastoral framework	$1,800
Islamic studies curriculum design	$2,400
Qur’an programme design (memorisation pathway)	$1,900
Bilingual / trilingual education framework	$2,800

Table 75 — Education services

14.3 Governance, policy & compliance

ITEM	PRICE
Governance charter	$2,400
Board handbook	$1,900
Constitution / statutes drafting	$3,600
Governance framework review	$1,800
Policy pack — core (10 policies)	$1,200
Policy pack — comprehensive (30 policies)	$3,200
Individual policy drafting	$180 each
Staff handbook	$980
Student handbook	$890

ITEM	PRICE
Parent handbook	$780
Safeguarding framework	$1,600
Code of conduct	$520
Risk register & management framework	$1,400
Business registration & incorporation support	$640 + statutory fees
Trademark filing support (per class, per jurisdiction)	$480 + statutory fees
Regulatory licensing support	From $1,600
Data protection compliance	Division 10
Financial authority matrix	$940
Delegation of authority framework	$1,100
Internal audit framework	$2,100

Table 76 — Governance, policy & compliance

14.4 Marketing & growth services

ITEM	PRICE
Marketing strategy	$1,900
Brand positioning workshop	$1,400
Admissions marketing campaign	From $2,200
Social media management	$640/mo
Content marketing retainer	$980/mo
Paid advertising management	15% of spend, min $340/mo
Email marketing programme	$420/mo
Public relations retainer	$1,200/mo
Press release writing & distribution	$340
Crisis communications support	$2,400 + $180/hr
Event management	From $1,800
Open day / admissions event package	$2,600
Alumni engagement programme	$1,600
Fundraising / development strategy	$2,900
Donor management system	Division 2

ITEM	PRICE
Grant writing	$1,400/application

Table 77 — Marketing & growth services

DIVISION 15 — STROMEX ACADEMY

Training and certification. Target gross margin: 65–80%. Free where it drives adoption; paid where it confers credential value.

ITEM	PRICE
Product documentation	**Free**
Video tutorial library	**Free**
Community forum	**Free**
Live onboarding webinar	**Free**
Admin training (up to 10 staff, remote)	$340
Admin training (on-site, full day)	$980
Teacher training (per cohort of 25)	$620
Advanced administrator training	$780
Train-the-trainer programme	$1,900
Digital literacy programme (per staff member)	$46
Cybersecurity awareness (per staff member)	$22
Redeployment training after automation	**Free** (capped per contract, Volume I §7.1)
StromeX Certified Administrator	$180/candidate
StromeX Certified Implementer	$340/candidate
StromeX Certified Partner Engineer	$520/candidate
StromeX Certified Designer	$290/candidate
Certification renewal (2-yearly)	$95
Custom course development for a client	From $2,400
Learning content production (per finished hour)	$890
Academy licence for an institution (unlimited internal use)	$980/yr

Table 78 — Stromex Academy

DIVISION 16 — MARKETPLACE, LICENSING & WHITE-LABEL

16.1 Marketplace

A store where institutions buy, and third parties sell: templates, policies, curricula, question banks, ERP modules, automation packs, AI agents, design assets, book templates, certificate designs, brand kits, plugins, themes, fonts, icon sets, and professional services.

ITEM	TERMS
Listing	Free
StromeX revenue share	**20%** of the sale (18% for certified partners, 15% above $50k lifetime sales)
Payout	Monthly, $50 minimum
Featured placement	$180/mo
Certification / quality review of a listing	$140 one-time
Institution’s own private marketplace (sell to their own community)	$340 + $28/mo, StromeX takes 8%

Table 79 — Marketplace

Rev-share is deliberately below the 30% platform norm. A sustainable seller ecosystem is worth more than the extra ten points.

16.2 White-label & licensing

ITEM	PRICE
Partner white-label (partner’s brand, StromeX platform)	35% of end-customer revenue, min $420/mo
Reseller (StromeX brand, partner sells)	25–35% partner margin (Volume V)
OEM licence (embed StromeX modules in another product)	From $2,400/mo + per-seat
Source-available licence (customer may inspect & self-host)	2.2× standard subscription
Perpetual licence (government / air-gapped only)	4.5× annual, + 22%/yr maintenance
Territory licence (exclusive, per country)	Negotiated; requires volume commitment
Brand licensing (institutions using StromeX-designed IP)	Per agreement
API commercial tier (high-volume third-party use)	From $340/mo

Table 80 — White-label & licensing

DIVISION 17 — SUPPORT, SLA & MANAGED SERVICES

17.1 Support tiers

TIER	CHANNELS	FIRST RESPONSE	HOURS	PRICE
Community	Docs, forum	—	—	**Free**
Standard	Email, chat	24h	Business hours	**Free with any paid plan**
Priority	+ phone	4h	Extended	$89/mo
Premium	+ named contact	1h	12×5	$290/mo
Elite	+ dedicated engineer	30 min	24×7	$890/mo
Enterprise	+ on-site, TAM	15 min	24×7	From $2,400/mo

Table 81 — Support tiers

Standing rule (Volume I §6.1): when something *we built* is broken, the customer speaks to a human at no charge, on every tier including Foundation. Support tiers govern speed and scope for everything else — never access to help when we are at fault.

17.2 Service levels

SLA	UPTIME	CREDIT IF MISSED
Standard	99.5%	10% of monthly fee
Professional	99.9%	25%
Premium	99.95%	50%
Elite	99.99%	100%

Table 82 — Service levels

Credits are applied automatically from our own monitoring. **The customer never has to notice, claim, or argue.** Uptime is published on a public status page with full incident history.

17.3 Managed services

ITEM	PRICE
Managed infrastructure	$340/mo
Managed database administration	$290/mo
Managed security operations	$420/mo
Managed backup & recovery	$140/mo
Fully managed institution (“we run your ICT”)	From $1,900/mo
Outsourced ICT desk (per 250	$980/mo

ITEM	PRICE
users)	
Dedicated on-site engineer	$2,800/mo
Technical account manager	$1,200/mo
Quarterly business review	Included at Premium+

Table 83 — Managed services

CHAPTER 18 — CREDITS, TOKENS & CONSUMPTION UNITS

18.1 StromeX Credits

One prepaid unit powering all AI and metered consumption. **1 credit = $0.01.**

PACK	PRICE	CREDITS	BONUS
Trial	Free	2,000/mo	Renews monthly, forever
Small	$10	1,000	—
Medium	$50	5,250	+5%
Large	$200	22,000	+10%
Institution	$1,000	115,000	+15%
Enterprise	$5,000	600,000	+20%

Table 84 — StromeX Credits

Typical consumption:

ACTION	CREDITS
Short AI answer	1
Long AI answer with citations	4
Document summary (10 pages)	8
Marked homework script	6
Generated lesson plan	12
Translated page (AI-assisted)	9
Generated image	3
Transcribed audio minute	2
Voice agent minute	4
Deep research task	45
Generated exam paper	60
Drafted policy document	120

Table 85 — StromeX Credits

Credits never expire while the account is active. Unused credits are refundable within 12 months. Consumption is shown live, with a spend cap the customer sets themselves and a hard stop rather than an overage bill.

18.2 Verification tokens

Defined in Division 3.5 — Check $0.50 · Attest $1.00 · Seal $2.00 · Legalise $5.00.

18.3 The anti-bill-shock rules

82. Every metered service has a customer-set spending cap, defaulting to on.
83. Reaching the cap **stops the meter**; it does not generate an invoice.
84. Alerts at 50%, 80% and 95% of cap.
85. No service is ever cut off for exceeding a cap — the metered feature pauses, the institution keeps running.
86. First accidental overage in any 12-month period is forgiven, automatically, without being asked.

CHAPTER 19 — BUNDLES & SUITES

Bundles simplify. They never force. Every bundle is decomposable to its line items in the configurator, and a customer may always build their own.

19.1 Education suites

SUITE	CONTAINS	BUILD	MONTHLY	SAVING
School Starter	Standard website, core platform (300), student + parent portal, admissions, results	$2,100	$178	22%
School Professional	Professional website, platform (1,000), all portals, admissions, gradebook, exams, fees, certificates, ID cards, 2 AI agents	$6,400	$520	27%
School Premium	+ Premium website, LMS, library, transport, hostel, HR/payroll, analytics, mobile apps, 5 AI agents	$14,800	$1,140	31%
School Elite	+ Elite website, smart campus foundation, full AI suite, publishing allowance, Elite support	$38,000	$2,480	33%
University Suite	Higher-ed platform, faculties, programmes, credentials, research, alumni, finance	From $42,000	From $3,200	30%
Islamic Institution Suite	Bilingual site, Qur’an programme, memorisation, Arabic, madrasah records, waqf, certificates	$8,900	$690	29%

Table 86 — Education suites

19.2 Other sector suites

SUITE	BUILD	MONTHLY
Clinic / Hospital Starter	$9,400	$740
Mosque / Church Suite	$3,200	$240
SME Business Suite	$1,900	$148
Professional Firm Suite (law, accounting)	$6,800	$520
NGO Suite	$4,600	$340
Publisher Suite	$7,200	$560
Government Department Suite	From $48,000	From $3,800

Table 87 — Other sector suites

19.3 Cross-division bundles

BUNDLE	CONTENTS	PRICE
Identity Complete	Card design, 1,000 cards, printer, capture station, digital ID, verification portal	$9,200
Credential Complete	Certificate design (3 templates), signing keys, 2,000 certificates, verification portal, transcripts	$3,400
Brand Complete	Logo, identity system, guidelines, stationery, brand kit, website design direction	$4,900
Publish Complete	Editing, cover, layout, eBook, ISBN, UK print coordination for 500 copies	From $6,200
Launch Complete (new institution)	Registration, brand, website, platform, policies, curriculum, certificates, ID	From $22,000

Table 88 — Cross-division bundles

CHAPTER 20 — WORKED CONFIGURATIONS

Real assemblies at real prices, showing what an institution actually pays. Band A applied (×0.32) for the Nigerian examples, as those are the institutions this founding market actually contains.

20.1 A 220-student primary school in Ikorodu, Lagos (Band A)

The instinct is to sell the suite. The correct answer is to sell what they need.

LINE	LIST (BAND C)	BAND A
Standard website (8 pages)	$1,650	**$528**
Care plan	$45/mo	**$14.40/mo**
Core platform, Standard (300 students)	$89/mo	**$28.48/mo**
Student + parent portal	$1,040 + $68/mo	**$333 + $21.76/mo**
Admissions pipeline	$740 + $52/mo	**$237 + $16.64/mo**
Gradebook & report cards	$960 + $66/mo	**$307 + $21.12/mo**
Certificate platform + 1 template + 250 certificates	$848	**$271**
Verification portal	$420 + $32/mo	**$134 + $10.24/mo**
SMS module	$220 + $16/mo	**$70 + $5.12/mo**
Standard onboarding	$480	**$154**
Migration (220 records, clean)	$180	**$58**
One-time total	$6,318	**$2,022**
Monthly total	$279	**$117.76**
Year 1 total	—	**$3,435**
Per student per year	—	**$15.61**

Table 89 — A 220-student primary school in Ikorodu, Lagos (Band A)

Deliberately *not* sold: ERP, HR/payroll, hostel, transport, mobile apps, hardware, AI agents. They do not need them yet. When they do, the ladder in Volume I §6.5 applies.

20.2 An 1,800-student secondary school & college group (Band A)

LINE	BAND A
Premium website, 4 languages (setup + 60 pages translated)	$3,040 + $998
Core platform Premium (3,000)	$156.80/mo

LINE	BAND A
All portals (student, parent, staff, registrar, bursary, executive, alumni)	$1,299 + $87.68/mo
Academic suite (admissions, timetable, attendance, gradebook, exams, transcripts, curriculum)	$1,382 + $88.32/mo
Fee management, instalments, receipts, payments	$531 + $32.64/mo
HR & payroll	$416 + $28.80/mo + payslips
Credential suite (platform, 3 Premium templates, keys, identifiers, verification, 1,800 certs/yr)	$1,376 + $24.96/mo + $63/yr
Digital identity + 1,800 ID cards (premium PVC)	$362 + $13.44/mo + **$6,120 cards (band-invariant)**
Card printer + capture station + encoder	**$3,600 (band-invariant)**
5 AI agents (Receptionist, Admissions, Registrar, Bursar, Timetable)	$110.72/mo
Android + iOS apps	$1,472 + $28.16/mo
Analytics & executive reporting	$430 + $30.72/mo
Smart Entry (2 gates)	**$5,800 (band-invariant)**
Premium implementation	$1,536
Premium support	$92.80/mo
One-time total	**$27,932**
Monthly total	**$694.24**
Year 1 total	**$36,263**
Per student per year	**$20.15**

Table 90 — An 1,800-student secondary school & college group (Band A)

20.3 An author publishing an Islamic title, UK print (mixed bands)

LINE	PRICE
Manuscript assessment (Band A)	$93
Copy-edit, 62,000 words (Band A)	$337
Translation to Arabic, 62,000 words, literary (Band A)	$2,678
Premium cover design (Band A)	$250

LINE	PRICE
Premium Islamic interior typesetting, 280 pp (Band A)	$1,254
eBook conversion (Band A)	$109
ISBN + preflight (Band A)	$85
UK Premium print, 1,000 copies, case bound, foil (pass-through)	~$6,400
Coordination margin (18%)	$1,152
Wet proof	$340
Freight to Lagos + customs (cost + 12%)	~$1,450
Author brand kit + landing page (Band A)	$313
Retail distribution setup (Band A)	$154
Total	**≈$14,615**
Unit cost delivered	**≈$14.62/copy**

Table 91 — An author publishing an Islamic title, UK print (mixed bands)

20.4 A state ministry of education, 400 schools (Band A, Government tier)

Indicative only; government engagements are scoped, tendered and quoted individually.

LINE	INDICATIVE
Government Department Suite	$15,360 build
Multi-tenant platform, 400 schools / ~180,000 students	$8,400/mo
Statutory reporting & regulator connectors	$1,280 + $180/mo
Credential platform, state-wide verification	$2,100 + $640/mo
Analytics, benchmarking, executive dashboards	$1,900 + $420/mo
Data residency (Nigeria)	Included
Enterprise implementation, phased over 18 months	$86,000
Training, 400 school administrators	$18,400
Enterprise support	$2,400/mo
Year 1	**≈$266,000**

LINE	INDICATIVE
Per student per year	**≈$1.48**

Table 92 — A state ministry of education, 400 schools (Band A, Government tier)

That last line is the entire argument. It is the number that wins a tender, and it is true.

Price book governance

Version	2.0
Owner	Pricing Council (Volume I §8.4)
Review cadence	Quarterly
Changelog	Published internally each review; material changes notified to customers 90 days ahead
Grandfathering	Contract-term price protection (Volume I §6.3.6)

Table 93 — Price book governance

VOLUME

IV

Engineering, AI Architecture, Cloud & Security

How it is built · How it stays up · How it stays safe

AUTHORITY

Operational — CTO

EXECUTIVE SUMMARY

Volume IV governs construction. It sets ten engineering principles, the canonical stack, and the static-site pattern that four reference implementations converged on independently. It specifies multi-tenancy with the no-fork rule, the data architecture and its classification scheme, and — in the most consequential subsystem the group operates — the credential and verification architecture, including the signing model in which StromeX never signs on an institution’s behalf. It defines the AI architecture and its provider-independence doctrine, the stakes ladder that determines when a human must decide, the data rights and model-training policy, the performance budgets and degraded modes required by constrained environments, and internationalisation engineering. Chapters twelve to seventeen constitute the Security Bible, covering zero trust, identity, cryptography, secure development, resilience and incident response.

CHAPTER 1 — ENGINEERING PRINCIPLES

E1 — Boring beneath, ambitious above. Payments, authentication, credential signing, backups and audit logs use conservative, proven, widely-reviewed technology. Novelty budget is spent on the customer-visible frontier. A clever database is a liability; a clever admissions experience is an asset.

E2 — Build for the constrained case first. A 5-year-old Android on 3 Mbps with intermittent power is the reference environment. If it works there, it works everywhere. The inverse is never true.

E3 — One platform, many surfaces. Layers 1–4 of the ecosystem stack (Volume I §2.3) are shared by every sector. Forking them is a governed exception (Chapter 3), not an engineering convenience.

E4 — Reversibility over velocity. Every deployment rolls back. Every migration reverses. Every destructive user action has a recovery window. We ship faster *because* mistakes are cheap, not despite it.

E5 — Data is the customer's. Export is complete, open-format, self-service, free and immediate. This is enforced in code, in tests, and in the contract.

E6 — Correctness beats cleverness in the record. The system of record is the moat (Volume II §9). An elegant abstraction that risks a wrong grade, a wrong fee balance, or a wrongly-issued certificate is not elegant.

E7 — Everything is auditable. Who did what, to which record, when, from where, and what it was before. Not a feature — a property of the platform.

E8 — Observable before scalable. We do not optimise what we cannot measure, and we do not ship what we cannot see failing.

E9 — Composition over configuration over customisation. Prefer combining existing modules; then configuring them; then, last and reluctantly, writing bespoke code for one customer.

E10 — The build must be reproducible by a new hire on day two. If setup requires tribal knowledge, that is a defect with a ticket.

CHAPTER 2 — THE REFERENCE ARCHITECTURE

2.1 The shape

```
┌──────────────────────────────────────────────────────────────┐
│ SURFACES   Web · PWA · Android · iOS · Kiosk · Signage ·      │
│            USSD · Print · Card · Gate · API consumers         │
├──────────────────────────────────────────────────────────────┤
│ EDGE       CDN · WAF · DDoS · rate limit · geo-routing        │
├──────────────────────────────────────────────────────────────┤
│ BFF        Per-surface aggregation, response shaping, caching │
├──────────────────────────────────────────────────────────────┤
│ DOMAIN     Identity · Records · Academic · Finance ·          │
│ SERVICES   Credentials · Comms · Automation · Intelligence    │
├──────────────────────────────────────────────────────────────┤
│ PLATFORM   AuthN/Z · Audit · Search · Files · Events · Jobs · │
│            Notifications · Payments · Tenancy · Feature flags │
├──────────────────────────────────────────────────────────────┤
│ AI         Router · Agents · RAG · Guardrails · Evaluation    │
├──────────────────────────────────────────────────────────────┤
│ DATA       PostgreSQL · Object store · Vector · Search ·      │
│            Cache · Queue · Warehouse                          │
├──────────────────────────────────────────────────────────────┤
│ INFRA      Compute · Network · Secrets · Backup · Observ.     │
└──────────────────────────────────────────────────────────────┘
```

Figure 3 — The shape

2.2 The canonical stack

Chosen for hiring depth in our markets, operational simplicity, and longevity — not for novelty.

LAYER	CHOICE	WHY
Primary database	**PostgreSQL**	Relational integrity for a record system; JSONB where flexibility is genuinely needed; enormous operational knowledge base
API services	**Python (FastAPI)** for domain services	Already the MVP's stack; strong AI/ML adjacency; fast to hire for
Web	**Next.js / TypeScript / Tailwind**	Already the MVP's stack; SSR for performance on weak devices; huge talent pool
Static institutional sites	**Token-driven static build**	The pattern proven across four reference sites: partials + a page manifest + a build script; no framework, no runtime, near-zero hosting cost, trivially cacheable, survives a decade
Mobile	**Capacitor wrapper** first, native where justified	Already delivered; one codebase; native only where the platform genuinely requires it
Vector store	**Qdrant**	Already in the MVP

LAYER	CHOICE	WHY
Cache / queue	**Redis**	Already in the MVP
Search	**PostgreSQL FTS**, then a dedicated cluster at scale	Do not add an engine before the query volume justifies it
Containers	**Docker / Compose** → orchestrated at scale	Already in `infra/`
Edge	**Cloudflare**	Already the production topology

Table 94 — The canonical stack

The two-stack rule: the group runs at most two primary backend languages and two frontend frameworks at any time. A third requires CTO approval and a written retirement plan for one of the existing two.

2.3 The static-site pattern, promoted to a product

The four institutional reference sites converged independently on the same architecture, which is now the group standard for institutional web presence:

- —`partials/` — shared chrome as language-neutral templates carrying `{{t:key}}` translation tokens; **one file serves every language**
- —`pages/` — page body content only, one file per page per locale
- —`pages/manifest.json` — the registry: slug, output path, title, description, content file, `lang`, `dir`, `altHref`
- —`i18n/<code>.json` — one flat, dot-keyed dictionary per language
- —`i18n/locales.json` — the single source of truth for which languages exist
- —`scripts/build.js` — assembles partials + content into complete documents, deriving untranslated locales with translated chrome and a labelled notice rather than emitting a broken URL

Why this is the standard. It produces a four-language site whose marginal cost per additional language is a dictionary rather than a codebase; it has no runtime and therefore no runtime vulnerabilities; it is trivially CDN-cacheable, which is decisive on 3 Mbps connections; it costs almost nothing to host, which is what makes the Foundation tier affordable; and it will still build in ten years because it depends only on the language's standard library. This is the architectural expression of principle E1.

Never do: hand-edit built output. Regenerate.

CHAPTER 3 — MULTI-TENANCY & THE NO-FORK RULE

3.1 The tenancy model

Shared schema with a mandatory tenant discriminator, enforced at the data layer rather than trusted to application code, with row-level security as the backstop. Dedicated schemas or dedicated databases are available at Elite/Enterprise/Government tiers where residency, regulation or scale require it.

MODEL	WHO GETS IT	COST TO US
Shared schema, RLS-enforced	Foundation → Premium	Lowest
Dedicated schema	Elite, large multi-campus	Moderate
Dedicated database	Enterprise, regulated	High
Dedicated deployment	Government, air-gapped	Highest (Volume III §4.6)

Table 95 — The tenancy model

Non-negotiable: a tenant-scoped query that could return another tenant's row is a **P0 security incident**, not a bug. Every data-access path has a test that attempts cross-tenant access and asserts failure. This test suite may never be skipped in CI.

3.2 The no-fork rule

Forking platform layers 1–4 for a single customer requires written CTO approval, a documented justification, and a **sunset date**. Without a sunset date the answer is no.

The reason is arithmetic: breadth across twenty sectors is affordable only because the lower layers are shared (Volume I §2.3). Every fork multiplies the maintenance surface by the number of forks, and forks never get deleted unless someone was made responsible for deleting them on a date.

Legitimate alternatives, in order of preference: a feature flag, a configuration option, a tenant-scoped extension point, a plugin, a sector module, and only then a fork.

CHAPTER 4 — DATA ARCHITECTURE

4.1 The record principles

D1 — One institution, one person record. A student who becomes an alumnus who becomes a parent who becomes a staff member is one person with four relationships, not four records. This is harder and it is correct.

D2 — Events, not just state. Enrolment, role changes, fee movements, credential issuance and governance decisions are recorded as **events with a time**, from which state is derived. Auditability, reversibility and historical reconstruction all fall out of this for free, and none of them can be retrofitted.

D3 — Soft delete by default, hard delete on request. Nothing is destroyed accidentally; everything is destroyable deliberately, verifiably, and in fulfilment of a data subject's rights.

D4 — Every record carries provenance. Created by, modified by, source system, and confidence where derived.

D5 — Reference data is versioned. Curricula, fee structures, grading scales and policies change between years. A 2029 transcript must render under 2029's grading scale, not today's.

4.2 The canonical domain model

Validated against the 73-table credential-and-commerce schema in the group's reference implementation, which covers: users and roles; academic bodies, programmes, levels, courses, units and learning outcomes; enrolments and enrolment-integrity events; competencies, skills, assessments and marks; evidence items and versions; quizzes, exercises, vocabulary and self-checks; recordings, pronunciation targets and feedback; time-on-task and unit progress; awards, distinctions and award verifications; credential signatures, signing keys and issued documents; verifying institutions and institution checks; graduate profiles, profile shares and alumni chapters; payments, receipts, refunds, instalment plans, promo codes, scholarships, currencies and country payment routing; corporate accounts and seats; CPD records; live sessions; notifications.

```
PERSON ──┬── enrolment ──── PROGRAMME ── LEVEL ── COURSE ── UNIT
         │                                            └── LEARNING OUTCOME
         ├── role ──────── ORGANISATION UNIT
         ├── assessment ─── COMPETENCY / SKILL ── EVIDENCE
         ├── credential ─── ISSUED DOCUMENT ── SIGNATURE ── VERIFICATION
         ├── financial ──── INVOICE ── PAYMENT ── RECEIPT ── INSTALMENT
         └── identity ───── CARD · BIOMETRIC · CREDENTIAL · DEVICE
```

Figure 4 — The canonical domain model

4.3 Data classification

CLASS	EXAMPLES	HANDLING
Public	Published pages, verification results	CDN-cacheable
Internal	Aggregate statistics, non-identifying config	Authenticated
Confidential	Person records, grades, finance	Encrypted at rest, access-logged, least-privilege

CLASS	EXAMPLES	HANDLING
Sensitive	Health, safeguarding, discipline, biometrics	Additionally: field-level encryption, named-role access only, every read logged and reviewable
Children's data	Any record of a person under 18	**Strictest class in the group.** Minimisation, short retention, no behavioural profiling, no advertising use, parental access rights, and a standing prohibition on any secondary use

Table 96 — Data classification

4.4 Retention

Defaults, overridable upward by regulation and downward by customer instruction: operational logs 90 days · audit logs 7 years · financial records per jurisdiction (typically 6–7 years) · academic records **permanent by default** (a transcript must be producible in forty years) · biometric templates deleted on relationship end · marketing data 24 months from last engagement · AI conversation logs 12 months, or 30 days where they contain sensitive-class data.

CHAPTER 5 — THE CREDENTIAL & VERIFICATION ARCHITECTURE

The single most strategically important subsystem in the group (Volume II §9). Specified in detail because getting it wrong is unrecoverable — a credential system that is found to be forgeable destroys the institution's trust and ours simultaneously.

5.1 The four requirements

87. **Unforgeable** — a credential cannot be fabricated by anyone without the issuer's private key.
88. **Verifiable by anyone, free, forever** — the constitutional commitment (Volume I §5.3).
89. **Verifiable offline where possible** — a border officer with no connectivity should still get signal.
90. **Revocable** — issuance is not irreversible; a credential obtained fraudulently must be withdrawable and the withdrawal must be visible.

5.2 The identifier architecture

Every credential carries a **structured, human-transcribable identifier** designed with the institution (a service line — Volume III §3.1). It encodes issuing institution, document class, year, and sequence, with a check character. It must be readable aloud over a phone call without ambiguity, which rules out characters that collide when spoken or handwritten (0/O, 1/I/l, 5/S, 8/B).

5.3 The signing model

- Each issuing institution holds a distinct key pair. **StromeX never signs on an institution's behalf** — the institution is the issuer, in fact and in cryptography.
- Private keys live in a hardware security module or an equivalent managed key service. Key material is never in application memory longer than a signing operation, never in logs, never in backups in plaintext.
- Key generation, custody, rotation and revocation follow a documented ceremony with two-person control and a written record. This is a priced service (Volume III §3.1) because it is real work with real consequences.
- Compromise of one institution's key affects that institution only. There is no group-wide master key, deliberately — a single key whose compromise invalidates every credential the group ever issued is an unacceptable concentration of risk.

5.4 The verification surfaces

Four distinct surfaces are in production across the reference implementations — certificate, identity, receipt and graduation document — because they have genuinely different verification semantics and conflating them produces confusing results.

SURFACE	ANSWERS
Certificate verification	Was this award made, to this person, by this institution, on this date, and is it still valid?
Identity verification	Is this person currently who this card says they are, in this role?
Receipt verification	Was this payment actually received and applied?
Document verification	Is this multi-page document the sealed original, unaltered?

Table 97 — The verification surfaces

Every verification response states plainly: **valid**, **revoked** (with date and reason class), **not found**, or **superseded** — never an ambiguous result, and never a bare "invalid" that fails to distinguish a forgery from a typo.

5.5 The layered verification model

LAYER	MECHANISM	WORKS OFFLINE	STRENGTH
1	Human-readable identifier	Yes	Weak alone
2	QR → verification URL	No	Strong, convenient
3	Detached digital signature embedded in the document	Yes	Strong
4	Physical security features (guilloche, microtext, UV, foil)	Yes	Deters casual forgery
5	Optional public-ledger anchor	Yes	Independent of us existing

Table 98 — The layered verification model

Layer 5 exists for one reason and it is an honest one: it lets an institution's credentials remain verifiable even if StromeX ceases to exist. Offering it costs us a theoretical lock-in advantage and buys the customer real assurance. That trade is required by Volume I P5.

5.6 Revocation

A public, append-only revocation register. Revocation reasons are classified (issued in error, obtained fraudulently, superseded, withdrawn by institution) and the class is public while the narrative is not. Revocation requires a documented approval chain — one person must not be able to revoke a graduate's degree alone.

CHAPTER 6 — API & INTEGRATION STANDARDS

STANDARD	RULE
Style	REST + JSON. GraphQL only where a surface genuinely needs client-shaped queries
Versioning	URL-versioned (`/v1/`). A version is supported for **24 months** after its successor ships
Breaking changes	Never within a version. New version, dual-run, deprecation notice, telemetry-verified migration
Auth	OAuth 2.1 / OIDC; API keys for server-to-server; short-lived tokens; refresh rotation
Pagination	Cursor-based. Offset pagination is prohibited on any table that can exceed 100k rows
Errors	Structured: machine code, human message, remediation hint, correlation ID
Rate limits	Published, returned in headers, generous by default
Idempotency	Mandatory on every state-changing endpoint. Payments without idempotency keys are rejected
Webhooks	Signed, retried with exponential backoff, replayable for 7 days, with a delivery log the customer can see
Documentation	OpenAPI, generated from code, published, with runnable examples. **Free and public**
SDKs	JS/TS, Python, PHP. Open source. Generated where possible, hand-polished where not
Deprecation	Announced, dated, telemetry-monitored, and never enforced before the announced date

Table 99 — Api & Integration Standards

The dogfooding rule: every StromeX surface consumes the same public API a third party would. There is no privileged internal API. If it is awkward for us, it is awkward for them, and we will find out first.

CHAPTER 7 — THE AI ARCHITECTURE

7.1 The provider-independence doctrine

No StromeX capability may depend on a single AI provider. This is a commercial risk control (Volume II R7) as much as an architectural preference: provider pricing, availability, terms and model behaviour all change without our consent.

The MVP already implements multi-provider routing across Claude, OpenAI-compatible endpoints, DeepSeek and Perplexity, with a development provider for offline work. That router is the group standard and every AI capability goes through it.

```
Request → Policy gate → Router → Provider (primary)
                                     │           └ fallback chain
                                     ├ Cost / latency / capability routing
                                     ├ Self-hosted model (for sensitive classes)
                                     └ Deterministic fallback (never a blank failure)
```

Figure 5 — The provider-independence doctrine

Routing criteria, in order: data classification (sensitive-class work may be pinned to a self-hosted or residency-compliant model) → required capability → latency budget → cost. Cost is last, always. A cheaper model that gets a grade wrong is not cheaper.

7.2 Retrieval before generation

Institution-facing answers are **grounded in the institution's own records and documents**, retrieved and cited, not recalled from a model's training. The pattern:

```
Query → tenant-scoped retrieval → relevance filter → context assembly
      → generation with mandatory citation → citation validation
      → confidence assessment → response, or escalation
```

Figure 6 — Retrieval before generation

Citation validation is a real check, not a prompt instruction. A generated citation that does not resolve to a retrieved source is stripped, and if stripping leaves the answer unsupported the system says it does not know. Volume I §7.2.3 prohibits fabricated authority, and prompts alone do not enforce prohibitions.

7.3 The agent model

An agent is a **bounded role**, not a personality: a scope of data it may read, a set of actions it may take, a refusal boundary, an escalation target, and an audit trail. Agents are configured per institution, and the institution can see and change every one of those five things.

PROPERTY	RULE
Data scope	Explicit allow-list. An agent cannot read what its role would not read
Actions	Explicit allow-list. Write actions require a configured approval where they affect a person's record
Refusal boundary	Per-domain, enforced in the policy gate, not in the prompt

PROPERTY	RULE
Escalation	Every agent has a named human target. An agent with no escalation path may not be deployed
Audit	Every invocation logged: prompt, retrieved context IDs, model, output, action taken, cost
Kill switch	Per-agent, per-institution, immediate, available to the institution

Table 100 — The agent model

7.4 Evaluation

Agents are evaluated before release and continuously in production against: **factual accuracy** (against a held-out set of the institution's real records), **citation validity**, **refusal correctness** (does it decline what it must decline), **consistency** (repeated identical queries must not vary factually), **latency**, and **cost per task**.

A model or prompt change ships only when the evaluation suite is at or above the incumbent on accuracy, citation validity and refusal correctness. Latency and cost improvements do not license accuracy regressions.

CHAPTER 8 — AI GOVERNANCE & HUMAN-IN-THE-LOOP ENFORCEMENT

8.1 The stakes ladder

STAKES	EXAMPLES	RULE
Low	Draft an email, summarise a document, suggest a lesson activity	AI acts; human may review
Medium	Mark homework, draft a policy, propose a timetable, triage an enquiry	AI proposes; **human approves before it reaches the affected person**
High	Final grade, admission decision, disciplinary outcome, fee write-off, safeguarding classification, staff performance outcome	**AI may inform; a named human decides and is recorded as the decider**
Prohibited	Religious rulings, legal conclusions, medical diagnosis, immigration advice, credential issuance without human authorisation	**AI does not decide. It routes to a qualified human and says so to the user**

Table 101 — The stakes ladder

This ladder is enforced in the policy gate. A capability that would breach it cannot be configured into existence, and an institution cannot opt out of the High and Prohibited rows — not even by request, and not for a discount. Volume I P10 and §7.2.4.

8.2 Disclosure

Every AI surface is labelled as AI, in the user's language, at the point of interaction — not in a footer, not in a terms page. Where a deliverable is produced with AI assistance and human review, both facts are stated. Where a human reviewed it, that human is identifiable internally and accountable for it.

8.3 Faith and cultural content

Qur'anic text is **retrieved from verified sources and reproduced verbatim** — never generated, never paraphrased, never completed by a model. Translations and tafsir are attributed to their named scholars and editions. Questions of ruling route to qualified human scholars with the routing visible to the user. The same standard of fidelity applies to every tradition the group serves; there is no primary tradition and no set of afterthoughts (Volume I §7.5).

8.4 The prohibited-use enforcement

Prompt injection, jailbreak attempts and scope-escape attempts are detected, logged, and rate-limited. Repeated attempts against a sensitive-class agent trigger a human review of the account. Users are told when a request was declined and why, in plain language, without a lecture.

CHAPTER 9 — DATA RIGHTS & MODEL TRAINING POLICY

Stated as flatly as possible because ambiguity here is worth more to a competitor than any feature:

91. **Customer data is the customer's.** StromeX is a processor.
92. **We do not train third-party models on customer data.** Not with de-identification, not with aggregation, not with consent buried in terms. Where a provider's default terms would permit training, we contract out of it, and where we cannot, we do not route customer data to that provider.
93. **We do not sell, broker, licence or share customer data.** No exceptions for "partners", "analytics", or "research".
94. **Aggregate benchmarking is opt-in, anonymised, and never leaves the group** (Volume III Division 9).
95. **We may train our own models on our own operational data** — our documentation, our code, our support transcripts where the customer's content is removed — and on data a customer has explicitly and separately licensed to us, with the licence revocable and the benefit shared.
96. **Sub-processors are published**, with the data classes they receive, and customers are notified before a new one is added.
97. **Export is complete, open-format, self-service and free**, at any time, including after termination, for a defined post-termination window that is stated in the contract and is never shorter than 90 days.

CHAPTER 10 — PERFORMANCE & THE CONSTRAINED-ENVIRONMENT STANDARD

10.1 Budgets — release blockers, not aspirations

METRIC	TARGET	MEASURED ON
First contentful paint	< 1.5s	3 Mbps, mid-range Android
Largest contentful paint	< 2.5s	same
Time to interactive	< 3.5s	same
Interaction to next paint	< 200ms	same
Cumulative layout shift	< 0.1	same
API p50 / p95 / p99	120ms / 400ms / 900ms	production
Initial JS payload	< 180 KB gzipped	per route
Total page weight	< 900 KB	first load, incl. images
Fonts	≤ 4 files, subset, `font-display: swap`	—

Table 102 — Budgets — release blockers, not aspirations

10.2 The degraded-mode requirement

Every surface has a **defined behaviour under failure**, designed and tested, not emergent:

FAILURE	REQUIRED BEHAVIOUR
No network	Cached read access; queued writes; explicit “not synced” state; nothing silently lost
Slow network	Progressive rendering; skeleton states; no blocking spinners over 3s
No power (device)	Work is persisted continuously; nothing depends on a clean shutdown
Payment provider down	Queue and retry; offer alternative channel; never take money without recording it
AI provider down	Fall back through the chain; then a deterministic non-AI path; never a blank error
Our platform down	Static institutional site stays up (it has no runtime); status page states truth

Table 103 — The degraded-mode requirement

Offline-first is not a feature flag. Service workers, local persistence and sync-conflict resolution are architectural, and the reference implementations already ship them.

10.3 Data-cost awareness

In our founding markets users pay for data by the megabyte. A heavy page is a real cost transferred to a parent. Image budgets, lazy loading, and a “low-data mode” that ships text-first are therefore engineering requirements, not optimisations.

CHAPTER 11 — FRONTEND, DESIGN SYSTEM & INTERNATIONALISATION ENGINEERING

11.1 The design system

`StromeX Design System` — open source (Volume I §5.5) — is the single source of truth: tokens (colour, type, space, radius, motion, elevation), primitives, components, patterns, and per-sector theme layers. Divisions extend it; they do not fork it. A component used by three products is promoted into the system, and the third team to need it does the promotion.

Accessibility is enforced mechanically: contrast ratios checked in CI, keyboard traversal tested, focus visible always, `prefers-reduced-motion` honoured, semantic HTML before ARIA, and every interactive element reachable and labelled. WCAG 2.2 AA is the floor and it is verified, not asserted.

11.2 Internationalisation engineering

The group ships English, Arabic, Yorùbá and French today across its reference implementations. The standards that made that affordable:

98. **One dictionary per language**, flat and dot-keyed. Nested structures produce merge conflicts and orphaned keys.
99. **A locale registry as the single source of truth** for which languages exist and their direction.
100. **Language-neutral chrome templates** carrying translation tokens — one file per component, not one per language. Per-language copies are technical debt with a migration path, and the reference implementations record exactly this transition.
101. **Bidirectional layout is a property, not a mode.** Logical CSS properties (`margin-inline-start`, not `margin-left`) everywhere. Physical properties require an RTL counterpart in the same commit.
102. **Embedded LTR runs inside RTL text are wrapped explicitly** (`dir="ltr"` spans) for phone numbers, emails, addresses, acronyms and codes. The Unicode bidi algorithm does not reliably keep them in reading order, and this produces subtly wrong phone numbers that nobody notices until a parent cannot call the school.
103. **Script-specific typography is engineered, not fallen back to.** Arabic renders in Amiri/Cairo, Yorùbá in a face with correct diacritic support. A Latin font's fallback for Arabic is a defect.
104. **Untranslated pages are derived with translated chrome and a labelled notice**, never omitted. A language switcher that points at a 404 is worse than an honest partial translation.
105. **Machine translation is never shipped as finished work.** AI-assisted, human-reviewed, and disclosed (Volume I §7.2).
106. **Locale-correct formatting** for dates (including Hijri where relevant), numbers, currency, names and address order. A Nigerian date rendered US-style is a small error that costs real trust.

CHAPTER 12 — SECURITY BIBLE I: PRINCIPLES & ZERO TRUST

12.1 The security principles

S1 — Assume breach. Design so that a compromised component does not compromise the system. **S2 — Least privilege, always, including for us.** A StromeX engineer's default access to customer data is none. **S3 — Defence in depth.** No single control is load-bearing. **S4 — Secure by default.** The safe configuration is the one you get without asking. **S5 — Fail closed.** When an authorisation check cannot complete, deny. **S6 — Security is free** (Volume I §6.1). Patches, TLS, WAF, backups and audit logs are never a paid upgrade. **S7 — Children's data receives the strictest handling in the group**, without exception and without a customer having to request it. **S8 — Disclose fast and completely.** Reputational damage comes from concealment, not from incidents.

12.2 Zero trust, concretely

PRINCIPLE	IMPLEMENTATION
No implicit network trust	Every service-to-service call is authenticated and authorised; being “inside” grants nothing
Verify explicitly	Identity, device posture and context evaluated per request, not per session
Least privilege	Short-lived, scoped credentials; no long-lived shared secrets; no shared accounts, ever
Micro-segmentation	Network policy between services; databases reachable only from their owning service
Continuous verification	Session re-evaluation on privilege change, anomalous location, or new device
Encrypted everywhere	TLS 1.3 in transit including internally; encryption at rest by default
Assume compromise	Anomaly detection on access patterns; blast-radius limits designed in

Table 104 — Zero trust, concretely

12.3 Staff access to customer data

- Default access: **none**.
- Access requires a ticket, a stated reason, a time limit, and it is **logged and visible to the customer** in their audit log.
- Production database access requires two-person approval and is session-recorded.
- Access to sensitive-class data (health, safeguarding, biometrics, children's records) requires a named role, additional approval, and generates a customer-visible notification.
- Access reviews quarterly; departure revokes everything within one hour and this is tested.

CHAPTER 13 — SECURITY BIBLE II: IDENTITY & ACCESS

CONTROL	STANDARD
Password storage	Argon2id, per-user salt. Never anything else, never reversible
Password policy	Length over complexity; breached-password screening; no forced rotation without cause
MFA	Available on every tier including free; **mandatory** for admin, finance, credential-issuing and sensitive-data roles
Passkeys / WebAuthn	Supported and preferred; the recommended default as adoption allows
Sessions	Short-lived access tokens, rotating refresh tokens, server-side revocation, device list visible to the user
Token denylist	Immediate revocation capability — already implemented in the MVP
SSO	SAML 2.0 and OIDC for institutions
Authorisation	Role-based with attribute conditions; deny by default; evaluated server-side always
Privilege escalation	Requires approval; time-boxed; logged; auto-expiring
Service accounts	Scoped, rotated, never shared, never interactive
Account recovery	Verified out-of-band; recovery is the most-attacked path and is treated as such
Brute force	Progressive delays and lockout with a documented, non-punitive recovery route
Rate limiting	Per-identity and per-IP, at the edge and at the application

Table 105 — Security Bible Ii: Identity & Access

CHAPTER 14 — SECURITY BIBLE III: CRYPTOGRAPHY & DATA PROTECTION

CONCERN	STANDARD
In transit	TLS 1.3; HSTS with preload; modern cipher suites only; certificate transparency monitored
At rest	AES-256 volume and database encryption as the baseline
Field-level	Sensitive-class fields (health, safeguarding, biometric templates, bank details) encrypted at field level with separate key custody
Key management	HSM or managed KMS; documented rotation; two-person control for credential-signing keys (Chapter 5.3)
Signing	Per-institution key pairs; no group master key
Secrets	A secrets manager, never in source, never in environment files committed to a repository, never in logs. Scanned for in CI and in the repository history
Randomness	Cryptographically secure sources only
Hashing	SHA-256+ for integrity; Argon2id for passwords; never MD5 or SHA-1 for anything security-relevant
PII in logs	Redacted at the logging layer, not by convention
Backups	Encrypted, key-separated from the primary system
Biometrics	**Templates only, never raw images**; encrypted; deleted on relationship end; never leave the institution’s tenancy
Payment data	Never stored. Tokenised at the processor. PCI scope minimised deliberately
Quantum posture	Track NIST post-quantum standards; prioritise **credential signatures** for migration, since a certificate issued in 2027 may need to remain verifiable in 2067 and is therefore the group’s longest-lived cryptographic commitment (Volume IX, Chapter 7)

Table 106 — Security Bible Iii: Cryptography & Data Protection

CHAPTER 15 — SECURITY BIBLE IV: SECURE DEVELOPMENT

PRACTICE	RULE
Threat modelling	Required for any feature touching auth, payments, credentials, or sensitive-class data
Code review	Every change, by someone who did not write it. No exceptions for seniority or urgency
SAST / DAST	In CI, blocking on high severity
Dependency scanning	Continuous; critical vulnerabilities patched within 48h, high within 7 days
Supply chain	Pinned dependencies; lockfiles committed; provenance verified; a new dependency requires a justification
Secrets scanning	Pre-commit and in CI, including history
Input validation	Server-side always; client-side is UX, never security
Output encoding	Context-appropriate; templating auto-escapes by default
SQL	Parameterised queries only. String-built SQL is a blocking review failure
SSRF	**Allow-list outbound destinations; validate and re-validate redirects; block private address ranges.** This is called out specifically because a critical SSRF vulnerability was found, reproduced and fixed during the group's own independent MVP audit — the lesson is retained here rather than quietly forgotten
File uploads	Type and size validated; content-sniffed; stored outside the web root; served from a separate origin; scanned
Deserialisation	Never deserialise untrusted input into executable structures
Error handling	No stack traces, internal paths, or version strings to users
Security headers	CSP, HSTS, X-Content-Type-Options, Referrer-Policy, Permissions-Policy — verified in tests
Penetration testing	Annual minimum; before any government or enterprise go-live; after any architectural change to auth or credentials
Bug bounty	Public programme from Phase I; safe harbour stated; researchers paid and credited

Table 107 — Security Bible Iv: Secure Development

CHAPTER 16 — SECURITY BIBLE V: RESILIENCE, BACKUP & CONTINUITY

16.1 Backup

PROPERTY	STANDARD
Frequency	Continuous WAL archiving + daily full
Retention	30 days standard; 1 year and 7 years available
Geography	At least two regions; at least one outside the primary jurisdiction
Encryption	Always, with separate key custody
Immutability	Write-once retention on at least one copy — ransomware defence
Restore testing	**Monthly, automated, verified against a checksum and a functional smoke test.** An untested backup is not a backup and may not be described as one

Table 108 — Backup

16.2 Recovery objectives

TIER	RPO	RTO
Standard	24h	8h
Professional	4h	4h
Premium	1h	2h
Elite	15 min	1h
Enterprise	5 min	30 min

Table 109 — Recovery objectives

16.3 Continuity

Documented and **rehearsed**: regional failure, database failure, provider failure, payment-processor failure, AI-provider failure, key-person unavailability, office inaccessibility, and extended power or connectivity loss in a market. A plan that has never been rehearsed is a document, not a capability. Rehearsal cadence: at least annually, with findings recorded and acted on.

CHAPTER 17 — SECURITY BIBLE VI: INCIDENT RESPONSE

17.1 Severity

SEV	DEFINITION	RESPONSE
S0	Customer data exposed; credential system compromised; funds at risk	Immediate, all hands, executive informed within 15 min
S1	Platform down; auth broken; data integrity at risk	Immediate, on-call escalated
S2	Major feature broken; performance severely degraded	Same business day
S3	Minor defect with a workaround	Next release

Table 110 — Severity

Cross-tenant data access is automatically S0, regardless of how many records were involved, including one.

17.2 The response sequence

Detect → declare (anyone may declare; nobody is penalised for a false alarm) → assign an incident commander → contain → preserve evidence → **notify the customer** → eradicate → recover → verify → **publish**.

17.3 Notification commitments

AUDIENCE	TIMING
Affected customers	**Within 24 hours of confirmation**, before we fully understand it, saying what we do and do not yet know
Regulators	Per statute (NDPA, GDPR: 72 hours)
Public	With the post-incident review
Data subjects	Where required by law, or where the risk to them is material regardless of legal requirement

Table 111 — Notification commitments

17.4 The post-incident review

Published, blameless, and specific. It states what happened, the timeline, the impact including the number of records, the root cause, what we fixed, and what we changed so it cannot recur. It names no individual. It does not use the passive voice to obscure agency (Volume I §11.2.4).

We publish these even when nobody would have found out. The willingness to do so is the asset; the incident is just an event.

CHAPTER 18 — QUALITY, TESTING & THE DEFINITION OF DONE

18.1 The test pyramid

Unit (fast, numerous) → integration (real database, real dependencies) → contract (API compatibility) → end-to-end (critical journeys only) → performance (against Chapter 10 budgets) → security (the cross-tenant suite, auth suite, injection suite) → accessibility (automated, plus manual on new patterns) → **restore** (Chapter 16.1).

Mandatory suites that may never be skipped in CI: cross-tenant isolation, authentication and authorisation, payment idempotency, credential signing and verification, and data export completeness.

18.2 The Definition of Done

A change is done when **all** of the following are true:

107. It works on a mid-range Android on 3 Mbps.
108. It works in RTL and in at least one non-English locale.
109. It meets WCAG 2.2 AA, verified.
110. It is within the Chapter 10 performance budgets.
111. It has tests at the appropriate levels, and they pass.
112. It has been reviewed by someone who did not write it.
113. Its failure modes are defined and its degraded mode is implemented.
114. It is observable — logs, metrics, and an alert if it matters.
115. It is documented where a user or an integrator would look.
116. It exports. Any new record type is in the export.
117. It has a rollback, and the rollback has been considered rather than assumed.
118. Its audit events are emitted.
119. A migration, if any, has a tested reverse path.

“Done” is not “merged”. It is not “works on my machine”. It is not “we’ll add tests later”.

CHAPTER 19 — RELEASE ENGINEERING

Trunk-based development with short-lived branches · every commit builds and tests · staging mirrors production in shape if not scale · progressive rollout (internal → 5% → 25% → 100%) with automatic rollback on error-rate or latency regression · feature flags for anything user-visible and risky · database migrations are backward-compatible and deployed separately from the code that requires them · **never deploy on a Friday afternoon or during an institution's examination window, results period, or admissions deadline** — the customer's calendar constrains our release calendar, and this is a hard rule that has cost us less than one incident would have.

Release cadence: continuous for fixes, weekly for features, quarterly for anything requiring customer preparation. Institutions receive 30 days' notice of any change that alters a workflow they have trained staff on.

CHAPTER 20 — OBSERVABILITY

Structured logs with correlation IDs and redacted PII · RED metrics (rate, errors, duration) per service and USE metrics (utilisation, saturation, errors) per resource · distributed tracing across service boundaries · real user monitoring from the actual devices and networks our users have, not synthetic tests from a data centre · error tracking with grouping and ownership · a **public** status page with real uptime and full incident history · alerting that pages a human only for things a human must act on immediately, because an alert nobody acts on trains everyone to ignore alerts.

The customer-visible half: every institution sees its own audit log, its own uptime, its own SLA credit position (applied automatically — Volume III §17.2), and its own consumption against caps.

CHAPTER 21 — TECHNICAL DEBT & ARCHITECTURE DECISION RECORDS

21.1 ADRs

Every architecturally significant decision is recorded: context, options considered, decision, consequences, and what would make us revisit it. Stored in the repository, numbered, immutable once accepted, superseded rather than edited. A decision whose reasoning is undocumented will be re-litigated by someone who does not know it was already settled — that is the actual cost this prevents.

21.2 Debt

Debt is **registered, not tolerated silently**. Each item carries its cost (what it slows or risks), its interest (whether it worsens with time), and a decision: pay now, pay by a date, or accept permanently with a stated reason. **20% of engineering capacity is reserved for debt, tooling and reliability** and is not reallocated to features under delivery pressure. Under sustained pressure, scope is cut before this reserve is — because the reserve is what makes the next quarter possible.

VOLUME

V

Go-to-Market

Sales · Customer Success · Partners · Ecosystem

Operational

EXECUTIVE SUMMARY

Volume V begins from a binding constraint: paid acquisition does not work in the founding market, so growth must come from mechanisms whose cost does not scale with customers acquired. It identifies seven such engines, of which the most defensible is that every credential a StromeX institution issues is a permanent, cost-free advertisement seen by an employer or a university. It sets out the funnel, the sales motions, the sales playbook and its seven discovery questions, the discipline of public procurement, and — at its centre — the eleven-stage Customer Success Playbook with its health scoring and value ledger. It confronts the segment's unforgiving retention benchmarks directly, sets the partner network economics deliberately above the industry norm, and defines the platform ecosystem and community programmes.

CHAPTER 1 — THE GO-TO-MARKET DOCTRINE

1.1 The central constraint

Paid acquisition does not work in our founding market. A Nigerian private school's lifetime value at Band A pricing does not support a Western customer-acquisition cost, and the audience is not reliably reachable through the channels that paid acquisition uses. Any GTM plan that assumes we can buy growth is arithmetically wrong.

Everything in this volume follows from that single constraint. Our growth must come from mechanisms whose cost does not scale linearly with customers acquired:

ENGINE	WHY IT SCALES SUB-LINEARLY
The free tier	One build, unlimited users, no per-acquisition cost
Referral	Proprietors know proprietors; the network does the work
The reference deployment	One visible institution recruits its peers permanently
Partners	Their relationships, not our headcount
Content & SEO	Compounds; written once, found for years
The marketplace	Third parties bring their own customers
Free verification	Every credential we issue is an advertisement seen by an employer or a university

Table 112 — The central constraint

That last one deserves emphasis. **Every certificate a StromeX institution issues is a durable marketing asset.** When an employer scans a QR code and lands on a verification page, that employer has just learned that this school is serious and that StromeX exists. It costs us nothing per scan and it reaches exactly the audience that influences other institutions. This is the cheapest and most defensible distribution mechanism the group has, and it is a direct consequence of the constitutional decision to make verification free (Volume I §5.3).

1.2 The three GTM rules

120. **Give before asking.** Free tools, free audits, free assessments, free consultations. The Volume I §5 free constitution is a distribution strategy, not philanthropy.
121. **Sell the next step, not the ecosystem.** A 220-student school does not need an ERP. Selling them one produces churn, a bad reference, and a lost decade of expansion revenue (Volume III §20.1 shows the correct answer).
122. **Never sign what we cannot deliver.** Volume I §14.2. Delivery capacity is confirmed in writing before signature, by the person who will actually deliver it.

CHAPTER 2 — THE FUNNEL & THE FREE ENGINE

2.1 The funnel

```
AWARENESS      SEO · verification pages · referrals · community · press · partners
     ↓
FREE USE       Foundation tools · templates · calculators · audits · free tier
     ↓
IDENTIFIED     Account created for a tool that benefits from one
     ↓
QUALIFIED      Free readiness assessment or discovery consultation
     ↓
CONFIGURED     Self-serve configurator; running total visible before any contact
     ↓
CUSTOMER       Contract, onboarding, go-live
     ↓
ADOPTED        Measured usage, not measured signature
     ↓
EXPANDED       Next module, driven by the value ledger
     ↓
ADVOCATE       Reference, case study, referral, community
```

Figure 7 — The funnel

Note where qualification sits: **after** free use, not before. The free tier does the qualifying, at zero cost, more honestly than a discovery call.

2.2 The free-to-paid conversion mechanics

TRIGGER	NATURAL NEXT STEP
Free timetable optimiser hits its 30-class ceiling	Paid timetable module
Free attendance register hits 200 students	Core platform, Standard
Free certificate template downloaded repeatedly	Credential platform + verification
Free SEO audit shows real problems	Technical SEO implementation
Free policy generator used for 10 policies	Policy pack or governance consultancy
Free QR generator used on printed certificates	Verified credentials — “these QR codes could actually verify”
Free readiness assessment reveals data chaos	Migration and implementation

Table 113 — The free-to-paid conversion mechanics

The rule that makes this honest: every ceiling is stated up front, on the tool, before use. A limit discovered at the moment of frustration is a trap (Volume II §4.2), and traps convert once and poison the well.

2.3 Free-tier economics governance

Foundation costs are borne centrally, never charged to divisions (Volume I §8.2), with a hard annual ceiling set by the Executive. Per-tool unit cost is monitored monthly. When the ceiling is under pressure the response order is fixed: **reduce generosity of quantity before reducing quality, and never reduce the entrenched commitments** (free verification, free export, free security, free help when we broke it).

CHAPTER 3 — THE SALES MOTIONS

MOTION	SEGMENT	CYCLE	COST TO SERVE	WHO
Self-serve	Individuals, micro-institutions, SMEs	Minutes–days	~0	Product
Low-touch	Owner-led institutions (the Phase I core)	2–8 weeks	Low	One AE, remote + occasional visit
Consultative	Committee-led institutions, groups	3–9 months	Medium	AE + solutions lead + delivery lead
Enterprise	Multi-site, corporates, large universities	6–18 months	High	Named team, executive sponsor
Procurement	Government, donor programmes	6–24 months	Very high	Bid team, compliance, references
Partner-led	Any, via a certified partner	Varies	Lowest	Partner owns it; we enable

Table 114 — The Sales Motions

The Phase I concentration rule: self-serve and low-touch only, with consultative for lighthouse accounts. Enterprise and procurement motions open in Phase II, once the reference base makes them winnable. Bidding for a ministry contract without twenty referenceable schools is a donation of proposal costs (Volume II §2.3).

CHAPTER 4 — THE SALES PLAYBOOK

4.1 Discovery — the seven questions

Before any proposal, we must be able to answer these in the customer's own words:

123. What is the institution trying to become in three years?
124. What breaks most often today, and who feels it?
125. How many hours a week does the current process consume, and whose hours?
126. What has been tried before, and why did it fail? *(This is the single most predictive question. An institution abandoned by a previous vendor buys differently and needs different reassurance.)*
127. Who decides, who influences, who can veto?
128. What does the money look like — budget cycle, approval threshold, cash rhythm?
129. What would make this a failure in their eyes, in twelve months?

A proposal written without answers to all seven is a guess, and guesses lose to competitors who asked.

4.2 The proposal standard

Every proposal contains, in this order: the institution's own stated objective in their words · what we understood · what we propose, module by module, at Volume III prices · **what is explicitly excluded** · the delivery timeline with the customer's own dependencies named and dated · what we need from them · the total, the annual, and the three-year cost · what happens if they want to leave · the named people who will do the work.

Exclusions are mandatory. Most disputes in this industry originate in something both parties assumed the other was doing.

The total does not move after signature unless the customer changes scope in writing. A vendor whose price rises after commitment has taught the customer never to trust a vendor's price again, and that damage outlives the contract.

4.3 Objection handling

Volume II §3.3 carries the five standing objections and their answers. Two additional rules:

— **"It's too expensive" is almost never about price.** It is about predictability, or about not believing the value. Respond with the value ledger and the unit economics (per student per year), not with a discount. Volume III §20.1's *$15.61 per student per year* is the entire argument.

— **When the honest answer is "you don't need this yet", say it.** It costs one small contract and buys a decade. The proprietor tells other proprietors that StromeX talked them *out* of spending money — which is a more powerful referral than any case study.

4.4 Qualification and disqualification

We walk away when: the institution cannot afford the recurring commitment and would be harmed by trying · the timeline is impossible and the customer will not move it · the scope requires a sector we have no domain expert in (Volume I §14.6) · the work is on the Volume I §7.4 declined list · delivery capacity is not available and cannot be created honestly.

Walking away is a recorded, respected outcome. A salesperson who disqualifies well is more valuable than one who signs badly, and compensation is structured so that this is true in their pay as well as in this document.

4.5 Compensation principles

Commission is paid on **delivered and retained** revenue, not on signature. A contract that churns in year one claws back. Commission on services is weighted below commission on recurring software, because Volume I §6.2′s portfolio rule requires it. There is no commission accelerator that would make it rational to oversell a customer.

CHAPTER 5 — PROPOSALS, TENDERS & GOVERNMENT PROCUREMENT

5.1 The absolute rules

No bribery. No facilitation payments. No "processing fees". No bid-rigging. No shell bidders. Ever, in any market, regardless of local norm or of what a competitor is doing. (Volume I §6.6.) Losing a contract for this reason is a correct outcome and is reported as a win in the integrity record.

If an official requests a payment, the engagement is escalated to the Executive immediately, documented, and — if the request stands — we withdraw and say why in writing.

5.2 The bid/no-bid gate

Bid only when: we meet the mandatory criteria without misrepresentation · we have referenceable delivery at comparable scale · capacity is confirmed for the delivery window · the payment terms are survivable (public-sector payment delay is a real and frequently fatal risk to suppliers our size) · the evaluation is transparent enough that a good bid can win.

5.3 The government value argument

The winning number in a public-sector tender is almost never the total. It is the **per-beneficiary cost**. Volume III §20.4′s ≈*$1.48 per student per year* for a 400-school state deployment is the argument, and it is true, which is why it can be defended under scrutiny.

The second argument is auditability: a ministry's fear is not cost, it is a scandal. Published pricing, complete audit logs, published incident history and free public verification are precisely the properties that make a procurement officer safe.

CHAPTER 6 — THE CUSTOMER SUCCESS PLAYBOOK

The most under-invested function in most software companies and the one that determines whether the Volume II §9 moat is ever actually built. Net revenue retention is the metric this chapter exists to move, and Chapter 12 explains why our benchmark is harder than most.

6.1 The eleven stages

Stage 1 — Discovery

Owner: Marketing/Product. Goal: they find us without us paying for it. They arrive via a free tool, a verification page, a referral, a search result or a partner. **Success:** they get value in the first session without talking to anyone.

Stage 2 — Consultation

Owner: Sales. Goal: understand, not pitch. Free discovery consultation (60 min, physical or remote) and free readiness assessment. The assessment is the most valuable thing we do pre-contract and it is deliberately free (Volume III §2.6) because charging for it means fewer institutions do it, and skipping it is the leading cause of failed implementations. **Success:** they leave with a written honest assessment they could act on **even if they never buy from us**.

Stage 3 — Proposal

Owner: Sales. Goal: a document they can defend to their board. Per Chapter 4.2. **Success:** the buyer can explain the proposal to a trustee without us in the room.

Stage 4 — Onboarding

Owner: Delivery. Goal: time-to-first-value, measured in days. A named delivery lead from day one. A kick-off that establishes: what go-live means, what the customer must provide and by when, who their champion is, and what we will measure. **The first-value rule:** something real must work within **14 days** — a live site, a working admissions form, the first batch of verified certificates. An institution that has waited eight weeks to see anything has begun to doubt, and doubt in week eight is far more expensive than scope in week one.

Stage 5 — Migration

Owner: Delivery. Goal: their history arrives intact. The most technically and emotionally fraught stage. Rules: **parallel run before switch-off, always** · reconciliation reports the customer signs off · nothing is switched off until its replacement is proven · a documented reverse path.

Stage 6 — Training

Owner: Academy. Goal: staff competence, not attendance. In their language, on their devices, with their data. Role-specific, not generic. Recorded for staff who join later. **Redeployment training is free where StromeX automation changed someone's role** (Volume I §7.1) — this is the single highest-return line in the entire GTM budget, because staff resistance is the largest cause of failed deployments and this converts the most resistant person into the most invested one.

Stage 7 — Go-live

Owner: Delivery. Goal: an uneventful day. Never during an examination window, results period, or admissions deadline. Hypercare for 14 days: elevated support, daily check-in, a named human. A go-live that needed heroics is reviewed as a planning failure (Volume I §13.1.9).

Stage 8 — Adoption

Owner: Customer Success. Goal: usage, not signature. **Adoption is measured, and low adoption is our problem, not the customer's.** Tracked: active users by role, feature usage against what they bought, workflows completed, support ticket themes. A module bought and unused is a renewal we are going to lose in eleven months, and it is visible now. **The unused-module rule:** if a module is unused for 60 days, we intervene — train, reconfigure, or **proactively remove it and stop charging for it.** Charging for something demonstrably unused is exactly the extractive behaviour Volume I §6.1 exists to prevent.

Stage 9 — Support

Owner: Support. Goal: resolution, and pattern detection. Volume III §17. The standing rule: when something we built is broken, the customer speaks to a human free, on every tier including Foundation. Support tickets are read as product intelligence — three institutions asking the same question is a documentation defect; ten is a product defect.

Stage 10 — Renewal & Expansion

Owner: Customer Success. Goal: earned, not extracted. Chapter 7.

Stage 11 — Advocacy

Owner: Marketing. Goal: they recruit for us. Reference calls, case studies, community contribution, speaking, referral. **Compensated honestly** — referral credits, not undisclosed inducements. No customer appears in our marketing without written permission and an accurate description of what we actually did (Volume II §8).

6.2 Health scoring

Every account carries a health score from measured signals, not from the account manager's feelings:

SIGNAL	WEIGHT	WHY
Active users vs. licensed	High	The leading indicator of everything
Modules used vs. purchased	High	Directly predicts non-renewal
Core workflow completion	High	Are they actually running on us?
Support ticket sentiment & recurrence	Medium	Repeated same-theme tickets = unresolved friction
Champion still employed	High	Champion departure is the most under-weighted churn signal in this industry
Payment behaviour	Medium	Late payment often precedes churn, and often signals distress we could help with

SIGNAL	WEIGHT	WHY
Value ledger trend	High	Are we actually delivering measurable gain?
Executive engagement	Medium	Does anyone senior know we exist?

Table 115 — Health scoring

Red health triggers an intervention within 5 business days, owned by a named person, with a written plan. Not an email — a plan.

6.3 The value ledger

Volume I §4.4. Maintained continuously per customer: staff-hours returned, fee collection rate change, admissions cycle time, certificate fraud incidents prevented, print cost avoided, error rates. Shown at every renewal **including the periods where the numbers are unimpressive.** A renewal conversation with no value ledger is a renewal we have not earned, and presenting only the good quarters teaches the customer that our numbers are marketing.

CHAPTER 7 — RETENTION, EXPANSION & THE RENEWAL DOCTRINE

7.1 The honest benchmark problem

Published 2026 B2B SaaS benchmarks put median net revenue retention at roughly **108%**, but segmented by contract value the picture is unforgiving for a business like ours: accounts below ~$25k ACV show median NRR around **97%**, mid-market around **108%**, and enterprise (>$100k ACV) around **118%** (Growthspree, Digital Applied).

Our Phase I customers sit squarely in the sub-$25k band, where the median company *shrinks* its existing customer base year over year. This is the most important uncomfortable fact in this volume, and the financial model in Volume VIII is built on it rather than around it.

Our target is 105–112% NRR in Phase I, rising toward 115%+ as ACV grows in Phase II. Beating the segment median by 8–15 points is achievable but it is not automatic, and it requires exactly three things:

130. **A genuine expansion ladder** — the Volume I §6.5 module ladder is the mechanism. Institutions grow into more modules year over year, which is the only reliable source of expansion revenue in a market where we cannot raise prices aggressively.
131. **Gross retention above 92%** — which in this market means surviving fee-collection stress, proprietor changes and FX shocks. Chapter 7.3.
132. **Adoption as a managed function, not a hope** — Chapter 6.2.

If we cannot demonstrate NRR above 105% by the end of Phase I, the Volume VIII growth scenarios do not hold and the plan must be revised rather than the number defended.

7.2 The renewal doctrine

- —**No auto-renewal without notice.** 60 days' notice, in plain language, stating the amount.
- —**No price increase mid-term.** At renewal, capped at 12% annually without documented, communicated scope increase (Volume I §6.3.6).
- —**Cancelling is easier than subscribing.** Measured, reported, and a defect if it is not true.
- —**Downgrade is self-serve.** "Contact sales to downgrade" is a prohibited pattern.
- —**We proactively remove what they don't use** (Chapter 6.1, Stage 8).
- —**We tell them when a cheaper configuration would serve them better.** This costs revenue in the quarter and is the single most effective retention action available.

7.3 Churn: causes and countermeasures

CAUSE	FREQUENCY IN OUR MARKET	COUNTERMEASURE
Cannot afford it	High	Downgrade path offered before churn; Band A pricing; instalment terms; honest qualification at Stage 2
Never adopted it	High	Stage 8 adoption management; unused-module removal
Champion left	High	Multi-threading — never fewer than three relationships in any account above Standard
Proprietor/leadership	Medium	Executive re-onboarding within 30

CAUSE	FREQUENCY IN OUR MARKET	COUNTERMEASURE
change		days of any leadership change
Delivery failure	Medium	Volume I §14.2; capacity confirmed before signature
Outgrew us	Low	Should not happen; if it does, it is a product roadmap input, urgently
Competitor	Low–Medium	Usually a symptom of one of the above, not a cause; recorded honestly

Table 116 — Churn: causes and countermeasures

Every churn is logged with a real reason and reviewed monthly. “Price” is not accepted as a reason without evidence (Volume II §5.2.5).

7.4 The win-back rule

An institution that leaves keeps its data export, its verification pages stay live for issued credentials, and it is treated with the same courtesy as a customer. **Credentials already issued remain verifiable forever regardless of whether the institution is still a customer** — because the graduate did nothing wrong, and holding a graduate’s verifiable degree hostage to their school’s subscription would be exactly the data hostage-taking Volume I P5 forbids. This is also, incidentally, the most effective win-back mechanism we have.

CHAPTER 8 — MARKETING

8.1 The mix

CHANNEL	ROLE	PHASE
SEO & content	The primary engine. Compounds. Answers the questions institutions actually search	I onward
Free tools	Distribution disguised as generosity (Volume I §5.3)	I onward
Verification pages	Passive, permanent, high-credibility exposure to employers and universities	I onward
Referral programme	The highest-converting channel in an owner-led market	I onward
Reference deployments	One visible institution recruits its peers	I onward
Community & events	Proprietor associations, education fairs, faith networks	I onward
Partner co-marketing	Their audience, shared cost	II
Developer relations	Ecosystem seeding	II
Press & thought leadership	Credibility with ministries and enterprise	II
Paid acquisition	Only in Band C markets where LTV supports it	III
Annual summit	The ecosystem's centre of gravity	II onward

Table 117 — The mix

8.2 Content strategy

We publish what an institution needs whether or not they buy from us: how to run an admissions cycle, how to write a safeguarding policy, how to prepare for accreditation, how to move records off paper, how certificate verification actually works, what school software should cost. **The honest guide to pricing in our own category is published, including where we are more expensive than a competitor and why.**

This works for a specific reason: our buyers search for problems, not for products. A proprietor searching "how to stop certificate forgery" is worth more than one searching "school management software", is cheaper to reach, and arrives already trusting us.

8.3 Marketing rules

Every claim substantiated (Volume II §7.4 — five proof points, and no sixth until a sixth is true) · no fabricated testimonials, reviews, case studies or statistics, ever (Volume I §7.2.3) · no fear-selling · no competitor disparagement by name · no pre-announcing unshipped products (Volume II §8) · all pricing in marketing matches Volume III exactly · every customer story approved in writing and accurate about what we actually did.

CHAPTER 9 — THE PARTNER NETWORK

9.1 The strategic logic

The most fragmented part of the competitive landscape — freelancers and small agencies (Volume II §5.1, Class 5) — is simultaneously the largest untapped distribution network in our market. They have the relationships. They lack product, continuity, security posture and scale. **We do not compete with them at the bottom of the market; we recruit them.**

This converts our most numerous competitor into our distribution channel, and it is how the group reaches thousands of institutions without thousands of employees.

9.2 Partner tiers

TIER	WHO	MARGIN / SHARE	REQUIREMENTS
Referral	Anyone	10% of year-1 revenue	Register the deal
Reseller	Agencies, ICT firms	25–35%	1 certified person, 2 deals/yr
Delivery Partner	Implementation firms	30–40% of services	2 certified implementers, quality audit
Solution Partner	Sector specialists	35–45%	Domain expertise, 5 references
White-label Partner	Established brands	65% of end revenue to partner	Volume commitment, quality audit
Distributor	Country/regional	Negotiated	Market commitment, local entity, support capability

Table 118 — Partner tiers

9.3 What we give partners

Certification and training (Volume III Division 15) · deal registration with real protection · demo environments · co-branded and white-label materials · pre-sales support · technical escalation · a partner portal with pipeline, resources and payouts · leads routed by geography and specialism · co-marketing funds at Solution tier and above.

9.4 What we require

Certified staff · adherence to Volume III pricing (partners set their own services margin but do not undercut published software prices, which would destroy the transparency wedge for everyone) · the Volume I §6.6 prohibited-practices standard · customer satisfaction above a threshold · annual recertification · **no misrepresentation of what StromeX does** — a partner overselling capability damages us and is grounds for termination.

9.5 The partner covenant

We will not compete with a partner on their registered deal. We will not take a partner's customer direct. We will not change partner economics without 180 days' notice. If we acquire a company that competes with a partner, we say so immediately and offer a transition.

A partner network that fears the vendor does not invest in the vendor. The economics above are deliberately more generous than the industry norm for the same reason the marketplace rev-share is (Volume III §16.1): a sustainable partner ecosystem is worth more than the extra points.

CHAPTER 10 — THE PLATFORM ECOSYSTEM

10.1 Why we open the platform

A closed platform grows at the rate we can hire. An open one grows at the rate the world can build. The ecosystem exists so that capabilities we would never prioritise — a Kano-specific fee-collection integration, a Ghanaian examination-board connector, an Urdu madrasah module — get built by people closer to the need than we will ever be.

10.2 The ecosystem surfaces

SURFACE	WHAT THIRD PARTIES BUILD	TERMS
REST API	Anything	Free; commercial tier at high volume
Webhooks & events	Reactive integrations	Free
SDKs (JS, Python, PHP)	Client libraries	Free, open source
Design system	Consistent UI on top of us	Free, open source
Plugins	In-product extensions	Marketplace, 20% rev-share
Mini-apps	Embedded experiences inside portals	Marketplace
AI agents	Custom agents for niches	Marketplace
Connectors	Third-party system bridges	Marketplace
Themes & templates	Design assets, document templates	Marketplace
Sector modules	Whole verticals we have not entered	Solution Partner terms
Embedded/OEM	StromeX modules inside another product	Volume III §16.2

Table 119 — The ecosystem surfaces

10.3 The developer promise

Documentation is free and public · the same API we use is the API you get (Volume IV §6, dogfooding rule) · versions are supported 24 months after supersession · breaking changes never happen within a version · rate limits are published and generous · a free sandbox with realistic seed data · **we will not clone your successful extension and bundle it** — if a marketplace extension becomes essential, we acquire it or partner, we do not copy it. This last commitment is what makes the ecosystem investable by anyone other than us, and breaking it once would end the ecosystem permanently.

10.4 Marketplace quality

Every listing is reviewed before publication: does what it claims, handles data per Volume IV §9, does not exfiltrate customer data, states its pricing clearly, has a support contact. Listings that break customer trust are removed and the seller is barred. **We are accountable to the customer for the marketplace even where we did not build the extension** — an institution has one relationship, with us.

CHAPTER 11 — COMMUNITY, ACADEMY & ADVOCACY

11.1 StromeX Academy

Volume III Division 15. Free documentation, video library, community forum and onboarding webinars; paid training and certification where a credential confers real value. Certifications: Certified Administrator, Certified Implementer, Certified Partner Engineer, Certified Designer.

The certification must be worth something in the labour market or it is not worth selling. That means it must be genuinely difficult, independently verifiable (using our own credential infrastructure — we verify our own certificates with the same system we sell), and recognised by employers because certified people are demonstrably better. A certification that everyone passes is a receipt, not a credential.

11.2 Community programmes

PROGRAMME	PURPOSE	PHASE
Community forum	Peer support; reduces support load; surfaces product truth	I
Open-source community	Design system, SDKs, exporters	I
Educator community	Curriculum, templates, question banks shared between institutions	I
Campus ambassadors	Students at universities we serve	II
Student ambassadors	Secondary students; the next generation of developers	II
Developer community	Ecosystem builders	II
User conferences (regional)	Customers meet each other; the strongest retention event there is	II
The annual summit	The ecosystem's centre of gravity; customers, partners, developers, government	II
Research fellows	Volume IX, Chapter 3	II

Table 120 — Community programmes

11.3 The advocacy principle

An advocate is a customer who recruits for us because the product and the relationship earned it. Advocacy cannot be bought, and attempting to buy it (paid testimonials, undisclosed inducements, review incentives) is prohibited and would destroy the only asset — credibility — that makes the referral engine work in the first place.

Referral credits are legitimate, disclosed, and modest. The advocacy itself must be free.

CHAPTER 12 — GTM METRICS

Reviewed monthly by the Executive. Every metric has a named owner.

12.1 Acquisition

Free-tier monthly active institutions · free-to-identified conversion · identified-to-qualified · qualified-to-customer · CAC by motion and by channel · **CAC payback in months** (target <12 for low-touch, <18 for consultative) · organic traffic and ranked keywords · verification-page sessions (the passive engine) · referral share of new customers (target: >35% by end of Phase I) · partner share of new customers (target: >30% by end of Phase II).

12.2 Delivery

Time to first value (target <14 days) · time to go-live vs. committed · implementation margin vs. Volume I §6.2 band · go-lives requiring heroics (target: zero) · migration reconciliation exceptions.

12.3 Retention & expansion

Gross revenue retention (target >92%) · **net revenue retention** (target 105–112% Phase I, >115% Phase II — Chapter 7.1) · logo churn with real reasons · adoption rate per module · unused-module count · health score distribution · red accounts with an active written plan (target: 100%).

12.4 Trust

Cancellation clicks vs. sign-up clicks (must be fewer) · support first-response and resolution times by tier · incidents published within commitment · SLA credits applied automatically · **customer-reported "would recommend"**, collected honestly and reported including the bad quarters.

12.5 The metric we refuse to optimise

Signature volume disconnected from adoption. A quarter of strong bookings and weak adoption is a bad quarter reported as a good one, and it is the precise mechanism by which software companies destroy their own next three years. Bookings are reported alongside 90-day adoption, always, in the same table.

VOLUME

VI

The Creative Division

Design · Publishing · Print · Media · The Craft Standard

AUTHORITY

Operational — Creative Director

EXECUTIVE SUMMARY

Volume VI explains why a technology group owns a creative division: craft is the differentiator competitors cannot copy cheaply, it closes the loop on the institution's physical reality, and it is the reason institutions refer one another. It sets the craft standard — work indistinguishable from a good London or New York studio that a Lagos printer can reproduce correctly on the first attempt — and the anti-slop rules that govern AI-assisted production, including the prohibition on inventing facts a client has not published. It specifies institutional identity and heraldry, security-document design and its layered defences, publishing practice, the exacting standards for Islamic and Arabic typesetting, rights and licensing, international print coordination, audio and motion, and educational publishing.

CHAPTER 1 — WHY A TECHNOLOGY GROUP OWNS A CREATIVE DIVISION

Three reasons, in ascending order of importance.

1. It is the differentiator competitors cannot copy cheaply. Volume II §6 Wedge 5: in a market where "good enough" is the norm, finished work is a moat. Software companies do not have art directors. Design agencies do not have credential infrastructure. The combination is rare because it is operationally awkward — which is exactly what makes it defensible.

2. It closes the loop on the institution's physical reality. An institution's certificate, ID card, prospectus, signage and textbook are the parts of the transformation that people *touch*. A beautiful portal and an ugly certificate is an incoherent institution, and the certificate is the part that goes out into the world and gets judged.

3. It is the reason for referral. Nobody photographs a database. Proprietors show other proprietors the prospectus, the certificate, the ID card. Craft is our most efficient marketing channel, and it is measured in Volume V §12.1 as the referral share of new customers.

There is a fourth, quieter reason: the founder's instruction to give institutions in our markets access to UK and US production standards they cannot reach directly. That is a genuine access gap, and closing it is worth doing on its own terms.

CHAPTER 2 — THE CRAFT STANDARD

2.1 The standard, stated

Every piece of creative work leaving StromeX must be indistinguishable in quality from work produced by a good London or New York studio — and must be reproducible by a printer in Lagos on the first attempt.

Both halves are binding. The first without the second is a portfolio piece that fails in production. The second without the first is why our market is full of mediocre work.

2.2 The eleven qualities

Volume I §12.1 applies to every creative surface: premium, elegant, minimalist, fast, reliable, beautiful, professional, accessible, human, modern, **timeless**.

Timeless is the demanding one, and in print it is even more binding than on screen. A website can be redesigned in three years. A crest, a certificate template, a textbook series and a signage system cannot. Anything with a life over five years is designed against the fashion of its decade, not with it.

2.3 The anti-slop rules

Written explicitly because the failure mode of AI-assisted creative production is a specific, recognisable mediocrity:

133. **No stock-photo-and-gradient generic.** If the work could belong to any institution, it belongs to none.
134. **No template that has not been made specific.** A template is a starting point that must be earned away from.
135. **No AI output shipped unedited.** Generated imagery, layout and copy are raw material. A human art director is accountable for what ships.
136. **No filler.** Lorem ipsum never reaches a client. Placeholder content is labelled as placeholder, visibly (the reference implementations do exactly this — a labelled placeholder block rather than an invented fact).
137. **No invented facts.** Volume I §7.2.3. A prospectus does not claim an accreditation the institution does not hold, a student outcome it cannot evidence, or a partnership that does not exist. **Where the client's ambitions imply information they have not published, we say so plainly in a labelled placeholder rather than fabricating it.** This standard is already established in the group's reference work and it is non-negotiable, including when the client asks.
138. **No borrowed identity.** Reference material is a mood board, never a source to imitate. A supplied "inspiration" crest informs the design *vocabulary* — the formal register of seals, ledgers, folios, chapter marks — without borrowing its name, its claims, or its colour story.
139. **No unlicensed assets.** Chapter 9.
140. **No detail left to the printer.** Bleed, trim, colour profile, stock, finish, binding and imposition are specified by us.

2.4 The three-look test

Before any creative work ships:

- —**At arm's length** — does the composition hold?
- —**At reading distance** — is the typography right?

—**At 200%** — is the craft real, or does it fall apart?
Most work that fails, fails the third.

CHAPTER 3 — THE AI-AND-HUMAN PRODUCTION MODEL

3.1 The model

The founder's brief was that our blend of AI and human effort should be **seamless**. Seamless is a quality standard, not a concealment instruction, and Volume I §7.2 governs: we say "AI-assisted, human-reviewed", and we make it true.

STAGE	AI DOES	HUMAN DOES
Research & reference	Gathers, summarises, finds precedent	Judges relevance, verifies facts
Concept	Generates volume and variation	Selects the direction and says why
Copy	Drafts, restructures, adapts register	Edits, fact-checks, owns the voice
Layout	Applies grids, flows text, produces variants	Sets the grid, resolves the hard spreads, judges rhythm
Imagery	Generates, extends, retouches	Art-directs, licenses, verifies
Translation	First pass	Reviews and signs off; literary and religious work is human-first
Production	Preflight, imposition checks, asset generation	Approves, press-checks
Approval	—	**A named human is accountable for everything that ships**

Table 121 — The model

3.2 Why this produces a structural advantage

This is Volume II §6 Wedge 4 in practice: heavy AI leverage on *production*, deliberate human investment on *judgement*. It produces developed-market quality at emerging-market cost, which is the margin advantage funding the whole plan.

And it is temporary. Volume II §9 assesses this advantage as largely gone by around 2032, as every competitor adopts the same model. The correct response is not to defend it but to spend it: use the margin window to build the durable moats — the system of record, the credential infrastructure, the partner network — while it exists.

3.3 Where AI does not lead

- **Qur'anic and scriptural text** — reproduced from verified sources, never generated (Volume IV §8.3).
- **Literary and religious translation** — human-first; AI assists the human, not the reverse.
- **Anything asserting a fact about a real institution or person** — human-verified.
- **Final art direction** — a machine can produce a hundred options; only a person can say which one is right and defend it.
- **Voice, for a named human narrator** — synthetic voice may never impersonate a real person, and a "human-narrated" audiobook is narrated by a human.

CHAPTER 4 — STROMEX STUDIO: DESIGN PRACTICE

4.1 The process

Brief → research → strategy → concept (3–5 genuine directions, not one plus two weak decoys) → refinement → system → production → handover.

The decoy rule: presenting deliberately weak options to steer a client toward a preferred one is manipulation and is prohibited. Every option presented must be one we would be willing to build.

4.2 Handover standard

Every design engagement hands over: source files in an open or industry-standard format, all exported variants, a specification document, licences for every asset used, and — for identity work — the full brand kit per Volume I §10.1 (app icon, monochrome, embossed, reversed, digital variants). The client owns the work outright on final payment. **We do not hold source files as leverage.**

4.3 Revisions

Stated in the quote, always. Standard: 2 rounds. Premium: 3. Elite: unlimited within the agreed direction. A change of *direction* is a new brief and is quoted as one — stated up front, not discovered at invoice.

CHAPTER 5 — INSTITUTIONAL IDENTITY & HERALDRY

Schools, universities, ministries, mosques and hospitals need identity systems with a formal register most commercial branding does not use: crests, seals, ledgers, folios, chapter marks, mottoes.

The rules:

141. **Heraldic devices carry meaning.** Charges, tinctures and ordinaries mean things. We use them correctly or we design a modern mark instead. A crest assembled from decorative clip-art is worse than no crest.
142. **Never claim what is not held.** No crown implying royal patronage that does not exist, no laurel implying an award not won, no “Est. 1954” for an institution founded in 2017, no city or country claim the institution cannot make.
143. **The motto is in the institution’s own language and their own words**, translated accurately, checked by a native speaker.
144. **Design for the seal first.** If it works blind-embossed at 25mm in one colour, it works everywhere. If it only works as a full-colour render, it is an illustration, not a crest — the same reasoning that selected Candidate D in Volume I §10.1.
145. **Deliver the full system**: full colour, one colour, reversed, embossed/blind, engraved line art, digital icon, watermark, and a print specification.

CHAPTER 6 — SECURITY DOCUMENT DESIGN

Certificates, transcripts, ID cards, receipts, licences. The most consequential creative work the group does — a forgeable credential destroys the institution's trust and ours together.

6.1 The dual mandate

A security document must be **beautiful enough that a graduate frames it** and **hard enough to forge that an employer can rely on it.** Designers who treat security features as decoration and security engineers who treat beauty as optional both produce failures.

6.2 The design layers

LAYER	PURPOSE
Guilloche	Fine mathematical line-work; extremely difficult to reproduce by scan-and-print; the primary visual security element
Microtext	Legible only under magnification; degrades visibly under photocopying
Cartouche	The framed identifier field — designed so the credential number is unambiguous and transcribable (Volume IV §5.2)
Seal	Foil-blocked, embossed or blind-debossed; tactile evidence a scan cannot carry
Substrate	Cotton stock, watermark, UV fibres, controlled deckle
UV / IR elements	Invisible under normal light
Registered ground	A background vector whose registration to the trim is itself a check
QR	Layer 2 of the Volume IV §5.5 verification model — placed where it survives framing
Signature blocks	Real, authorised signatories with a documented authority map

Table 122 — The design layers

6.3 The specification is part of the deliverable

Every security document ships with a **press specification**: stock weight and composition, colour profile and ink sequence, foil type and temperature, emboss depth and die spec, trim and bleed, imposition, varnish, and the acceptance criteria for a press check. Handing a printer a PDF and hoping is not a deliverable.

6.4 Governance

A security template is **frozen** on approval. Changes require a documented amendment with a new version identifier, because a credential's design is part of its verifiability — an employer comparing a 2031 certificate against a 2028 one must be able to establish which templates were valid when. Freeze declarations, revocation records and version registers are maintained as institutional records, not as design files.

CHAPTER 7 — STROMEX PRESS: PUBLISHING PRACTICE

7.1 What we publish

Institutional (prospectuses, yearbooks, handbooks, governance volumes, commemorative histories) · educational (textbooks, workbooks, teacher guides, question banks) · religious (Qur'anic study aids, Islamic scholarship, devotional works, children's Islamic books) · trade (author services, general non-fiction, memoir) · academic (journals, monographs, research volumes) · children's (picture books, readers, activity books).

7.2 The editorial process

Assessment → developmental edit → structural edit → line edit → copy-edit → **fact-check** → design → typeset → proofread → **second proof** → preflight → proof → print → distribute.

Two proofreads are standard, not optional. The cost of a typo in a printed run of 2,000 is the run.

7.3 Editorial standards

Volume I §11 governs voice. Additionally: house style documented per imprint · every factual claim verified and its source recorded · quotations checked against the original edition, never against a secondary source · names, dates, titles and honorifics verified with the person or institution · numbers checked twice · **the author's voice preserved** — editing improves clarity, it does not homogenise.

7.4 The author covenant

Clear contracts in plain language · rights and royalties stated without ambiguity · the author owns their work · transparent royalty accounting (Volume III §13.6) · no rights grab in a services agreement — an author paying us for editing does not thereby licence us anything · honest assessment even when it costs us the job · we say when a manuscript is not ready.

CHAPTER 8 — ISLAMIC & ARABIC PUBLISHING

The founder's brief named premium Islamic book design specifically. It is also the area where error is least forgivable, so the standard is the highest in the division.

8.1 The absolute rules

146. **Qur'anic text is reproduced, never generated, never paraphrased, never completed by a model.** It is set from a verified source with established provenance, and the source edition is recorded in the colophon.
147. **Translations and tafsir are attributed** to their named translator/scholar and edition. An unattributed translation is not publishable.
148. **No novel religious ruling is issued.** Where a text requires a ruling, a qualified human scholar provides it and is named (Volume I §7.5).
149. **Scholarly review before publication** for any work of religious instruction, by a reviewer qualified in the relevant tradition. If we do not have such a reviewer, we do not publish the work.
150. **Conventions differ between traditions and schools.** Where they differ, we ask; we do not assume, and we do not silently pick one.
151. **Handling standards** — the physical and typographic conventions for Qur'anic text (placement, ornamentation, what may appear on a facing page, binding and disposal guidance) are observed.

8.2 The typographic standard

Arabic typesetting is not Latin typesetting reversed. Getting it wrong is immediately visible to a literate reader and permanently damaging to credibility.

REQUIREMENT	STANDARD
Typeface	A genuine Arabic face designed for the register — Amiri for literary and Qur'anic work, a modern face for interface and contemporary text. Never a Latin font's fallback
Harakat (vocalisation)	Correctly positioned, correctly spaced, not colliding with ascenders or with the line above
Ligatures & contextual forms	Full shaping; a face and engine that handle initial/medial/final/isolated forms correctly
Justification	Kashida-aware where the tradition calls for it; never letter-spaced like Latin
Line spacing	Set for Arabic's vertical extent, not inherited from a Latin grid
Bidirectional runs	Latin, numeric and coded runs explicitly directed (Volume IV §11.2.5)
Footnote apparatus	Designed for the density Islamic scholarship actually uses — multiple apparatus streams, verse references, transliteration
Transliteration	One system, chosen and stated, applied consistently, with full diacritics
Facing-page layouts	Arabic and translation in register, spread by spread
Page furniture	Running heads, folios and section marks in the correct script and direction

Table 123 — The typographic standard

8.3 Why this is a business, not only a duty

There is a real and underserved market: institutions and authors across West Africa, the Gulf and South Asia who want their work produced to a standard that currently requires a Beirut, Cairo or Istanbul house — and who cannot easily reach one. Doing this properly, at Band A and Band B prices, with UK print available, is a defensible position that very few can occupy, because it requires simultaneous competence in Arabic typography, Islamic scholarship, international print and modern production. The group has the first three in its reference work already.

CHAPTER 9 — RIGHTS, LICENSING & ATTRIBUTION

ASSET	RULE
Fonts	Licensed for the actual use (desktop, web, app, ePub, broadcast) and the actual volume. Licences recorded per project. Font piracy is a firing matter, not a mistake
Images	Licensed or original. Model and property releases where people or private property appear. Children’s images require documented parental consent, always, in every jurisdiction
Illustration	Commissioned with clear rights transfer, or licensed
Music & audio	Licensed for the medium and the territory
AI-generated assets	Provenance recorded; the provider’s terms verified as permitting commercial use; **never used where a real person’s likeness or a living artist’s identifiable style is imitated**
Quotations	Within fair dealing, or permission obtained and recorded
Translation rights	Cleared before work starts
Client-supplied material	Client warrants their rights in writing; we verify what we reasonably can
Our work	Client owns the deliverable on final payment. We retain portfolio rights unless the client requests otherwise, and confidential work is never shown

Table 124 — Rights, Licensing & Attribution

The standing instruction: if the licence cannot be produced, the asset does not ship. “We found it online” is not a licence and never has been.

CHAPTER 10 — INTERNATIONAL PRINT COORDINATION

10.1 What this service actually is

The founder identified a genuine gap: institutions and authors in our markets cannot practically access UK and US production standards. The barriers are not price alone — they are specification literacy, colour management, freight, customs, minimum runs, payment mechanisms and the inability to attend a press check.

StromeX takes responsibility for the outcome, not merely for placing the order. Volume III §13.3: coordination margin 18%, minimum $240, with print cost passed through at cost and disclosed.

10.2 The tier decision

TIER	WHERE	USE IT FOR	DO NOT USE IT FOR
Local	Nigeria / regional	Workbooks, exam papers, newsletters, high-volume text	Anything with critical colour or premium finishing
Regional Premium	UAE / Turkey / India	Quality colour at moderate cost, medium runs	Very short runs (setup dominates)
UK Premium	United Kingdom	Prestige titles, case binding, foil, specialist finishing	Price-sensitive volume work
US Premium	United States	Trade distribution, large runs, POD network	Small runs shipped to Africa (freight dominates)
Fine Press	UK / EU specialist	Limited editions, hand-finishing, archival stock	Anything on a deadline

Table 125 — The tier decision

We recommend the cheapest tier that meets the requirement, and we say so even when a higher tier earns us more. The coordination margin is a percentage, so recommending UK print for a workbook would pay us more and would be advice we could not defend. Volume I §6.1 governs.

10.3 The coordination discipline

Specification written by us · at least two quotes for any run above $2,000 · paper dummy before commitment on any premium title · **wet proof for any title where colour matters**, and we say plainly when a digital proof is insufficient · press check attended for Elite work · freight, insurance, duties and clearance quoted before commitment, not discovered · a landed-cost figure given to the client, not an ex-works one.

The honest disclosure: print coordination is our lowest-margin service line (22–35%, Volume I §6.2) and it is operationally the most exposed — currency movement, freight volatility, customs delay and press error are all real. We do it because it closes the loop for the customer and because nobody else in our market will. It is not a growth engine and it must never be sold as though it were.

CHAPTER 11 — AUDIO, MOTION & MEDIA

11.1 Audio

STANDARD	RULE
Narration	AI narration is disclosed as AI. Human narration is by a human. A synthetic voice never impersonates a real person
Qur'anic recitation	By a qualified human reciter, always. Never synthesised
Mastering	Loudness-normalised to platform standards; checked on phone speakers, not only on studio monitors
Accessibility	Transcripts and captions for everything
Languages	Recorded by native speakers of the target variety, not approximated

Table 126 — Audio

11.2 Motion

Motion explains, never entertains (Volume I §12.2.3). Institutional motion work has a life of years, so it follows the timeless rule: no effects tied to a rendering fashion. Everything is delivered captioned and works with sound off, because most of it will be watched that way.

11.3 Photography

Institutional photography is the fastest way to make a school look either serious or cheap. Standards: real people from the institution, not stock · consent documented, and for children documented from parents · natural environments over staged ones · a consistent grade across a set · delivered in an organised, tagged library the institution can reuse, not as a folder of files.

CHAPTER 12 — EDUCATIONAL PUBLISHING

12.1 The bespoke-textbook proposition

The founder's brief named specialised textbooks for individual schools. This is a real and unusually sticky product: a school with its own textbook series, carrying its own brand, aligned to its own scheme of work, has a differentiator its competitors cannot copy and a switching cost that is genuinely earned.

Requirements before we accept the work: the institution's curriculum must be documented (we can produce it — Volume III §14.2) · alignment to the national or examination-board syllabus must be verified · subject-matter review by a qualified teacher of that subject · pedagogical structure designed by an instructional designer, not merely content laid out · assessment and answer material produced with it · a revision cycle agreed, because a textbook that is never revised becomes an embarrassment.

12.2 The instructional design standard

Content organised by learning outcome, not by chapter convenience · explicit prior-knowledge assumptions · worked examples before independent practice · retrieval practice and spacing built into the structure, not appended as end-of-chapter questions · differentiated pathways for stronger and weaker learners · accessibility (type size, contrast, line length, dyslexia-considerate setting) · culturally grounded examples — a Nigerian maths textbook uses naira, Nigerian names and Nigerian contexts, and the fact that this needs saying is itself the market gap.

12.3 The digital-print pairing

Every educational title is designed to exist in both forms from the start: print for the classroom, digital for access, with the same content model behind both. Retrofitting a print book into a digital one produces a bad PDF, which is what most of this market currently has.

CHAPTER 13 — CREATIVE OPERATIONS

13.1 Roles

Creative Director (final quality authority) · Art Director (per project) · Designers · Typesetters · Editors · Copywriters · Translators · Illustrators · Motion designers · Photographers · Production manager (print, freight, customs) · Scholarly reviewers (retained per tradition/discipline) · Subject reviewers (retained per subject).

Scholarly and subject reviewers are **retained, named and accountable**. A pool of unnamed reviewers is how errors reach print.

13.2 Quality gates

GATE	WHO	BLOCKS
Concept approval	Creative Director	Direction
Content accuracy	Editor + fact-checker	Every claim
Scholarly review	Retained reviewer	Religious and doctrinal content
Subject review	Retained subject expert	Educational content
Design review	Art Director	Craft
Accessibility review	Designer + checklist	Type, contrast, structure
Rights clearance	Production	Every asset (Chapter 9)
Preflight	Production	Technical print readiness
Client approval	Client, in writing	Final
Press check	Production	Elite work

Table 127 — Quality gates

No gate is skipped for a deadline. A deadline met with an error in a printed run of 2,000 has not been met.

13.3 Capacity and honesty about it

Creative capacity is finite and cannot be surged without quality loss. Volume I §14.2 applies with full force: we do not accept creative work we cannot staff. The correct response to overcapacity is a later date or a referral to a partner, never a rushed job — because in this division, a rushed job is the thing the client shows other people.

VOLUME

VII

Industry Ecosystems

Sector solutions · The Enterprise Division

AUTHORITY

Strategic

EXECUTIVE SUMMARY

Volume VII sets out how an ecosystem is built and the five conditions that must all hold before one may open — of which a domain expert already hired is the first and enthusiasm is not one. It then treats each sector in turn across three waves: education and higher education, faith institutions, publishing, professional services and small business in the first; healthcare, government, financial services, development, hospitality and property in the second; and the large, slow industrial ecosystems in the third. Each chapter states the institution's real problem, its buyer, its regulatory surface, its ecosystem stack, its AI agents and its hard boundaries. The volume closes with the Enterprise Division and the retention economics that argue for moving up-market over time.

CHAPTER 0 — HOW AN ECOSYSTEM IS BUILT

0.1 The entry gate

An ecosystem may not open until **all five** are true:

152. A **domain expert is hired** — someone who has actually done the work inside that sector (Volume I §14.6).
153. The **previous wave's ecosystems are deep enough to leave** — ≥25 paying institutions, ≥110% NRR, a reference willing to take calls, a partner who can deliver without headquarters (Volume I §2.4).
154. The **regulatory surface is understood and documented**, not assumed.
155. A **chapter of this volume is drafted and ratified.**
156. **Delivery capacity exists** for the first ten customers without borrowing it from an existing ecosystem.

Enthusiasm is not one of the five.

0.2 The shared foundation

Every sector reuses layers 1–7 of the Volume I §2.3 stack. Only layer 8 — vocabulary, workflows, compliance, sector apps — is new. **This is what makes twenty sectors affordable for an engineering organisation sized for three,** and it is why the no-fork rule (Volume IV §3.2) is enforced as strictly as it is.

The per-sector build is therefore, in practice: a vocabulary map, a set of role definitions, a compliance profile, a handful of sector-specific record types, the sector's own workflows, its regulator connectors, and its AI agents. Typically 15–25% of a full platform, not 100%.

0.3 The chapter template

Each chapter states: the institution's real problem · the buyer · the regulatory surface · the ecosystem stack · the AI agents · the physical layer · what we do **not** do · the entry motion.

CHAPTER 1 — K-12 EDUCATION

Wave 1. The founding ecosystem. Reference Implementations №1–4.

1.1 The market

Nigeria's Federal Ministry of Education reports **107,017 private schools in the 2024/25 school year — roughly 50% of the country's 215,033 schools — enrolling 8,297,861 learners (21% of total enrolment), against 31,827,244 in public schools** (Federal Ministry of Education EMIS).

That is the single most important number in this corpus. Applying the Volume II §2.2 method to it — rather than treating it as a TAM headline — the reachable segment is the urban and peri-urban fee-charging private schools with the willingness and ability to pay a recurring subscription. That is a minority of the 107,017, but even a low-single-digit percentage of it, at the Volume III §20.1 configuration, is a substantial business, and it is reachable through referral density rather than paid acquisition.

The honest counterweight: a large share of Nigeria's private schools are very small, very price-sensitive, and genuinely fragile as recurring-revenue customers. Volume II R2 (collection failure) is not a theoretical risk in this segment; it is the base case for the bottom half of it. Qualification at Volume V §6.1 Stage 2 exists precisely to decline institutions that would be harmed by committing.

1.2 The institution's real problem

Not "we need a website". The actual problems, in the order proprietors raise them: parents cannot see anything without calling the school · fees are collected in cash and reconciled by hand · admissions consumes weeks of a registrar's life every year · results take days to compile and are transcribed by hand at least twice · certificates are photocopied and forged · staff records live in a cupboard · the proprietor cannot answer "how many students do we have, and how many have paid?" without a two-day exercise.

1.3 The buyer

Proprietor or Director signs (Volume II §3.1). Cares about prestige, parent perception and fee collection. Kills deals over price and over fear of disruption during term. The champion is usually the ICT coordinator or a young registrar. The blocker is usually the bursar whose manual process is being replaced — and Volume I §7.1′s free redeployment training is aimed directly at them.

1.4 The regulatory surface

State ministry of education registration and inspection · national curriculum compliance · examination board interfaces (WAEC, NECO, and Cambridge/Edexcel for international streams) · child protection and safeguarding obligations · the Nigeria Data Protection Act, with children's data at the strictest handling class (Volume IV §4.3) · fee regulation where it applies · health and safety inspection.

1.5 The ecosystem stack

LAYER	MODULES
8 Sector	Admissions · enrolment · classes & cohorts · timetable · attendance · gradebook · examinations · report cards · transcripts · curriculum & schemes of work · learning outcomes · competencies · evidence · discipline & pastoral · safeguarding · library · hostel · transport · catering · clinic · SEN · alumni

LAYER	MODULES
7 Intelligence	Enrolment forecasting · attrition risk · fee-default risk · comprehension-gap detection · timetable optimisation · inspection readiness
6 Automation	Admissions pack · fee pack · certificate pack · result-publication pack · attendance pack · safeguarding pack · board pack (Volume III §7.2)
5 Records	People · finance · assets · documents · governance decisions
4 Identity	Student/staff/parent identity · ID cards · biometrics · SSO · verifiable credentials
3 Experience	Multilingual website · student, parent, staff, registrar, bursary, executive, alumni portals · mobile apps · kiosks · print
2 Platform	Auth · payments · messaging (SMS/WhatsApp/push) · search · files · audit
1 Infra	Cloud, backup, security, observability

Table 128 — The ecosystem stack

1.6 The AI agents

Receptionist · Admissions Officer · Registrar · Bursar · Teacher (per subject) · Tutor (per student) · Homework Marker · Curriculum Designer · Examination Officer · Librarian · Timetable Optimiser · Parent Assistant · Student Companion · Inspection Readiness Assistant · Arabic Tutor · Qur'an Memorisation Coach.

Hard boundaries in this sector: final grades, admission decisions, disciplinary outcomes and safeguarding classifications are **High stakes** — AI informs, a named human decides and is recorded as the decider (Volume IV §8.1). No exceptions, and no institution may configure them away.

1.7 The physical layer

Smart Entry (gates, readers) · Smart Attendance (kiosks) · Smart Identity (card printer, capture station, encoder) · CCTV · PA and bell · signage · interactive classrooms · bus tracking · power resilience (inverter/UPS — non-optional in this market, and a school without it will blame us for downtime that is not ours).

1.8 What we do not do

We do not teach. We do not employ teachers on the institution's behalf. We do not assume the institution's regulatory obligations. We do not warrant examination outcomes. We do not sell an ERP to a 40-student school (Volume V §1.2.2).

1.9 Entry motion

Free tools → free readiness assessment → website + core platform + admissions → the Volume I §6.5 ladder. Referral density is the growth engine; the reference deployment is the sales asset.

CHAPTER 2 — HIGHER EDUCATION

Wave 1.

The real problem. Accreditation defensibility, records integrity across decades, credential fraud at scale, fragmented departmental systems, alumni relations that never got built, and research administration on spreadsheets.

The buyer. Registrar, Vice-Chancellor, Bursar. Committee segment — 3–9 month cycles, budget-bound. Kills deals over data-migration risk, which is the correct thing to fear.

Regulatory surface. National regulator (NUC in Nigeria; equivalents elsewhere) · programme accreditation · qualifications frameworks · research ethics · student data protection · international student compliance.

Additional stack over K-12. Faculty/department/school hierarchy · programme and level architecture · credit accumulation and transfer · modular course registration · academic bodies and external relations · senate/board governance registers · research management and outputs · CPD records · thesis and dissertation workflow · graduate register and graduate profiles · alumni chapters · corporate accounts and sponsored seats · scholarships and bursaries · convocation and graduation document production · institutional repository.

The group's reference credential schema already models programme definitions, levels, units, learning outcomes, competencies, evidence, awards, distinctions, credential signatures, issued documents, verifying institutions, graduate profiles and alumni chapters — this is the ecosystem those tables were built for.

AI agents. Registrar · Research Assistant · Librarian · Admissions Officer · Curriculum Designer · Examination Officer · Compliance Officer · Executive Analyst.

Why this ecosystem matters disproportionately. A university's credentials are checked by employers, foreign universities and immigration authorities worldwide. Every verification is a Volume V §1.1 marketing event, at the highest-credibility end of the market. One university is worth more distribution than fifty schools.

What we do not do. Set academic policy. Award qualifications. Adjudicate academic misconduct. Guarantee accreditation outcomes.

CHAPTER 3 — FAITH INSTITUTIONS

Wave 1. Mosques, churches, madrasahs, Islamic schools, seminaries, endowments.

The real problem. Community records nobody maintains, donations recorded on paper, programme scheduling by word of mouth, no way to show donors where money went, memorisation and study progress untracked, and endowment (waqf) assets undocumented.

The buyer. Imam, pastor, board of trustees, or the endowment’s administrator. Kills deals over anything culturally careless — this sector’s single largest vendor risk is not price, it is a supplier who gets the register wrong.

The stack. Community and membership register · programme and class scheduling · prayer/service times and displays · donations, zakat and sadaqah with **designation tracking** (a donor who gave for a well must be able to see the well) · waqf and endowment asset records · madrasah records · Qur’an memorisation and tajweed tracking · recitation recording and review · Arabic instruction · Islamic certificates · event management · volunteer coordination · burial and welfare records · multilingual site (Arabic-first where appropriate) · signage · PA.

AI agents. Receptionist · Librarian · Arabic Tutor · Qur’an Memorisation Coach · Translator · Secretary.

The hard boundary. Volume I §7.5 and Volume IV §8.3 apply at their strictest. **No StromeX system issues a religious ruling.** Qur’anic text is reproduced from verified sources, never generated. Questions of ruling route to a named qualified human, visibly. This is enforced in the policy gate and is not configurable.

Commercial note. This ecosystem has lower ACV and higher trust value than any other. It is served at or near cost in many cases, deliberately, and is a standing candidate for the Volume III §0.4 non-profit policy. Its return is reputational and referral-based, and that return is real.

CHAPTER 4 — PUBLISHING

Wave 1. Detailed practice in Volume VI.

The real problem. Publishers in our markets cannot access international production standards; rights and royalties are administered on spreadsheets; distribution is fragmented; and print economics are opaque.

The buyer. Publisher, Managing Director. Kills deals over quality of finish.

The stack. Title and catalogue management · manuscript and editorial workflow · rights and permissions register · contributor and contract management · royalty calculation and statements · ISBN and metadata management · production tracking · print procurement (Volume VI §10) · inventory and warehousing · distribution and retail feeds · digital storefront · subscription and serial management · author portal · reader analytics.

AI agents. Editor · Fact-Checker · Translator · Marketing Assistant · Customer Support.

Why it belongs in the group. Publishing is where the Creative Division's capability becomes a *platform business* rather than a services business — and educational publishing (Volume VI §12) feeds directly back into the education ecosystem as bespoke textbooks, which is a cross-sell with unusually high switching cost.

CHAPTER 5 — PROFESSIONAL SERVICES

Wave 1. Law firms, accounting firms, consultancies, architects, engineers, medical practices as businesses.

The real problem. Time and billing leakage, matter/engagement records scattered across email, document version chaos, client onboarding compliance, and no visibility of practice profitability.

The buyer. Managing Partner. Owner-led to committee depending on size.

The stack. Client and matter management · engagement letters · conflict checking · time recording and billing · trust/client account handling (regulated — jurisdiction-specific) · document management with version control · document assembly from templates · e-signature · deadline and limitation tracking · KYC/AML onboarding · knowledge base and precedent library · CPD records · practice analytics · client portal · secure client messaging.

AI agents. Legal Research Assistant · Contract Analyst · Secretary · Customer Support · Translator · Compliance Officer.

The hard boundary. AI does not give legal, accounting, tax or medical advice (Volume IV §8.1, Prohibited row). It supports a qualified professional who remains accountable. Marketing for this sector states this explicitly.

Regulatory surface. Bar/professional body rules on client confidentiality, client accounts, conflicts and advertising · AML obligations · professional indemnity implications of any system we supply.

CHAPTER 6 — SME & RETAIL

Wave 1. The self-serve volume ecosystem.

The real problem. Get found, get paid, get organised. In that order, and rarely anything more sophisticated.

The buyer. The owner. Self-serve. Decides in minutes. Churns instantly if it is complicated or if the monthly cost is not obviously worth it.

The stack. Website and online presence · online store · bookings and appointments · invoicing and payments · customer records (light CRM) · inventory · POS · staff and rota · basic accounting integration · WhatsApp Business integration (in our markets this is the actual customer channel, not a nice-to-have) · marketing tools · business email · domain.

AI agents. Receptionist · Customer Support · Marketing Assistant · Sales Development.

Strategic role. This ecosystem is not primarily a profit centre in Phase I. It is the **free-tier engine's natural home** and the volume top of the funnel: hundreds of thousands of micro-businesses using free tools, a small percentage converting, and — critically — a large number of those owners also being school proprietors, mosque trustees, clinic owners and NGO directors. The SME funnel feeds every other ecosystem.

CHAPTER 7 — HEALTHCARE

Wave 2. Requires a clinical domain hire before entry — non-negotiable.

The real problem. Paper records, unfindable patient histories, no continuity between visits, billing and insurance friction, drug stock-outs, and regulatory reporting done by hand.

The buyer. Medical Director, Chief Medical Director, Hospital Administrator. Kills deals over clinical risk and downtime — a system that is down during an emergency is not a software problem, and this sector's tolerance is correctly near zero.

The stack. Patient records · appointments and queue management · triage · clinical notes and encounters · orders and results (lab, imaging) · pharmacy and dispensing · inventory with expiry tracking · theatre scheduling · ward and bed management · billing and insurance claims · staff rota and credentialing · referrals · public health and regulatory reporting · patient portal · consent management.

AI agents. Patient Coordinator · Clinical Documentation Assistant · Receptionist · Bursar · Compliance Officer.

The hard boundary, stated as strongly as this corpus can. No StromeX system diagnoses, prescribes, triages autonomously, or makes a clinical decision. Clinical documentation assistance means transcribing and structuring what a clinician said — it does not mean suggesting a diagnosis. This is the Prohibited row of Volume IV §8.1 and it is enforced in the policy gate, not in a disclaimer.

Regulatory surface. National health regulator · professional council registration · patient data protection at the sensitive class · medical device regulation (which certain software features can inadvertently trigger — this must be assessed by counsel before any clinical-decision-adjacent feature ships) · insurance and NHIS-equivalent interfaces · retention obligations measured in decades.

Uptime. Elite-tier RPO/RTO minimum (Volume IV §16.2), offline-capable clinical surfaces, and power resilience specified as part of every deployment.

CHAPTER 8 — GOVERNMENT & PUBLIC SECTOR

Wave 2. The largest contracts, the slowest cycles, the highest integrity exposure.

The real problem. Ministries cannot see what their own institutions are doing, records are unauditable, service delivery is invisible to the citizen, and procurement produces systems nobody uses.

The buyer. Permanent Secretary, Director-General, Commissioner. Procurement segment: 6–24 months, formal tender. Kills deals over anything resembling scandal — which is why Volume V §5.1 is absolute and why published pricing, complete audit logs and published incident history are the actual competitive weapons here.

The stack. Citizen-facing service portals · case and workflow management · licensing and permits · registries (business, land, vehicle, vital records) · inspection and compliance management · grants and disbursement · public procurement · asset and facilities management · workforce management · document and records management with statutory retention · FOI request handling · multi-institution reporting and dashboards · inter-agency data exchange · public open-data publication.

Education ministry variant (the natural first entry, given Wave 1): multi-school oversight, per-school dashboards, enrolment and attendance aggregation, teacher deployment, examination results at state level, school inspection, capitation and grant disbursement, and state-wide credential verification.

AI agents. Case Officer · Compliance Officer · Executive Analyst · Translator · Customer Support (citizen-facing).

Requirements unique to this sector. In-country data residency, technically enforced (Volume III §4.6) · source-available or on-premise options where sovereignty is required · full audit defensibility · accessibility to the strictest standard, because a government service must serve everyone · local-language coverage including minority languages · offline and USSD access for citizens without smartphones · procurement-compliant documentation.

The integrity commitment. Volume I §6.6 and Volume V §5.1. No facilitation payments, ever. An engagement where a payment is requested is escalated, documented, and withdrawn from if the request stands. **We will lose contracts for this and we will report those losses as integrity wins.**

The value argument. Volume III §20.4: ≈$1.48 per student per year across a 400-school state deployment. Per-beneficiary cost is what wins a public tender, and ours is defensible under scrutiny because it is true.

CHAPTER 9 — FINANCIAL SERVICES

Wave 2. Microfinance, cooperatives, insurance brokers, fintechs, SACCOs, bureaux de change. ***Not*** *deposit-taking banks in Phase II.*

The real problem. Regulatory reporting consumes disproportionate capacity; loan and policy administration is manual; KYC is a paper exercise; and reconciliation is a permanent fire.

The buyer. COO or CTO. Kills deals over security review failure — this sector will audit us properly, which is good for the group.

The stack. Customer onboarding and KYC/AML · account and portfolio management · loan origination, servicing and collections · savings and cooperative contributions · insurance policy administration and claims · agent and branch networks · commission management · regulatory reporting · reconciliation · credit scoring inputs (as inputs to a human decision) · customer portal and mobile · fraud monitoring.

AI agents. Loan Officer Assistant · Underwriting Assistant · Customer Support · Compliance Officer · Executive Analyst.

Hard boundary. AI does not approve or decline credit, does not set premiums, and does not make an underwriting decision. It assembles, summarises and flags for a human who decides and is recorded. Beyond the Volume IV §8.1 ladder, this is also the position most financial regulators are converging on, and building it any other way would be building a compliance problem.

What we decline. Predatory lending, payday-style products, and any product whose economics depend on the borrower failing (Volume I §7.4).

CHAPTER 10 — NGO & DEVELOPMENT

Wave 2.

The real problem. Donor reporting consumes programme capacity; beneficiary data is scattered; impact cannot be evidenced; and multi-donor compliance is contradictory.

The buyer. Country Director, Programme Director. Cares about cost per beneficiary and donor audit survivability.

The stack. Beneficiary registry with consent and minimisation by design · programme and project management · results and indicator frameworks (logframe/theory of change) · monitoring, evaluation and learning · field data collection (offline-first, non-negotiable) · grant management and donor reporting · disbursement and cash-transfer tracking · procurement and logistics · volunteer management · safeguarding case management · impact dashboards · donor portal.

AI agents. Executive Analyst · Translator · Compliance Officer · Secretary.

The specific obligation. Beneficiary data belongs to vulnerable people who often cannot meaningfully consent and have no recourse. Data minimisation, short retention, no secondary use, and a genuine deletion path are not compliance items here — they are the ethical core of serving this sector. Volume IV §4.3 sensitive class applies to every beneficiary record by default.

CHAPTER 11 — HOSPITALITY

Wave 2. Hotels, restaurants, event venues, tourism.

Stack. Property management · reservations and channel management · front desk and housekeeping · food and beverage · POS · table and event booking · guest profiles and loyalty · staff rota · procurement and stock · maintenance · revenue management · guest messaging · review management · digital signage · smart room access (Volume III Division 11).

AI agents. Receptionist · Customer Support · Marketing Assistant · Bursar.

Note. Lower strategic priority than its revenue suggests — it is competitive, has weak switching costs, and does not feed other ecosystems. It is included because the SME ecosystem naturally produces these customers, not because we would enter it deliberately.

CHAPTER 12 — REAL ESTATE & CONSTRUCTION

Wave 2.

Stack. Property and unit registry · listings and marketing · tenancy and lease management · rent collection and arrears · facilities and maintenance requests · service charge accounting · project management · bill of quantities and cost control · site progress and photo record · subcontractor management · procurement and materials · plant and equipment tracking · health and safety records and incident reporting · document control and drawing registers · handover and snagging · client portal.

AI agents. Contract Analyst · Procurement Officer · Executive Analyst · Customer Support.

Physical layer. Site access control, plant tracking, environmental sensors, drone progress capture (Volume III Divisions 11–12).

CHAPTER 13 — WAVE 3 ECOSYSTEMS

Phase III (2036–2040). Specified at outline depth; each requires a full chapter and a domain hire before entry.

ECOSYSTEM	THE REAL PROBLEM	DISTINCTIVE STACK ELEMENTS	DISTINCTIVE AI
Manufacturing & Industry	No visibility of production, quality or downtime	Production planning · MES · quality management · maintenance · traceability · OEE · supply chain · warehouse	Maintenance predictor · Quality analyst · Procurement Officer
Agriculture & Agritech	Yields, inputs and offtake undocumented; finance inaccessible without records	Farm and plot registry · input distribution · extension services · yield and harvest records · cold chain · offtake and market linkage · farmer credit records · weather and advisory	Extension advisor · Yield analyst · Loan Officer Assistant
Logistics & Transport	Fleet, cargo and cost per trip invisible	Fleet · route planning · dispatch · consignment tracking · proof of delivery · fuel and maintenance · driver management · customs documentation	Route optimiser · Dispatch assistant · Compliance Officer
Media & Broadcast	Rights, archives and scheduling unmanaged	Content and rights management · scheduling · archive and MAM · production workflow · advertising sales · audience analytics · distribution	Editor · Fact-Checker · Archivist · Translator
Energy & Utilities	Metering, billing and outage response manual	Asset and network registry · metering · billing · outage management · field service · regulatory reporting · customer portal · prepaid/token vending	Demand forecaster · Outage analyst · Customer Support
Telecommunications	Subscriber, network and revenue systems siloed	Subscriber management · provisioning · billing and mediation · network inventory · field operations · dealer networks · regulatory reporting	Customer Support · Churn predictor · Executive Analyst

ECOSYSTEM	THE REAL PROBLEM	DISTINCTIVE STACK ELEMENTS	DISTINCTIVE AI
Research Institutions	Grants, outputs and integrity administration manual	Grant lifecycle · ethics approval · research data management · outputs repository · collaboration · equipment booking · researcher profiles · integrity records	Research Assistant · Fact-Checker · Compliance Officer
Smart Cities	Services, assets and citizens on unconnected systems	Integrated citizen services · asset registry · sensor networks · traffic · waste · public safety coordination · permits · civic engagement · open data	Case Officer · Anomaly detector · Executive Analyst

Table 129 — Wave 3 Ecosystems

Standing caution for Wave 3. Every ecosystem in this table is served today by large, entrenched, well-capitalised specialists. The group enters these only where the *emerging-market* variant is genuinely under-served, where the shared platform gives a real cost advantage, and where a credible domain hire is in place. Entering because a sector is large is precisely the over-extension failure mode Volume II R4 identifies.

CHAPTER 14 — THE ENTERPRISE DIVISION

14.1 What it is

A cross-sector division serving organisations whose complexity exceeds any single ecosystem: multi-site groups, conglomerates, multinationals, and organisations spanning several of the sectors above (a group that owns schools, a clinic and a farm is a real and common shape in our markets).

14.2 What it does differently

DIMENSION	STANDARD DELIVERY	ENTERPRISE DELIVERY
Scoping	Configurator	Discovery engagement, architecture design
Tenancy	Shared schema	Dedicated schema, database or deployment (Volume IV §3.1)
Contract	Standard terms	Negotiated MSA, bespoke SLA, security schedule
Delivery	Templated	Named team, programme management, phased
Support	Tiered	Named TAM, 24×7, on-site option (Volume III §17)
Integration	Connectors	Bespoke bridges to whatever they already run
Governance	Quarterly review	Steering committee, executive sponsor both sides
Compliance	Standard posture	Their audit, their pen test, their questionnaire

Table 130 — What it does differently

14.3 The enterprise disciplines

Never sign what we cannot deliver (Volume I §14.2) — applied hardest here, because an enterprise failure is a public failure. Delivery capacity is confirmed in writing by the person who will deliver it, before signature, without exception and without regard to the quarter.

Phase everything. No enterprise programme goes live as a single event. Sequence by risk: lowest-risk highest-visibility first, so the organisation sees value while the hard parts are still being built.

Multi-thread the relationship. Never fewer than five relationships in an enterprise account. Champion departure is the leading cause of enterprise churn (Volume V §7.3).

Own the integration burden. An enterprise runs systems we did not build and will not replace. Our willingness to integrate rather than demand replacement is a decisive advantage over competitors who require a rip-and-replace, and it is consistent with Volume II §5.1 Class 1: integrate, never fight.

14.4 The enterprise economics

Enterprise ACV supports a median NRR around 118% against ~97% for sub-$25k accounts (Growthspree, 2026). **This is the single strongest argument in the corpus for moving up-market over time** — not abandoning the founding market, but ensuring that by Phase II a meaningful share of revenue sits in the ACV band where retention economics actually compound.

Volume VIII, Chapter 5 builds this into the financial model explicitly: the growth and transformational scenarios depend on the enterprise and government mix rising, and the conservative scenario is what happens if it does not.

VOLUME

VIII

Expansion, Finance & the Roadmap

The three phases · The three scenarios · Capital · Geography · Acquisitions

AUTHORITY

Strategic — Board

EXECUTIVE SUMMARY

Volume VIII carries the canonical three-phase roadmap — Foundation, Race and Global Scale — from which every date in the corpus derives, together with the gates that a phase must pass before the next may open. It builds

the unit economics from the ground up, states plainly where the founding market's favourable ratios will compress and why, and models the group's future as three scenarios rather than one target: a conservative path that is a good outcome and not a failure, a growth path the company should plan against, and a transformational path that alone reaches the stated ambition. It sets the capital strategy and the constitutional condition on capital, the allocation framework, the international expansion sequence and its entry checklist, the acquisition doctrine, and the conditions that would make the plan wrong.

CHAPTER 1 — THE 20-YEAR MASTER ROADMAP

Ratified as the canonical phase structure of the group. Every other volume's dates derive from this chapter.

1.1 The three phases

PHASE	YEARS	NAME	GOVERNING QUESTION
I	2027–2030	**Foundation**	*"Will this decision make StromeX stronger in 2031?"*
II	2031–2035	**Race**	*"How do we scale without compromising quality, trust or customer satisfaction?"*
III	2036–2040	**Global Scale**	*"How do we become durable enough to outlive our founders?"*

Table 131 — The three phases

Beyond 2040 the corpus plans in decades rather than phases (Volume IX, Chapter 9 — the 100-Year Plan).

1.2 PHASE I — FOUNDATION (2027–2030)

> **Mission: build quietly. Perfect the products. Build trust instead of chasing growth.**

This phase prepares for scale. It does not maximise revenue, and it is not judged on revenue.

Objectives

Engineering & product Build world-class engineering capability · take every core product to genuine production quality (Volume IV §18.2 definition of done, applied without exception) · build the StromeX Design System to open-source quality · build the AI infrastructure with provider independence proven, not asserted · build StromeX Cloud to the point where we run our own customers on it · build internal automation tools so the group's own operations are the first proof of the automation thesis.

Institutional Establish standard operating procedures across every function (Volume IX, Chapter 6) · build the knowledge base and this Editorial Bible into a genuine operating manual · obtain legal registrations in Nigeria and secure trademarks in the markets of Phase II · secure intellectual property where it is worth securing and open-source the rest deliberately (Volume I §5.5) · put governance in place before it is needed, not after.

Market Develop partnerships with schools and institutions · pilot every product with real users, on real data, with real consequences · build case studies from real deployments with written permission · recruit a high-quality core team · create documentation, APIs, SDKs and developer resources to a standard a stranger can build against · refine pricing through actual customer feedback rather than through theory · invest heavily and consistently in SEO and educational content, because it compounds and because paid acquisition does not work in this market (Volume V §1.1).

The financial posture

Phase I optimises for **product quality, customer satisfaction, reputation and recurring revenue mix** — explicitly **not** for rapid profitability and explicitly not for headline growth.

A smaller number of delighted customers is worth more than a larger number of dissatisfied ones, and in a referral-driven market this is not a platitude — it is the growth mechanism. Ten institutions that recommend us are worth more than a hundred that do not.

What Phase I deliberately does not do

Enter a new country · enter Wave 2 sectors · open enterprise or government sales motions · raise institutional capital before the unit economics are demonstrated · hire ahead of demonstrated demand · pre-announce anything · take a contract to make a quarter.

1.3 PHASE II — RACE (2031–2035)

> **The company expands aggressively, having earned the right to.**

Objectives

Nigeria-wide leadership in education technology · expansion into fintech and business automation · build enterprise sales capability · pursue government partnerships now that the reference base makes them winnable (Volume V §3) · scale cloud infrastructure services · large-scale digital transformation engagements · open international offices (Chapter 9) · marketing at scale · begin strategic acquisitions (Chapter 11) · establish a formal, funded R&D division (Volume IX, Chapters 2–3).

The major product launches

PRODUCT	WHAT IT IS
StromeX Education OS	The full Wave 1 education ecosystem as a single coherent platform
StromeX Finance OS	Financial services ecosystem (Volume VII §9) plus StromeX Pay at scale
StromeX Government Suite	Public-sector ecosystem with sovereignty, residency and audit posture
StromeX Business Cloud	The SME and enterprise cross-sector platform
StromeX AI	The agent family as an independent product line
StromeX Marketplace	The third-party ecosystem at scale
StromeX Payments	Payments as infrastructure, not a module
StromeX Identity	Identity and credentials as a standalone, sellable platform
StromeX IoT	Smart campus, smart facility, sensor and access infrastructure
StromeX Analytics	Intelligence as a product
SpaceTalk	The communication operating system (Volume X)

Table 132 — PHASE II — RACE (2031–2035)

The Phase II discipline

Growth in this phase is the objective, and the constitution is the constraint. The specific failure mode to guard against is **delivery quality collapse under growth** (Volume II R5) — which is why partner certification, templated delivery and the written capacity confirmation before signature all become load-bearing in exactly this phase rather than earlier.

1.4 PHASE III — GLOBAL SCALE (2036–2040)

> **Become a globally recognised technology company.**

Geography

United States · United Kingdom · Europe · Middle East · Asia · Latin America, sequenced per Chapter 9.

Priorities

AI research at a level that produces original contribution, not only application · robotics integration where it creates real customer value · enterprise cloud at scale · digital identity as global infrastructure · smart cities · healthcare technology · financial infrastructure · educational infrastructure.

The Phase III question

The governing question shifts again in this phase, and the shift matters: from *how do we grow* to **how do we become durable enough to outlive our founders.** A company that reaches global scale and remains dependent on one person has not succeeded; it has deferred a failure. Volume II R3 (key-person dependency) is a Phase I risk that becomes a Phase III existential one if it was never actually addressed.

CHAPTER 2 — THE PHASE GATES

A phase does not end because its years elapsed. It ends when its gate is passed. Running Phase I into 2032 because the gate was not met is a correct decision; entering Phase II in 2031 with the gate unmet is how companies destroy themselves.

Gate I → II (assessed from 2030)

#	CRITERION	THRESHOLD
1	Paying institutions	≥ 250
2	Net revenue retention	≥ 105% (Volume V §7.1)
3	Gross revenue retention	≥ 92%
4	Recurring share of revenue	≥ 55%
5	Blended gross margin	≥ 60%
6	CAC payback	≤ 15 months
7	Referral share of new customers	≥ 35%
8	Certified delivery partners	≥ 15
9	Reference customers willing to take calls	≥ 10
10	Executive layer capable of operating without the founder for 60 days	Demonstrated, not asserted
11	Security: independent penetration test passed, no unresolved high findings	Yes
12	Free-tier monthly active institutions	≥ 5,000

Table 133 — Gate I → II (assessed from 2030)

Gate 10 is the one most likely to be quietly skipped and the one that matters most. It is assessed by the founder actually being absent for 60 days.

Gate II → III (assessed from 2035)

Revenue ≥ the Chapter 5 growth-scenario floor · recurring share ≥ 65% (Volume I §6.2 portfolio rule) · NRR ≥ 115% · operations in ≥ 3 countries with local leadership · ≥ 2 ecosystems beyond education at Volume I §2.4 depth · partner-delivered revenue ≥ 30% · a functioning board with independent directors · audited accounts · Rule of 40 ≥ 30.

CHAPTER 3 — UNIT ECONOMICS

The foundation of every number in this volume. Assumptions are stated so they can be attacked.

3.1 The reference customer — a Band A Professional-tier school

Derived from the Volume III §20.1–20.2 worked configurations.

METRIC	VALUE	ASSUMPTION
Year-1 revenue	$3,435	Volume III §20.1, small school
Recurring revenue (year 2+)	$1,413/yr	Monthly total × 12
Expansion (module ladder)	+18%/yr, years 2–5	Volume I §6.5 ladder at historical attach rates
Blended gross margin	62%	Software 82%, services 48%, weighted by mix
CAC (low-touch, referral-heavy)	$290	Loaded sales cost ÷ customers; assumes ≥35% referral share
CAC payback	~4.9 months	On year-1 gross profit
Gross annual churn	9%	Band A, sub-$25k ACV; deliberately pessimistic vs. our 92% GRR target
Expected lifetime	~8 years	1 ÷ (churn − expansion offset), capped conservatively
LTV (gross profit)	**≈ $11,400**	Discounted at 15%
LTV : CAC	**≈ 39 : 1**	

Table 134 — The reference customer — a Band A Professional-tier school

That ratio looks implausible and requires explanation, because a number that good is usually a modelling error. It is high for three real reasons and one caveat:

157. CAC is genuinely low because acquisition is referral- and free-tier-driven rather than paid (Volume V §1.1).
158. Delivery labour is local and Band A pricing is set against local cost, so margin survives the discount.
159. The module ladder produces expansion revenue that partially offsets churn.

The caveat that matters: this ratio is only achievable *at low volume, in the founding market, with the founder's network doing much of the referral work.* It will compress substantially as the group scales beyond the network, enters Band B and C markets where CAC is far higher, and takes on the delivery cost of institutions that are not near an office. **Chapter 5′s scenarios assume LTV:CAC compresses to roughly 8:1 by 2035 and 4–5:1 by 2040**, which is still healthy and is a far safer planning basis than the Phase I figure.

3.2 The other reference customers

CUSTOMER	YEAR-1	RECURRING	GM	CAC	PAYBACK
SME (self-serve, Band A)	$180	$148/yr	84%	~$12	<2 mo
Large school group (Band A)	$36,263	$8,331/yr	58%	$2,400	~5 mo
University (Band A)	~$115,000	~$34,000/yr	55%	$14,000	~11 mo
Government (state, Band A)	~$266,000	~$150,000/yr	52%	$48,000	~15 mo
Band C school (UK/US)	$14,600	$6,100/yr	66%	$4,200	~14 mo

Table 135 — The other reference customers

The pattern to notice: CAC payback lengthens dramatically up-market and in Band C, while NRR improves dramatically (Volume VII §14.4). This tension — cheap-to-acquire customers that retain poorly, expensive-to-acquire customers that retain well — is the central financial dynamic of the whole plan, and Chapter 5′s scenarios differ mostly in how successfully the group navigates it.

CHAPTER 4 — THE REVENUE MODEL

4.1 The engine mix over time

Volume I §6.2's portfolio rule requires recurring, high-margin engines (subscription, consumption, cloud, licensing) to exceed **65% of group revenue by 2031**.

ENGINE	2027	2030	2035	2040
Subscription software	22%	38%	46%	50%
Consumption (AI, cloud, verification)	3%	12%	17%	20%
Cloud infrastructure	2%	6%	8%	9%
Licensing & white-label	0%	2%	5%	7%
Recurring subtotal	**27%**	**58%**	**76%**	**86%**
Implementation	24%	15%	8%	4%
Professional services	18%	10%	5%	3%
Creative services	16%	8%	4%	2%
Publishing & print	9%	4%	2%	1%
Hardware	6%	4%	3%	2%
Marketplace & financial services	0%	1%	2%	2%

Table 136 — The engine mix over time

Assumption: services and creative revenue keep growing in absolute terms throughout — they simply grow more slowly than software, so their share falls. ***Services are never deliberately shrunk****; they win deals nobody else can win and they create the relationship (Volume I §6.2).*

4.2 Gross margin trajectory

	2027	2030	2035	2040
Blended gross margin	52%	62%	71%	76%

Table 137 — Gross margin trajectory

Driven by mix shift, not by price increases. Assumes cloud gross margin improves as we move from rented to owned-and-optimised infrastructure, and that the AI inference cost per task continues to fall — the latter is an assumption about the industry, not about us, and is the single most exposed line in this table.

CHAPTER 5 — THE THREE SCENARIOS

The founder's stated target is a group valued at $15–40B within 10–20 years. This chapter takes that seriously by modelling what would actually have to be true — and by stating plainly which scenario the group should plan against.

> **The primary objective of StromeX is not a valuation.** It is: build products customers genuinely value, achieve world-class quality, create a trusted brand, generate sustainable cash flow, expand methodically, invest consistently in R&D, and reinvest profits wisely. Those are the things the company can directly influence. A very large valuation is a *possible consequence* of doing them exceptionally well over two decades — it is not the mission, and it must never become the operating target, because optimising directly for valuation reliably produces the behaviours the Volume I constitution prohibits.

5.1 Shared assumptions across all three scenarios

- Phase structure per Chapter 1; gates per Chapter 2.
- Nigeria remains the founding market; no country entry before Phase II.
- Recurring mix reaches ≥65% by 2031 (Volume I §6.2).
- The free tier is maintained at its constitutional commitments throughout.
- No revenue from data sale, advertising, or any prohibited practice.
- Figures are **group revenue in nominal USD**, not ARR alone (StromeX is not a pure-SaaS business, and reporting only ARR would flatter it).
- 2027 is Phase I year one, with revenue in the low single-digit millions at most.

5.2 The scenarios

SCENARIO A — CONSERVATIVE: "Sustainable Regional Leader"

The most probable good outcome. This is not failure.

YEAR	REVENUE	INSTITUTIONS	COUNTRIES	HEADCOUNT
2030	$6M	320	1	60
2035	$32M	1,900	3	260
2040	$90M	5,500	5	640
2046	$170M	9,000	6	1,000

Table 138 — The scenarios

Assumptions: NRR stabilises at ~105% · Nigeria and West Africa only until late Phase II · one modest developed-market entry (UK) that takes longer and costs more than planned · government contracts won but slowly · recurring mix reaches 70% · gross margin 68% by 2040 · profitable from ~2032 · minimal external capital. **Implied CAGR 2030→2040: ~31%. Indicative valuation at 4–5× revenue: $680M–$850M (2046).**

Why this is a good outcome. A profitable, independent, $170M-revenue company employing a thousand people, serving nine thousand institutions across six countries, with a genuinely trusted brand and no external control, is a serious achievement and a fine life's work. Any plan that treats this as failure will make bad decisions to avoid it.

SCENARIO B — GROWTH: "Multinational Technology Company"

The scenario the group should plan and staff against.

YEAR	REVENUE	INSTITUTIONS	COUNTRIES	HEADCOUNT
2030	$14M	700	1	120
2035	$160M	6,500	5	900
2040	$700M	22,000	9	3,200
2046	$1.8B	45,000	14	7,000

Table 139 — The scenarios

Assumptions: NRR reaches 115% by 2033 and holds · successful UK and US entry in Phase II · at least three ecosystems beyond education at depth · government revenue reaches 20% of the mix · partner-delivered revenue exceeds 35% · marketplace and licensing become material · recurring mix 82% · gross margin 74% by 2040 · Rule of 40 sustained above 40 from 2034 · two to three growth financing rounds plus selective acquisitions. **Implied CAGR 2030→2040: ~48%** — demanding, sustained for a decade, and achieved by real companies but not by most. **Indicative valuation at 6–8× revenue: $10.8B–$14.4B (2046).**

SCENARIO C — TRANSFORMATIONAL: "Global Platform Company"

The only scenario in which the $15–40B target is reached. Possible; not probable.

YEAR	REVENUE	INSTITUTIONS	COUNTRIES	HEADCOUNT
2030	$25M	1,200	1	200
2035	$400M	14,000	7	2,200
2040	$2.2B	60,000	14	9,000
2046	$5.5B	130,000	20+	20,000

Table 140 — The scenarios

Assumptions — all of which must hold simultaneously: NRR ≥ 125% · StromeX Identity and credential verification become *de facto* infrastructure across multiple national education systems · SpaceTalk (Volume X) achieves genuine network effects and becomes a platform rather than a product · the marketplace becomes a significant independent economy · government becomes the largest single revenue line through multi-country national deployments · at least two developed markets reach material scale · several successful acquisitions integrate cleanly · recurring mix 86% · gross margin 78% · Rule of 40 above 50 · substantial growth capital available at each stage without loss of constitutional control. **Implied CAGR 2030→2040: ~57%,** sustained for a decade. **Indicative valuation at 6–8× revenue: $33B–$44B (2046).**

5.3 The valuation arithmetic, stated plainly

Working backwards from the target rather than forwards from optimism:

TARGET VALUATION	AT 5× REVENUE	AT 6×	AT 8×	AT 10×
$15B	$3.0B revenue	$2.5B	$1.88B	$1.5B
$40B	$8.0B revenue	$6.7B	$5.0B	$4.0B

Table 141 — The valuation arithmetic, stated plainly

Therefore: the $15–40B ambition requires group revenue somewhere between roughly $1.9B and $8B, depending entirely on the multiple the group earns. Scenario B reaches $1.8B by 2046 — which touches $15B *only* at a premium 8×+ multiple. Scenario C reaches the band comfortably. Scenario A does not approach it.

5.4 What determines the multiple

Public SaaS median EV/Revenue sat at roughly **8.5× NTM revenue as of mid-2026**, having recovered from a ~4.5× trough in Q1 2023 but remaining well below the ~16× 2021 peak; other measures of the same market put the median EV/TTM materially lower, around 3.3×, and private SaaS medians around **4–5× ARR**. Companies with **NRR above 120% and a Rule of 40 above 50 reach 7–9×** (ValueAdd VC, Aventis Advisors, L40). The median Rule of 40 score across actively traded SaaS companies is only **28%, with just 20% exceeding 40** (Eqvista).

Three consequences the group must internalise:

160. **Multiples are volatile and outside our control.** The same company was worth 16× in 2021 and 4.5× in 2023. Any plan whose success depends on a specific multiple is a plan that depends on the weather. **We control revenue, margin, retention and growth. We do not control the multiple, and we will not manage the company as though we do.**
161. **A mixed-margin group is valued below a pure-SaaS comparable.** Services, print and hardware drag the blended multiple. This is a real and permanent cost of the completeness strategy (Volume II §6, Wedge 2) — and it is a cost worth paying, because completeness is what wins the customers in the first place. But it must be stated rather than wished away.
162. **The premium band is earned by two specific numbers: NRR and Rule of 40.** Everything in Volume V Chapter 7 exists to move the first. Everything in Volume I §6.2′s portfolio rule exists to make the second possible.

5.5 The standing instruction on this chapter

Anyone in this company who cites the $15–40B figure without the qualification in §5.3 is misrepresenting the company. Anyone who dismisses the ambition because it is improbable is misunderstanding the job. **We plan against Scenario B, prepare for Scenario A, and build the company so that Scenario C remains structurally possible.**

CHAPTER 6 — VALUATION LOGIC, HONESTLY STATED

Beyond the multiple, the group's value in any transaction rests on assets that are not on a balance sheet:

ASSET	WHY A BUYER OR INVESTOR VALUES IT
The system of record	Ten-year institutional data that cannot be replicated by capital (Volume II §9)
Credential infrastructure	Every credential we have ever issued must remain verifiable; whoever owns that owns a permanent utility
The free-tier install base	Distribution that cost no acquisition spend and cannot be bought
Partner and marketplace ecosystem	Third parties whose businesses depend on the platform
Trust	The only asset that takes a decade to build and a day to lose, and the one that makes government and enterprise revenue possible at all
Emerging-market-native architecture	Genuinely hard for an incumbent to retrofit (Volume II §6, Wedge 7)

Table 142 — Valuation Logic, Honestly Stated

And the liabilities an honest assessment must state: concentration in volatile currencies · dependence on a founder through Phase I · a services and hardware mix that dilutes the multiple · exposure to foundation-model providers whose pricing and terms we do not control · and a customer base with structurally weaker retention economics than a developed-market equivalent.

CHAPTER 7 — CAPITAL STRATEGY

7.1 The sequence

STAGE	WHEN	SOURCE	USE	CONSTITUTIONAL CONDITION
Bootstrap	2027–2029	Services revenue funding product	Build the platform; prove unit economics	None
Seed / strategic angel	2029–2030	Angels with sector or market knowledge	Reach Gate I; hire the executive layer	Terms must not touch Volume I Ch. 5 or 7
Series A	2031	Institutional VC, Africa or global	Phase II launch; enterprise motion; first country entry	Board seat acceptable; constitutional veto not
Series B/C	2032–2035	Growth equity	International expansion; acquisitions; R&D division	—
Debt	2033+	Working-capital and hardware-inventory facilities	Fund receivables and hardware without dilution	Never to fund losses
Pre-IPO / late growth	2037–2039	Crossover funds, strategic investors	Scale, acquisitions	—
Public listing or independence	2040+	Decision, not assumption	—	Chapter 7.3

Table 143 — The sequence

7.2 The bootstrap doctrine of Phase I

Phase I is funded primarily by services revenue. This is deliberate and it has a cost: it is slower. It buys three things worth the delay — the unit economics get proven with real money before anyone bets on them; the constitution is not negotiated away at a moment of weakness; and the founder retains the ability to run Phase I on Phase I's terms rather than on an investor's growth expectations.

Do not raise institutional capital before Gate I. Capital raised against an unproven model buys pressure to grow before the product deserves it, which is precisely the failure mode Phase I exists to avoid.

7.3 The constitutional condition on capital

If a term sheet requires amending Volume I Chapter 5 (the free constitution) or Chapter 7 (ethics), we decline the term sheet. (Volume I §14.9.)

This will cost the group money at least once. It is the single most important sentence in this chapter, because a company that will trade its constitution for capital does not have a constitution — it has a marketing document.

Related conditions: no investor gets a veto over the entrenched provisions · no investor may compel entry into a declined sector (Volume I §7.4) · no investor may compel the sale or monetisation of customer data.

7.4 On listing

An IPO is a financing decision, not an achievement. It is appropriate only if the group needs public capital, can sustain quarterly scrutiny without degrading the twenty-year posture, and can protect the entrenched provisions through the listing structure. **Remaining private and profitable is a legitimate permanent outcome** and is the base case in Scenario A.

CHAPTER 8 — CAPITAL ALLOCATION & INVESTMENT FRAMEWORK

8.1 Allocation of free cash flow, by phase

USE	PHASE I	PHASE II	PHASE III
Research & development	35%	25%	20%
Infrastructure & platform	15%	15%	15%
Hiring ahead of demonstrated demand	10%	15%	12%
Sales & marketing	12%	22%	18%
International expansion	0%	12%	15%
Strategic acquisitions	0%	5%	12%
Emergency reserve	25%	15%	12%
Distributions	3%	5%	8%

Table 144 — Allocation of free cash flow, by phase

8.2 The emergency reserve

Never below six months of fully-loaded operating expense, in hard currency, in a jurisdiction outside the primary operating market.

This is not conservatism for its own sake. It is a direct response to Volume II R1 (FX volatility), R2 (collection failure) and R9 (political shock). A company operating in naira-denominated markets with dollar-denominated costs and a customer base whose ability to pay is genuinely fragile needs a reserve that survives a bad year — and the years in which a reserve is needed are precisely the years in which raising capital is hardest.

The reserve may be drawn only on Board decision, and drawing it triggers a mandatory plan to restore it.

8.3 The investment test

Every material investment answers, in writing, the same five questions the whole corpus is held to:

163. **Why is this included?** What does the group get that it does not have?
164. **What measurable value does it create?** In what units, by when?
165. **What risks does it introduce?** Including the risk of distraction.
166. **What assumptions underpin it?** Which one, if wrong, kills it?
167. **How will success be measured?** And who is accountable for measuring it?

An investment case that cannot answer all five is not declined for being unattractive — it is declined for being unfinished.

8.4 The R&D commitment

R&D is funded as a **fixed percentage of revenue, not as a residual after other budgets are met.** A residual R&D budget is the first thing cut under pressure and the last thing recovered, and a company that stops investing in R&D during a hard year has traded its next decade for its current quarter. The percentage is set in §8.1 and may be reduced only by Board decision.

CHAPTER 9 — INTERNATIONAL EXPANSION

9.1 The sequence and its logic

WAVE	MARKETS	PHASE	WHY HERE, WHY NOW
Home	Nigeria	I	The founding market; proof; referral density; 107,017 private schools
1	Ghana, Kenya	II (2031–32)	Anglophone, similar education structures, credential fraud is a live issue, low regulatory delta
2	UK	II (2032–33)	Diaspora bridge; Band C revenue; credibility that reflects back into African markets; **and geographic risk diversification** (Volume II R9)
3	UAE, Saudi Arabia	II (2033–34)	Arabic-first capability is already built and is a genuine differentiator; large education and government spend; Islamic publishing and institutions are a natural fit
4	South Africa, Egypt, Tanzania, Rwanda	II–III (2034–36)	Regional depth; Rwanda and Egypt as government-partnership entries
5	United States	III (2036–37)	Largest market, hardest to win; entered only with scale, references and capital already in place
6	Canada, EU (France, Germany)	III (2037–38)	French capability already built; EU data-residency posture already required by Phase II
7	Pakistan, Indonesia, Malaysia, Bangladesh	III (2038–39)	Large Muslim-majority education markets where the Islamic-education capability is decisive
8	Latin America (Brazil, Mexico), India	III (2039–40)	Large, competitive, entered last and only through partnership or acquisition

Table 145 — The sequence and its logic

9.2 The three entry modes

MODE	USE WHEN	COST	SPEED
Partner-first	The market is reachable through an existing distribution network	Lowest	Fastest
Acquisition	A credible local player exists and integration is realistic	Highest	Fast
Direct entry	The market is strategically essential and no partner or target is adequate	High	Slowest

Table 146 — The three entry modes

Default to partner-first. Direct entry into a market where we have no relationships, no local leadership and no reference customers is the most reliable way to spend two years learning what a partner would have told us in a month.

9.3 The localisation obligation

A market is not entered by translating the website. Entry requires: language (properly — Volume IV §11), currency and local payment rails, local legal entity and tax registration, data residency where required, local employment compliance, curriculum or sector-standard alignment, local support hours, local pricing band (Volume I §6.4), and **local leadership with genuine authority**. A country run remotely by headquarters is a country we will lose.

CHAPTER 10 — THE MARKET-ENTRY CHECKLIST

Every item must be complete and signed off before the first customer contract in a new country. Executive and Board approval per Volume I §8.3.

Legal & regulatory ☐ Entity incorporated ☐ Tax registration ☐ Employment compliance reviewed ☐ Data protection regime mapped and implemented ☐ Sector regulator requirements documented ☐ Trademark filed ☐ Contracts localised by local counsel ☐ Sanctions and export-control screening ☐ Anti-corruption exposure assessed (Volume I §6.6)

Commercial ☐ Market sized by the Volume II §2.2 method with dated inputs ☐ Price band assigned ☐ Competitors mapped ☐ At least three reference prospects identified ☐ Partner or acquisition target assessed ☐ Payment rails live ☐ Local currency handling ☐ Collection risk assessed

Product ☐ Language complete ☐ Curriculum/sector standards mapped ☐ Local integrations built (payments, examination boards, regulators) ☐ Data residency in place if required ☐ Accessibility to local standard ☐ Offline/low-bandwidth verified on local networks

Operational ☐ Local leader hired, with authority ☐ Support hours covered ☐ Delivery capacity confirmed in writing ☐ Escalation path defined ☐ Local banking ☐ Insurance ☐ Office or remote model decided

Financial ☐ Entry budget approved ☐ Break-even modelled with stated assumptions ☐ FX exposure hedged or accepted explicitly ☐ **Exit criteria defined** — what would make us withdraw, and by when

That last item is the one most often omitted and the most valuable. A market entry with no pre-agreed failure condition will be defended indefinitely by the people who championed it.

CHAPTER 11 — ACQUISITION STRATEGY

11.1 The doctrine

Acquire capability, market access, or a team — never size. An acquisition that adds revenue without adding one of those three adds integration risk, cultural dilution and management distraction in exchange for a number.

Acquisitions begin in Phase II (§8.1 allocates no acquisition capital in Phase I) and become material in Phase III.

11.2 The target categories

CATEGORY	WHAT WE WOULD BUY	WHY
Design & creative studios	Craft capability, art direction, a book of institutional clients	Volume VI capacity is the hardest to hire and the slowest to build
AI teams	Applied research capability, agent tooling	Capability, not models
Fintech / payments	Licences, rails, compliance infrastructure	Licences take years; buying them is often the only realistic path
EdTech (regional)	Installed base, local relationships, market entry	The fastest legitimate route into a new country
Publishing houses	Catalogue, rights, editorial capability, distribution	Volume VI §7 at scale
Cybersecurity	Capability and credibility for government and enterprise	Buys the posture Volume VII §8 requires
Cloud / infrastructure	Capacity, regional presence, residency	Improves cloud gross margin and sovereignty options
Analytics / data	Capability	Volume III Division 9
Local vendors (Class 3)	Customers and continuity	Often the graceful outcome for a competitor that is failing (Volume II §5.1)

Table 147 — The target categories

11.3 The acquisition test

Every target must pass **all seven**:

168. **Strategic fit** — it advances a named objective in this corpus, not merely an adjacent opportunity.
169. **Cultural fit** — would we hire these people? Do they share the Volume I §7 ethics? *A team that would not pass our ethics review is not acquirable at any price.*
170. **Integration plan exists** — written before the offer, naming who integrates it and by when.
171. **The team stays** — key people committed, or the acquisition is of assets and priced as such.
172. **Clean diligence** — legal, financial, technical, security, IP ownership, customer contracts, and data-handling practices. **A target that has been selling or mishandling customer data is not acquirable**, regardless of its numbers.
173. **We can afford to be wrong** — no single acquisition may risk the group.

174. **The alternative was considered** — build, partner or buy, compared explicitly. Buying because it is faster is a valid answer only when the comparison was actually made.

11.4 Post-acquisition

Constitution applies from day one — the acquired team is briefed on Volume I before anything else · customers are told what changes and what does not, in the first week · **existing customer prices are honoured for their term** (Volume I §6.3.6 applies to inherited contracts) · products are integrated onto the platform or explicitly sunset with a stated date and a migration path, never left to drift · **and if the acquisition competes with an existing partner, we say so immediately and offer a transition** (Volume V §9.5).

11.5 The standing caution

Most acquisitions in most companies destroy value. The group's default answer is **build or partner**. Acquisition is the exception, requires an explicit case against the alternatives, and is reviewed 24 months later against the case that justified it — with the review published internally whether it flatters the decision or not.

CHAPTER 12 — FINANCIAL CONTROLS & REPORTING

Controls. Segregation of duties · documented approval thresholds and a delegation-of-authority matrix · dual authorisation above threshold · monthly bank and revenue reconciliation · fixed-asset register · **no cash payments to officials, ever** (Volume I §6.6) · expense policy with published limits · related-party transactions disclosed to the Board · annual external audit from the point the group exceeds $10M revenue · an internal audit function independent of finance above $25M (Volume I §8.6).

Reporting cadence. Weekly: cash, collections, pipeline. Monthly: divisional P&L against margin band, revenue by engine, headcount, ARR/NRR/GRR, CAC payback, Rule of 40. Quarterly: full management accounts, Pricing Council report and changelog, risk register review, ethics docket. Annually: audited accounts, strategy review, corpus amendment cycle.

The reporting honesty rule. Bookings are always reported alongside 90-day adoption, in the same table (Volume V §12.5). A board pack that shows bookings without adoption is not a board pack, it is a sales presentation.

CHAPTER 13 — WHAT WOULD MAKE THIS PLAN WRONG

A plan that cannot state its own falsification conditions is not a plan. Each of these is monitored, with a named owner, and any one of them triggering requires the plan to be revised rather than defended.

#	CONDITION	WHAT IT WOULD MEAN	DETECTED BY
1	**NRR stays below 105% through Phase I**	The expansion ladder does not work; Scenarios B and C are unavailable; the group is a Scenario A business and should be run as one	Volume V §12.3, monthly
2	**CAC payback exceeds 18 months in the home market**	Referral density is weaker than assumed; growth requires capital we planned not to need	Monthly cohort analysis
3	**Gross retention falls below 88%**	Institutions cannot sustain the recurring commitment; the Band A pricing model needs restructuring, not defending	Monthly
4	**Free-tier cost per converted customer exceeds CAC of paid channels**	The free engine is a cost centre rather than a distribution strategy	Quarterly free-tier unit economics
5	**AI inference cost per task stops falling**	The Chapter 4.2 margin trajectory breaks; consumption pricing must be restructured	Quarterly
6	**A well-funded competitor replicates the transparency and completeness wedges**	The Volume II §6 advantage compresses faster than the durable moats can form	Competitive review, quarterly
7	**Delivery quality degrades as volume grows**	Phase II must be slowed; this is the failure that ends companies like ours	Adoption rates, health scores, go-lives requiring heroics
8	**The founder remains structurally indispensable at Gate I**	Phase II cannot open. Gate 10 exists precisely for this	Gate I assessment, by absence
9	**FX movement makes Band A revenue unviable in USD terms**	Pricing bands, not the strategy, must change — but the model's dollar figures become unreliable	Monthly

#	CONDITION	WHAT IT WOULD MEAN	DETECTED BY
10	**A security incident involving children's data**	Existential for trust. Volume IV Chapters 12–17 exist to prevent it, and Volume I §4.3 governs the response	Continuous

Table 148 — What Would Make This Plan Wrong

The standing instruction: when one of these triggers, the correct response is to change the plan. The incorrect response — and the common one — is to change the measurement.

VOLUME

IX

The Institution

Innovation · Research · Talent · Operations · The 100-Year Plan

AUTHORITY

Constitutional & Strategic

EXECUTIVE SUMMARY

Volume IX defines StromeX as an institution rather than as a business — the parts that must outlive any product, market or founder. It establishes the innovation laboratory with its annual production targets and its kill discipline, the research institute and the independence clause that permits it to publish findings unfavourable to the company, and the measurement constitution together with the anti-metrics the group refuses to optimise. It sets the talent system from hiring through promotion, leadership development, entry programmes and performance review; the standard operating procedure library; the future-technology watch list and its adoption test; and the sustainability commitments. It closes with the 100-Year Plan, the values that must never change, and the Living Constitution Directive that governs every future addition to this corpus.

CHAPTER 1 — WHY THIS VOLUME EXISTS

Volumes I–VIII describe a company: what it sells, to whom, how it builds, where it goes, what it charges. A company of that description could be excellent and still be finished in thirty years — because companies die of the things this volume addresses: they stop inventing, they stop learning, they stop measuring honestly, they hire badly, their operations decay into tribal knowledge, and they optimise the present until they have no future.

An institution is a company that has solved for its own succession — of ideas, of people, of purpose. This volume is the difference between building a business and building something that lasts.

CHAPTER 2 — STROMEX LABS: THE INNOVATION LABORATORY

2.1 The mandate

A permanent, funded, protected department whose job is to produce things the operating divisions would never produce, because operating divisions are correctly optimised for this quarter's delivery.

Funded as a fixed percentage of revenue (Volume VIII §8.1: **35% of allocated free cash flow in Phase I, 25% in Phase II, 20% in Phase III**), never as a residual. Established formally in Phase II; run informally by the founding team throughout Phase I.

2.2 The annual production target

OUTPUT	ANNUAL TARGET	GATE TO THE NEXT STAGE
New product ideas	100	Written one-pager: problem, who has it, why us, why now
Internal experiments	50	Ideas that survive the one-pager review
Prototypes	20	Experiments that showed a real signal
Production launches	10	Prototypes with a customer willing to pay or depend on it

Table 149 — The annual production target

The funnel shape is the point: **90 of every 100 ideas are supposed to die**, and killing them fast is the department's core competence. An innovation lab with a high hit rate is not being ambitious enough.

2.3 The 100-idea discipline

The 100 ideas are generated deliberately, not opportunistically, and come from named sources so the pipeline never depends on inspiration:

SOURCE	CONTRIBUTION
Support ticket themes	What customers repeatedly cannot do
Lost-deal reasons	What we could not offer
The unused-module register	What we built that nobody wanted, and why
Field observation	What staff actually do that the software does not know about
Partner requests	What the ecosystem keeps asking for
Research Institute findings	Chapter 3
Future-technology watch	Chapter 7
Employee innovation time	§2.4
Adjacent-sector transfer	A solved problem in one ecosystem, unsolved in another
Constraint inversion	"What if bandwidth were free?" / "What if it had to work on paper?"

Table 150 — The 100-idea discipline

2.4 Employee innovation time

Every employee may direct **10% of their time** to self-chosen work that plausibly benefits StromeX. Inspired by long-standing practice at several research-led technology companies, with the failure mode of those programmes explicitly designed out:

- —**Time is protected, not theoretical.** A manager who reclaims it is in breach; this is a reviewed behaviour.
- —**Output is presented, not filed.** A quarterly internal showcase. Presenting a failure is a legitimate and celebrated outcome.
- —**No approval needed to start.** Approval is needed to *ship*, which is a different gate.
- —**Credit is individual and permanent.** Whoever had the idea is named on it, forever, in the record.
- —**It is not overtime.** It comes out of the working week, not out of a life.

2.5 The kill discipline

Every experiment declares its kill criterion **before it starts**: what result would end it, and by when. An experiment without a kill criterion is not an experiment, it is a hobby with a budget.

Killed projects are documented — what we tried, what happened, what we learned, what would make us revisit it. The kill record is one of the most valuable documents the group produces, because the second person to have an idea deserves to know what happened to the first.

2.6 The innovation boundary

Labs works within the constitution. An innovation that would require amending Volume I Chapter 5 or 7 is not an innovation, it is a proposal to become a different company, and it goes through Volume I §9.1 rather than through Labs.

CHAPTER 3 — STROMEX RESEARCH INSTITUTE

3.1 Why a commercial company runs a research institute

Three reasons, and only the third is really about credibility:

175. **Our markets are genuinely under-researched.** There is very little rigorous published work on what actually works in West African education technology, on credential fraud economics, on institutional digitisation in low-bandwidth environments, or on Islamic education technology. We have data nobody else has and questions nobody else is asking.
176. **Research improves the product.** A learning-science finding about spacing and retrieval changes how the memorisation engine should work. That is a product decision that should be made from evidence rather than from intuition.
177. **Credibility with the institutions that matter.** Ministries, universities, donors and regulators take published research seriously in a way they do not take marketing seriously. A company that has published a rigorous study on examination integrity is a different kind of supplier in a tender.

3.2 Research domains

DOMAIN	EXAMPLE QUESTIONS
Artificial intelligence	How should AI assist assessment without displacing judgement? What does grounding actually reduce hallucination by, measured?
Education & learning science	What spacing schedules work for Qur'anic memorisation specifically? What predicts dropout in West African secondary schools?
Educational technology	What determines whether school software is adopted or abandoned?
FinTech	How do fee-instalment structures affect collection rates and completion rates?
Psychology & behaviour	How do institutional buyers in low-trust markets actually evaluate suppliers?
Human-computer interaction	Interface design for low digital confidence and shared devices
Digital transformation	What distinguishes institutions that transform from those that install software?
Islamic education technology	An almost entirely unstudied field where we have unusual access and obligation
Public administration technology	What makes government digital services actually get used?
Digital identity & credentials	The economics of credential fraud; what verification actually changes

Table 151 — Research domains

3.3 Research standards

Non-negotiable, because bad research published by a commercial company is worse than no research:

178. **Pre-register the question and the method.** Analysis decided after seeing the data is not a finding.

179. **Publish negative results.** A study showing our own product did not help is exactly the study that makes the positive ones credible.

180. **Never publish research that cannot survive being wrong for us.** If the finding would embarrass the company, that is a reason to publish it carefully, not a reason to bury it.

181. **Consent and ethics first.** Institutional data used in research requires informed institutional consent; children's data requires the strictest handling (Volume IV §4.3) and in most cases should not be used at all.

182. **Anonymise and aggregate.** No institution or individual identifiable without explicit, specific, revocable permission.

183. **External review.** Serious work is reviewed by qualified people outside the company before publication.

184. **Separate research from marketing.** A white paper written to sell something is a brochure and must be labelled as one. The Institute does not produce brochures.

185. **Open access.** Published freely. Data published where consent and privacy permit.

3.4 Outputs

Peer-reviewed papers where the work merits it · practical reports for practitioners, written in plain language · open datasets where ethically permissible · annual "State of Institutional Digitisation" report for our markets · methodology and instrument publications so others can replicate · research fellowships for academics in our markets (Volume V §11.2).

3.5 The independence clause

The Research Institute may publish findings unfavourable to StromeX products, and no commercial function may block it. Suppression requires a Board decision and the fact of suppression is recorded in the corpus. This clause is what makes everything else the Institute publishes worth reading.

CHAPTER 4 — THE MEASUREMENT CONSTITUTION

4.1 The principle

Everything of consequence is measured; every measure has an owner; no measure is optimised at the expense of the thing it was meant to represent.

That last clause is the whole chapter. A metric that becomes a target stops being a good metric, and the specific ways this happens in a company like ours are predictable enough to name in advance.

4.2 The measurement domains

DOMAIN	PRIMARY MEASURES	OWNER
Customer satisfaction	Would-recommend, CSAT after support, adoption rate, value-ledger trend	Customer Success
Feature usage	Active use by module, unused-module register, workflow completion	Product
Revenue	By engine, by ecosystem, by geography, by tier	Finance
Profitability	Gross margin vs. band per division, contribution per customer segment	Finance
Retention	GRR, NRR, logo churn with real reasons, cohort curves	Customer Success
Support quality	First response, resolution, reopen rate, ticket themes	Support
Employee	Retention, internal mobility, time-to-productivity, engagement	People
Reliability	Uptime, error rate, latency percentiles, SLA credits applied	Engineering
Security	Incidents by severity, time to patch, access-review completion, pen-test findings	Security
Marketing	Organic traffic, ranked terms, referral share, verification-page sessions	Marketing
Conversion	Free→identified→qualified→customer, by channel and cohort	Growth
Delivery	Time to first value, go-lives requiring heroics, implementation margin	Delivery
Innovation	Ideas→experiments→prototypes→launches, kill-rate, time-to-kill	Labs
Ethics & integrity	Ethics reviews, refusals, integrity losses, prohibited-practice reports	Executive

Table 152 — The measurement domains

4.3 The named anti-metrics

Measures we track but explicitly refuse to optimise, with the reason each would corrupt:

ANTI-METRIC	WHY WE REFUSE
Bookings without adoption	Manufactures a good quarter out of a bad year (Volume V §12.5)
Support ticket volume (minimised)	Punishes customers for asking and hides product defects
Time on app	We are not an attention business (Volume I §14.3)
Headcount	A cost, not an achievement (Volume I §14.5)
Lines of code / tickets closed	Measures motion, not value
Free-tier signups (as a headline)	Trivially inflatable and disconnected from value delivered
Valuation	Volume VIII §5 — not controllable, and optimising for it corrupts everything upstream
Any customer metric the customer would dispute	If they would not recognise the number as true, it is not true

Table 153 — The named anti-metrics

4.4 Reporting honesty

Every metric reported with its **denominator**, its **period**, and its **trend** — never a bare number · bad quarters reported as prominently as good ones · a metric definition may not be changed without restating history and saying that it was changed · **when a target is missed, the plan changes, not the measurement** (Volume VIII §13).

CHAPTER 5 — THE TALENT SYSTEM

5.1 The hiring philosophy

Hire for judgement. Skills are teachable at pace; judgement is not (Volume I §13.1.8). The most predictive interview question in this company is *"tell me about a decision you got wrong, and what you changed."* A candidate who cannot answer it either has not made decisions or has not examined them.

5.2 The hiring standard

RULE	REASON
Structured interviews, same questions, scored against a rubric	Unstructured interviews measure similarity to the interviewer
Work sample over whiteboard puzzle	Predicts the job; puzzles predict puzzle skill
Paid work samples above 2 hours	Unpaid extended work is an unfair transfer from candidate to company
At least one interviewer from outside the hiring team	Prevents team-shaped hiring
Reference checks actually made	Not a formality
No hiring on urgency	The wrong hire costs more than the vacancy — and this rule will be tested every time
Compensation benchmarked against remote-work rates, not local rates	Volume II R13: our competitors for talent are overseas employers, and pretending otherwise loses people
Written offer, no exploding deadlines	Pressure tactics in hiring predict pressure tactics in management

Table 154 — The hiring standard

5.3 The promotion framework

Promotion recognises demonstrated operation at the next level, not tenure, proximity or advocacy.

LEVEL	INDIVIDUAL CONTRIBUTOR	MANAGEMENT
1 — Learning	Executes defined work with support	—
2 — Contributing	Owns defined work independently	—
3 — Owning	Owns an area; others build on their work	Leads a small team
4 — Leading	Sets direction for an area; multiplies others	Leads a function
5 — Shaping	Changes how the company works; judgement trusted under uncertainty	Leads a division
6 — Institutional	Work outlives their tenure; the company is different because they were here	Executive

Table 155 — The promotion framework

Both tracks are equal at every level, in title, in compensation and in influence. A company where the only path upward is management loses its best engineers, designers and editors to management, where they are frequently worse and always scarcer.

The Volume I §13.3 criteria govern: judgement under uncertainty · work others build on · customers who ask for them by name · documents that outlive the project · mistakes surfaced early and learned from publicly · colleagues who are better for working alongside them.

5.4 Leadership development

Managers are trained before they manage, not after they struggle · every manager has a manager who reviews their team's retention and growth, not only their output · management is a **role, not a rank** — moving out of management is a lateral move, and is treated and paid as one · leadership candidates are identified two levels down and developed deliberately · **succession plans exist for every executive role from Phase II**, including the founder's (Volume VIII Gate I, criterion 10).

5.5 Entry programmes

PROGRAMME	SHAPE	PURPOSE
Graduate programme	18 months, three rotations, mentor, guaranteed role on completion	Build the next decade's leadership from our own markets
Internship	3–6 months, paid, real work, named mentor	Pipeline; never unpaid, never coffee-fetching
Apprenticeship	24 months, no degree required, structured curriculum	Talent that the credential system missed — of which our markets have an enormous amount
Returnship	6 months, for people re-entering after a career break	An under-recruited pool, disproportionately women
Campus & student ambassadors	Part-time, at institutions we serve	Volume V §11.2

Table 156 — Entry programmes

The apprenticeship deserves emphasis. A company whose entire business is that credentials should be verifiable and that institutions should be measured on outcomes cannot then hire exclusively on the basis of degrees. Building a genuine non-graduate path is both consistent and commercially smart, because that talent is available and under-competed.

5.6 Continuous learning

An annual learning budget per employee, defended · internal teaching expected at level 3 and above, because teaching is how understanding is verified · conference attendance with an obligation to report back · StromeX Academy certifications free to employees · **explicit time for learning, protected the same way innovation time is.**

5.7 Performance review

Twice yearly, no forced ranking, no stack ranking · self-assessment first, then manager, then calibration across managers to check for bias · **feedback is continuous and the review contains no surprises** — a review that surprises someone is a management failure, not an employee failure · compensation reviewed separately from development, because merging them makes honest development conversations impossible · underperformance is addressed early, specifically, with support and a clear standard, and if it does not resolve, quickly and with dignity.

5.8 The mentorship system

Every new employee has a mentor outside their reporting line from week one · mentors are trained and their contribution counts toward promotion · reverse mentoring from Phase II, where junior employees mentor executives on what the company actually feels like from below — which is information that otherwise never reaches the top of any organisation.

5.9 The culture protections

Sustainable pace enforced by managers, not by policy documents (Volume I §13.1.9) · bad news travels fastest and is rewarded, always (§13.1.4) · dissent recorded and preserved (§9.3) · concealment is the one issue that is not coached · **no one is indispensable, and anyone who has become indispensable is a risk we created and must fix.**

CHAPTER 6 — OPERATIONAL EXCELLENCE & THE SOP LIBRARY

6.1 The principle

Anything done more than three times is documented; anything documented and stable is automated.

This is not bureaucracy. It is the mechanism by which a company scales without its quality depending on who happens to be doing the work — and it is the mechanism by which we can credibly sell automation to institutions, because we are our own first customer.

6.2 The SOP library

Every function maintains standard operating procedures with an owner, a version, a review date, and a named escalation path.

FUNCTION	SOPS INCLUDE
Product development	Discovery · specification · design review · build · Volume IV §18.2 definition of done · release · post-launch review
Quality assurance	Test strategy · regression suites · the mandatory suites that may never be skipped · accessibility · performance · security testing · UAT with the customer
Design	Brief · research · concept · review gates · handover · asset licensing (Volume VI §9)
Customer support	Triage · severity classification · escalation · resolution · post-resolution follow-up · pattern reporting to product
Sales	Qualification · discovery (the seven questions) · proposal · pricing authority · contract · handover to delivery
Delivery	Kick-off · data readiness · migration · parallel run · training · go-live · hypercare · handover to success
Finance	Invoicing · collections · reconciliation · approvals · procurement · expense · payroll · month-end · reporting
Procurement	Requisition · quotes · supplier due diligence · approval · receipt · payment
HR	Requisition · interview · offer · onboarding · probation · review · promotion · grievance · exit
Legal	Contract review · IP · disputes · regulatory filings · data agreements · export/sanctions screening
Security	Access provisioning and review · vulnerability handling · incident response (Volume IV §17) · penetration testing · third-party assessment
Creative	Volume VI §13 quality gates
Partner	Onboarding · certification · deal registration · dispute resolution · termination

Table 157 — The SOP library

6.3 The SOP discipline

An SOP that has not been reviewed in 12 months is flagged · an SOP nobody follows is either wrong or unnecessary, and both are fixed rather than enforced · **the person who does the work owns the SOP**, not a process department · every SOP states what to do when the SOP does not fit, because a procedure with no exception path produces either paralysis or quiet non-compliance.

6.4 The internal automation mandate

The group automates its own operations first, and the reason is not efficiency — it is credibility and product truth. Every automation we sell should be one we run. An automation pack we would not use on ourselves is one we should not sell.

Priority order for internal automation: onboarding and offboarding · invoicing and collections · procurement approvals · reporting assembly · support triage · deployment and release · access provisioning and review · SOP compliance checks.

CHAPTER 7 — FUTURE TECHNOLOGIES

7.1 The evaluation doctrine

We adopt a technology when it creates measurable customer value, not when it becomes fashionable. The rolling watch list below is reviewed half-yearly by Labs and the CTO. Each entry carries a posture, and postures change on evidence.

POSTURE	MEANING
Deploy	In production, generating value
Build	Actively building
Pilot	Testing with real customers
Watch	Monitored; not invested in
Decline	Assessed and rejected, with a stated reason and a revisit condition

Table 158 — The evaluation doctrine

7.2 The watch list

TECHNOLOGY	POSTURE	ASSESSMENT
Applied AI / LLM agents	Deploy	Core to the group. Provider-independent by architecture (Volume IV §7.1)
Retrieval & knowledge grounding	Deploy	The mechanism by which AI becomes trustworthy in an institutional context
On-device / small models	Build	Directly serves the low-bandwidth, low-cost mandate. Strategically important, not merely interesting
Speech (ASR/TTS) & voice agents	Deploy	High value in low-literacy and multilingual contexts
Digital identity & verifiable credentials	Deploy	Volume IV §5. The group's strategic core
Quantum-safe cryptography	Build	**Prioritised for credential signatures.** A certificate signed in 2027 may need to verify in 2067; it is the group's longest-lived cryptographic commitment and therefore the first thing to migrate, ahead of transport security
IoT & sensor networks	Deploy	Smart campus, facilities, agriculture
Advanced analytics & forecasting	Deploy	Volume III Division 9
Offline-first & sync architectures	Deploy	Not a future technology in our markets — a present requirement
Robotics	Watch	Real value in manufacturing and

TECHNOLOGY	POSTURE	ASSESSMENT
		logistics (Wave 3). No customer value in education worth the cost today
Spatial computing / AR/VR	Watch	Genuine potential for laboratory simulation and vocational training; blocked on device cost in our markets. Revisit when a capable headset costs under $150
Autonomous agents (multi-step, unsupervised)	Pilot	Constrained by Volume IV §8.1. Useful for low-stakes internal work; not for institutional decisions
Satellite connectivity	Watch	Could materially change the rural-institution proposition. Revisit on cost per Mbps in our markets
Mesh networking	Watch	Interesting for campus and emergency scenarios; unproven at institutional scale
Blockchain / distributed ledger	Pilot (narrow)	**One legitimate use only: optional credential anchoring** (Volume IV §5.5, layer 5), because it makes credentials verifiable independent of our continued existence. Every other proposed use in our sectors has a simpler, cheaper, better answer, and we say so plainly
Digital twins	Watch	Facilities and manufacturing, Wave 3
Wearables	Watch	Limited institutional value today
Brain-computer interfaces	Decline	No credible institutional application; profound ethical exposure. Revisit condition: none currently foreseeable
Emotion recognition / affect detection	**Decline, permanently**	Scientifically contested, discriminatory in practice, and its obvious application — monitoring students and employees — is precisely the surveillance we refuse (Volume I §7.4). **This is a values decision, not a technical one, and it does not have a revisit condition**

Table 159 — The watch list

7.3 The adoption test

Before any watch-list technology moves to Pilot: what customer problem does it solve, in the customer's words? · what does the customer currently do instead, and why is that worse? · what does it cost to deploy in our markets, on our customers' devices and connections? · what does it cost if it fails? · what would make us abandon it? · does it comply with Volume I Chapter 7?

A technology that passes every question except the last does not get adopted.

CHAPTER 8 — SUSTAINABILITY & SOCIAL COMMITMENT

Commitments, not aspirations. Each has a measure and an owner, because a sustainability chapter without measures is a marketing document.

8.1 Environmental

COMMITMENT	MEASURE
Efficiency as a design property	Performance budgets (Volume IV §10.1) reduce compute, bandwidth and device energy simultaneously — the low-bandwidth mandate is also our largest environmental lever
Renewable-powered infrastructure where available	% of compute on renewable-powered regions, published
Hardware lifecycle responsibility	Take-back and recycling for hardware we supply (Volume III §11.5); refurbishment offered before replacement
Reduce shipping impact	Print locally where quality permits (Volume VI §10.2) — which is also the cheaper and better advice
Measure before claiming	We publish our footprint before we make any claim about reducing it, and we make no carbon-neutrality claim we cannot substantiate

Table 160 — Environmental

8.2 Digital inclusion

Low-bandwidth and low-spec support as an architectural requirement, not a charitable feature (Volume IV §10) · USSD and feature-phone access where the population requires it (Volume III Division 8) · offline-first everywhere · the free tier as genuine infrastructure for institutions that will never pay · **free credential verification forever** (Volume I §5.3) · pricing bands that reflect purchasing power rather than extracting from it (Volume I §6.4).

8.3 Accessibility

WCAG 2.2 AA as the enforced floor, AAA for body text where achievable, verified in CI (Volume IV §11.1) · every product usable by keyboard and screen reader · accessibility audits free for anyone who asks (Volume III §1.5) · the design system open-sourced so accessibility benefits beyond our customers · **accessibility never a paid tier.**

8.4 Ethical AI

The Volume I §7.2 honesty rules · the Volume IV §8.1 stakes ladder · the Volume IV §9 data rights, including the refusal to train third-party models on customer data · the Volume I §7.1 workforce commitment, with free redeployment training where our automation changed someone's role · **and the standing refusal to build emotion recognition or surveillance systems** (Chapter 7.2).

8.5 Community

PROGRAMME	COMMITMENT
Scholarships	Funded places on StromeX Academy certifications for students and

PROGRAMME	COMMITMENT
	teachers who cannot pay
Free tier for low-fee schools	Standing policy discount up to 40% (Volume III §0.4); free tier without limit for schools below a fee threshold
Open source	Design system, SDKs, exporters, typography toolkit — permanently free (Volume I §5.5)
Educational outreach	Free curriculum, policy and assessment resources for any institution, customer or not
Apprenticeships	A genuine non-graduate path into technology careers (Chapter 5.5)
Research	Published openly, including findings unfavourable to us (Chapter 3.5)
Local employment	Build teams in the markets we serve, at rates benchmarked internationally, not locally

Table 161 — Community

8.6 The honesty clause

We will not publish a sustainability claim we cannot evidence, will not describe a commercial decision as a social one, and will not count the free tier as philanthropy when it is also a distribution strategy — it is both, and saying so is more credible than pretending it is only the first.

CHAPTER 9 — THE 100-YEAR PLAN

The final strategic chapter of the corpus. Written for people not yet born, who will work at a company whose founders they never met.

9.1 What StromeX should stand for after 100 years

> **That institutions serving ordinary people deserve extraordinary tools — and that where you were born should not determine the quality of the systems that shape your life.**

If StromeX exists in 2127 and that sentence is still true of it, the company will have succeeded regardless of its size, its share price, or whether anyone remembers who started it.

9.2 The values that must never change

These are the entrenched provisions (Volume I §9.1). Amending any of them requires a Board supermajority, a published rationale and thirty days' notice — and any future leadership that finds them inconvenient should understand that the inconvenience is the point.

186. **Verification is free for the verifier, forever.** A person's credential belongs to them. Checking it must never have a price, and a graduate's degree is never held hostage to their institution's subscription.
187. **Customer data belongs to the customer.** Never sold, never brokered, never used to train third-party models, always exportable, free and complete.
188. **Children's data receives the strictest handling in the group** — never monetised, never profiled, never advertised against.
189. **No bribery, ever, in any market, for any contract.** Losing business for this reason is a correct outcome.
190. **We do not build surveillance, weapons, or systems designed to persecute.**
191. **High-stakes decisions about a person are made by a person**, and that person is named and accountable.
192. **We disclose our failures before we are asked**, with a timeline and a fix.
193. **The free tier is real**, and the boundary of what is free is stated plainly rather than discovered at the moment of frustration.
194. **We do not sell what we cannot deliver.**
195. **A price is honoured for its term**, and increases are announced, capped and explained.

9.3 The things that may and should change

Products. Markets. Technologies. Business models. Organisational structure. Pricing. The sectors we serve. The countries we operate in. The people who lead. **All of it should change, repeatedly, and a future leadership that preserves any of it out of reverence has misunderstood this document.** The corpus is a constitution, not a museum.

9.4 The four generational transitions

ERA	THE TRANSITION	THE RISK IT CARRIES
Years 1–20 (2027–2046)	Founder-led → institution-led	The founder never becomes dispensable, and the company ends

ERA	THE TRANSITION	THE RISK IT CARRIES
		when they do
Years 20–40	First professional leadership; the founding generation departs	The constitution is treated as the founder's preferences rather than as law, and is quietly abandoned
Years 40–70	Second and third leaderships; the founding markets mature	Success produces the complacency that makes the company a Class 2 incumbent — the very thing Volume II §5.1 describes us beating
Years 70–100	The company outlives its original technologies entirely	Nobody remembers why the entrenched provisions exist, and the reasoning was never written down

Table 162 — The four generational transitions

The mitigation for all four is the same, and it is this corpus. Not because a document has authority on its own, but because the reasoning is written down — and a future employee who disagrees can at least argue with the argument rather than guess at it. That is why Volume I §9.2 requires every amendment to state what becomes false if it is wrong, and why §9.3 preserves dissent permanently.

9.5 The legacy, stated concretely

In education: that a child in Ikorodu, Kano, Kumasi or Karachi learns in an institution whose systems are as good as any in London — and whose certificate is believed anywhere in the world without a phone call.

In technology: that the industry has a working example of a company built in an emerging market, to a global standard, without pretending to be from somewhere else, and without extracting from the people it served.

In society: that the millions of hours of administrative labour StromeX automated were returned to teaching, to care, to scholarship and to family — and that the people whose jobs changed were trained, redeployed and better off, because the company that automated their tasks took responsibility for it.

9.6 The final instruction

> **If StromeX ever has to choose between being large and being trusted, it chooses trusted.**
>
> That choice will present itself. It will be expensive. It will be argued for persuasively by intelligent people with good intentions and a real business case. The answer is in this line, and it was written down in 2026 precisely so that whoever faces it does not have to decide alone.

CHAPTER 10 — THE LIVING CONSTITUTION DIRECTIVE

Permanent standing instruction, applying to every author of this corpus — human or AI — in perpetuity.

10.1 The directive

> **Do not treat this Editorial Bible as a finished document. Treat it as a living constitution.**
>
> Whenever you identify a missing capability, governance principle, product category, revenue model, technology, operational process, or strategic opportunity that aligns with StromeX's mission and long-term sustainability, propose and incorporate it with clear justification.
>
> Distinguish evidence-based recommendations from aspirational ideas. Preserve internal consistency. Ensure every addition strengthens the company over decades rather than optimising for short-term gains.

10.2 The authorship standard

Every addition to this corpus, by anyone, must meet the same bar the corpus holds itself to:

Answer the five questions (Volume VIII §8.3): why is this included · what measurable value does it create · what risks does it introduce · what assumptions underpin it · how will success be measured.

Separate evidence from inference, explicitly. The corpus uses three registers and marks which is which:

REGISTER	MEANING	HOW IT APPEARS
Verified	Sourced, dated, and linked	Cited inline with a source
Synthesis	Well-established industry practice, reasoned from	Stated as practice, not as a finding
Judgement	Our reasoning from first principles	Stated as our position, with the reasoning exposed so it can be attacked
Projection	Estimates about the future	Labelled, with assumptions inline and scenarios where material

Table 163 — The authorship standard

Never fabricate. No invented statistics, no fictional interviews, no imaginary case studies, no synthetic quotes, no citations that do not resolve. Where the corpus needs a number it does not have, it says so and states the method by which the number should be obtained (Volume II §2.2 is the model).

Compare before recommending. State the alternatives considered and the trade-off accepted. A recommendation with no stated alternative is an assertion.

Do not inflate. Where forecasts are given, give conservative, expected and optimistic scenarios with their assumptions, and state which one the company should plan against (Volume VIII §5).

Preserve consistency. An addition that contradicts an existing chapter must either reconcile with it or amend it explicitly under Volume I §9. Silent contradiction is the mechanism by which constitutions become decoration.

Write for the reader who was not there. The person consulting this corpus in 2043 does not know what was obvious in 2027. Write the reasoning down, not just the conclusion.

10.3 The review obligation

Every volume carries a review date. An unreviewed chapter may not be cited as authority in a dispute (Volume I §9.2.5). A chapter unamended for twelve months is flagged for review — not because it is necessarily wrong, but because a constitution nobody has questioned in a year is a constitution nobody is reading.

VOLUME

X

SpaceTalk

The Communication Operating System

AUTHORITY

Product

EXECUTIVE SUMMARY

Volume X opens by stating what most documents in its category avoid: a general-purpose consumer messenger competing directly with the incumbents cannot be won, and attempting it would be the most expensive mistake available to the group. It then sets out the position that is winnable — institutional communication infrastructure that grows institution by institution rather than person by person, entering through relationships StromeX already holds. It establishes three wedges in verified identity, the institutional record and federation; specifies the architecture and the encryption position that reconciles user privacy with institutional obligation without concealment; answers the connectivity question honestly; and defines trust and safety, the business model, the roadmap and the risks, including the most likely one — that the product is started too early.

CHAPTER 1 — THE STRATEGIC POSITION

1.1 The honest starting point

The ambition is that SpaceTalk becomes the world's communication operating system. This volume takes that ambition seriously, which requires beginning with the thing most product documents in this category avoid:

A general-purpose consumer messenger competing head-on with WhatsApp cannot be won, and attempting it would be the single most expensive mistake in this corpus.

WhatsApp has around 2 billion monthly active users; WeChat close to 1.3 billion; Messenger 1.01 billion; Telegram around 900 million; Snapchat 800 million; Discord around 200 million (Priori Data, 2026). Messaging is the most network-effect-dominated category in software: the value of the app is entirely the people already on it, switching costs are social rather than technical, and a graveyard of extremely well-funded, technically excellent competitors — including from Google and Amazon — demonstrates that capital and engineering do not overcome this.

Volume I §14.4 applies: we do not compete on being cheapest, and we do not compete where the only path to winning is outspending an incumbent at their own game.

1.2 The position that is actually winnable

SpaceTalk is not a messenger that hopes to become institutional. **It is institutional communication infrastructure that happens to be excellent at messaging.**

The distinction determines everything:

A CONSUMER MESSENGER	SPACETALK
Grows person by person	Grows **institution by institution** — one school onboards 800 parents at once
Network is unbounded and must be won globally	Network is **bounded and already exists** — a school's community, a ministry's agencies, a hospital's staff
Identity is a phone number	Identity is **verified and role-bearing** (StromeX Identity)
Competes on features	Competes on **being the system of record for communication**
Monetises attention or subscription	Monetises institutional capability (Chapter 15)
Cold-start problem is existential	Cold-start problem is **solved by the parent company's customer base**

Table 164 — The position that is actually winnable

The cold-start advantage is the entire thesis. StromeX already sells to institutions whose communities must communicate: schools with parents, universities with students, hospitals with staff, mosques with congregations, ministries with agencies. Every institution that deploys SpaceTalk brings its whole community with it, in one act, because the institution — not the individual — makes the decision. No consumer messenger has ever had this, and it is not available to anyone who is not already inside institutions.

1.3 What SpaceTalk is

> **The communication layer of the institutional operating system: verified-identity messaging, calling, collaboration, workflow and commerce for communities that have a reason to be a community.**

And what it is not: not a social network · not an attention product · not ad-funded · not a WhatsApp replacement for the general public · not a walled garden (Chapter 5).

CHAPTER 2 — THE COMPETITIVE LANDSCAPE, HONESTLY READ

Studied for principles, not for imitation (Volume I §14.4). What follows is what each platform actually teaches.

PLATFORM	WHAT IT GOT RIGHT — THE TRANSFERABLE PRINCIPLE	WHAT IT LEFT OPEN
WhatsApp	Phone-number identity removed all onboarding friction; ruthless simplicity; works on weak devices and networks	No verified identity; groups do not scale to institutions; no records, no roles, no audit; businesses use it because nothing better exists locally
WeChat	Proved a messenger can be an operating system — mini-apps, payments, services all inside one surface. **The single most important precedent for SpaceTalk**	Closed ecosystem; state-adjacent governance; not exportable as a trust model
Telegram	Cloud-first sync, huge groups, channels, bot platform, fast client engineering	Weak default encryption posture; moderation difficulties at scale
Signal	Encryption done properly and published; minimal metadata; proof that privacy can be a product	Deliberately no institutional features; no records; no monetisation
Discord	Solved persistent community structure — servers, channels, roles, voice presence. The best community model in the category	Consumer-culture-bound; not credible in institutional or government contexts
Slack	Made workplace channels normal; the integration and app-directory model	Expensive per seat; poor on weak networks; message retention as a paywall is resented
Teams	Distribution through an existing enterprise relationship — **the same mechanism SpaceTalk will use**	Heavy client; poor performance on low-spec devices; adoption is often compliance rather than preference
Zoom / Meet	Reliability under bad networks; adaptive bitrate; joining without an account	Meetings as a separate universe from messaging
iMessage / FaceTime	Platform integration and quality as a moat	Single-vendor lock-in; excludes most of our markets
LINE / KakaoTalk	National-scale super-apps built on local cultural fit, not global ambition — **directly relevant to our regional strategy**	Not exportable beyond their home markets
Matrix / Element	**Open federated protocol with real government adoption** — Chapter 5	Client and UX quality historically behind commercial rivals; complexity
Rocket.Chat / Mattermost	Self-hosting and sovereignty as a product; source-available commercial model	Limited consumer-grade polish
IRC / Mumble /	Protocol longevity; low resource use;	No modern identity, media or mobile

PLATFORM	WHAT IT GOT RIGHT — THE TRANSFERABLE PRINCIPLE	WHAT IT LEFT OPEN
TeamSpeak	the value of open specification	story
Skype (historical)	The cautionary tale: an early, dominant, technically excellent product that lost by failing to keep the client good on mobile. **Category leadership is not permanent**	—
X / LinkedIn messaging	Messaging attached to an identity graph	Messaging is an afterthought to the feed
Beeper	Aggregation across networks as a user need — evidence that fragmentation is genuinely painful	Depends on bridges the incumbents may break

Table 165 — The Competitive Landscape, Honestly Read

2.1 The three lessons that shape SpaceTalk

196. **From WeChat:** a communication surface can absorb payments, services and commerce, and when it does, it stops being an app and becomes infrastructure. This is the ceiling of the ambition and it has been demonstrated once, in one market, which means it is possible and not yet globally settled.

197. **From Teams:** the reliable way to win institutional communication is to arrive through a relationship the institution already has. StromeX has that relationship.

198. **From Skype:** the position is lost by neglecting client quality on the devices people actually use — which, in our markets, means mid-range Android on unreliable networks (Volume IV §10).

CHAPTER 3 — THE THREE WEDGES

Wedge 1 — Verified identity. Every other platform's identity is a phone number or an email address. SpaceTalk's identity is the StromeX identity: verified, role-bearing, institution-issued, and cryptographically checkable (Volume IV §5). A message from the Principal is *provably* from the Principal. In markets where impersonation fraud — fake school accounts demanding fee payments, fake officials, fake examination results — is endemic and costly, this is not a feature. It is the reason to switch, and it is the one thing an incumbent messenger structurally cannot copy, because they have no issuing authority behind their users.

Wedge 2 — The institutional record. Communication in institutions is not chat; it is *record*. A fee notice, an approval, a safeguarding disclosure, a governance decision, an examination instruction — these need retention, audit, legal hold, search and export. Consumer messengers deliberately have none of this. Slack and Teams have some of it, at a price and a device weight our markets cannot carry. **SpaceTalk treats every message as a potential record and every record as auditable** — the same doctrine as Volume IV §4.2.

Wedge 3 — Federation and sovereignty. Chapter 5.

CHAPTER 4 — ARCHITECTURE

Engineering judgement, stated as such. Where a claim is a design intention rather than a demonstrated result, it says so.

4.1 Principles

Inherits Volume IV in full, with four additions specific to real-time communication:

C1 — Offline-first is the baseline, not a mode. Messages compose, queue and send when connectivity returns. State is local-first with server reconciliation. In our markets this is not a nicety; a messenger that fails when the network drops is not usable.

C2 — The client must stay small and fast. Volume IV §10.1 budgets apply with a stricter target: cold start under 2s and steady-state memory low enough for a 2GB device. The Skype lesson (Chapter 2) is that this is where the category is lost.

C3 — Encryption by default, with institutional obligations honestly reconciled. Chapter 4.3.

C4 — Protocol before product. The wire format and federation semantics are specified and documented before the client is built, because a communication system's protocol outlives every client that speaks it.

4.2 The shape

```
CLIENTS      Android · iOS · Web · Desktop · PWA · USSD/SMS bridge · Kiosk
                                 |
EDGE         Global PoPs · WebSocket gateway · TURN/STUN · media relay
                                 |
CORE         Sync · rooms & timelines · presence · push · media pipeline
             Identity & roles · permissions · retention & legal hold · search
                                 |
FEDERATION   Matrix-interoperable server-to-server (Chapter 5)
                                 |
PLATFORM     StromeX Identity · Pay · Cloud · AI · Automation · Records
                                 |
DATA         Event store (append-only) · object store (media) · index · cache
```

Figure 8 — The shape

The event store is the architectural core. A room is an append-only, causally-ordered timeline of events, from which all state is derived. This is the same doctrine as Volume IV §4.2 D2 (events, not state), and it is what makes federation, audit, legal hold, offline sync and eventual consistency tractable rather than bolted on. It is also the design that every serious federated messaging system has converged on, which is corroborating evidence rather than coincidence.

4.3 The encryption position

Default: end-to-end encryption for private and small-group conversation.

The institutional reconciliation, stated plainly rather than fudged: an institution with legal retention, safeguarding and audit obligations cannot operate a communication system whose content is permanently inaccessible to it. Pretending otherwise produces either a product institutions cannot lawfully use, or a hidden backdoor. Both are unacceptable.

SpaceTalk's answer:

CONTEXT	ENCRYPTION MODEL	DISCLOSED TO USERS
Private 1:1 and personal groups	End-to-end; the institution cannot read them	Yes, visibly, in the room
Institutional rooms (class, department, ministry)	Encrypted in transit and at rest; **the institution is a key holder and this is stated in the room**	Yes, visibly, permanently
Safeguarding and disclosure channels	Encrypted; access restricted to named safeguarding roles; every access logged and reviewable	Yes
Regulated/records channels	Retention and legal hold applied per the institution's obligations	Yes

Table 166 — The encryption position

The rule: the user is always told, in the room, who can read it. **There are no hidden key holders, ever, and there is no configuration in which a user believes a conversation is private when it is not.** An institution may not silently enable access to a room that was presented as private; changing a room's access model notifies every participant and is recorded in the room's timeline.

This is a harder position than "everything is E2E" and a much harder one than "trust us". It is the only one that is both lawful for institutions and honest to users, and Volume I §4.3 requires it.

CHAPTER 5 — THE FEDERATION DECISION

5.1 The evidence

Matrix — the open, federated communication protocol — has moved from an enthusiast project to demonstrable public-sector infrastructure. The European Commission has begun trialling an open-source communications system for internal use; Germany's Bundeswehr and its IT supplier BWI have deployed it; Switzerland's Swiss Post and Austria's healthcare system have adopted it; France's government runs Tchap and Visio within its La Suite workspace on Matrix components; and Sweden's public sector has gone live with Matrix-based federation (The Register, Element, Open Source For You).

The driver is digital sovereignty: governments want communication data inside their own borders, under their own control, interoperable with others but owned by nobody.

5.2 The decision

SpaceTalk will be federation-capable and Matrix-interoperable at the server-to-server layer.

Four reasons, in order of weight:

199. **It is the only credible route into government** (Volume VII §8). A ministry that has adopted a sovereignty posture will not buy a proprietary silo. Interoperating with the protocol their peers have standardised on converts our largest strategic obstacle into an advantage.
200. **It partially defeats the cold-start problem.** A federated SpaceTalk user can reach users on other federated servers. The network is not only ours, which is the single hardest problem in this category.
201. **It is consistent with the constitution.** Volume I P5 forbids data hostage-taking; a federated, exportable, interoperable protocol is that principle expressed in architecture. A customer can leave and take their communication with them — which is exactly why they will trust us enough to arrive.
202. **It is a durable differentiator against consumer incumbents**, none of whom will federate, because their entire commercial position depends on not doing so.

5.3 The trade-offs, stated

Federation is not free, and this corpus does not pretend otherwise:

COST	ASSESSMENT
Engineering complexity	Substantial. Federated state resolution, eventual consistency and cross-server permissions are genuinely hard
Abuse surface	Federation imports other servers' abuse problems. Requires server-level reputation, allow/deny lists and defederation capability from day one (Chapter 14)
Feature velocity	Federated features ship slower than proprietary ones. Some features may be ours-only
Client quality expectation	Matrix clients have historically lagged commercial rivals on polish. **This is precisely where StromeX's craft advantage applies** (Volume II §6, Wedge 5) — the protocol is proven, the experience is the gap, and closing that gap is a design problem we are unusually well placed to solve
Metadata	Federation reveals more metadata between servers than a closed

COST	ASSESSMENT
	system does. Must be disclosed, not minimised rhetorically

Table 167 — The trade-offs, stated

The chosen shape: SpaceTalk is a *first-class commercial client and server* on an open protocol — excellent experience, institutional capability, and StromeX-specific value on top (identity, records, payments, AI, automation), while remaining interoperable. This is the Volume I §5.5 open-source test applied to a product: **open what spreads the standard, hold what compounds the advantage.**

CHAPTER 6 — MESSAGING

CAPABILITY	NOTES
1:1, groups, communities, organisations	Four scales with genuinely different permission models, not one model stretched
Channels & broadcast	One-to-many with acknowledgement tracking — a fee notice needs to be *provably delivered*
Threads	Replies without fragmenting the room
Rich media	Images, video, documents, voice notes; **aggressive client-side compression by default** with an explicit “send original”
Voice notes	The dominant medium in much of our market; transcription and summary attached automatically
Collaborative notes	Shared documents inside the conversation
Scheduled & recurring messages	Fee reminders, prayer times, timetable notices
Disappearing messages	Per-room, disclosed; **disabled in rooms subject to retention obligations**, visibly
Read state & delivery receipts	Per-room configurable; off by default in large rooms
Reactions, pins, bookmarks, mentions	—
Polls, forms, approvals	Communication that produces a record (Wedge 2)
Translation	Inline, per-message, with the original always retrievable
Search	Across everything the user may see; institution-wide for authorised roles, logged
Message-level roles	A message can carry the sender’s verified role: *“Bursar, Sultan Hanafi Royal Schools”*
Announcements with acknowledgement	The parent tapped “read” — recorded, exportable, defensible

Table 168 — Messaging

Design constraints inherited from Volume IV §10: every feature works at 3 Mbps, on a 2GB device, in RTL, in four languages, and degrades explicitly rather than failing silently.

CHAPTER 7 — CALLING & REAL-TIME MEDIA

Built on WebRTC — the established, standardised, widely-implemented stack for browser and mobile real-time media. There is no credible reason to invent an alternative.

CAPABILITY	ENGINEERING POSITION
1:1 voice & video	WebRTC peer-to-peer where NAT traversal permits; TURN relay otherwise
Group calls	Selective Forwarding Unit; participant count scales with server capacity, not client capacity
Adaptive low-bandwidth mode	Simulcast + dynamic bitrate; **audio is never sacrificed to keep video** — video degrades to slideshow, then to audio-only, automatically and visibly. This is the most important media decision for our markets
Audio-only default on weak networks	Detected, applied, and announced to the user rather than silently
Screen sharing	With region selection and a bandwidth-appropriate frame rate
Live transcription	Chapter 9
Live translation	Chapter 9; latency and accuracy limits disclosed
Recording	**Only with in-call consent from all participants, and only where lawful in the jurisdiction.** Recording state is permanently visible in-call
Meeting summaries	Chapter 9; always attributed as AI-generated and always editable
Dial-in / PSTN bridge	Via telecom partners, per market; a real cost, priced through
Large broadcast / webinar	One-to-many streaming rather than conferencing above a threshold

Table 169 — Calling & Real-Time Media

CHAPTER 8 — THE CONNECTIVITY QUESTION, ANSWERED HONESTLY

The brief asked whether calls could work without data. This chapter answers directly, because an honest engineering answer is worth more than an optimistic one.

8.1 The direct answer

No. An IP-based voice or video call requires a data connection. There is no architecture, protocol or optimisation that removes this — the audio has to travel over something, and for an internet-based application that something is IP over cellular data, Wi-Fi, or satellite. Any product claim to the contrary would be false, and Volume I §7.2 forbids it.

What *can* be done is reduce the data requirement dramatically, and provide non-voice paths that genuinely work with no data at all.

8.2 The options, assessed

OPTION	DOES IT WORK WITHOUT DATA?	ASSESSMENT	POSTURE
Aggressive codec & bitrate optimisation	No — but reduces need to ~10–16 kbps for intelligible speech	The highest-value, lowest-risk work. Modern low-bitrate speech codecs make voice viable on very poor connections	**Build — Phase II**
Wi-Fi calling	No — Wi-Fi is data	Useful where Wi-Fi exists; no help in the rural case	Build
Cellular voice (PSTN) bridge	Yes, for the recipient	Real, works today, and is how you actually reach someone with no data. Requires telecom partnership and per-minute cost	**Build — Phase II**
Operator partnership / zero-rating	Data is used but not charged to the user	**The realistic answer to "calls without data" as users experience it.** Commercially complex, regulator-sensitive (net-neutrality rules vary by jurisdiction), and dependent on operator willingness — but it is how this problem is genuinely solved in emerging markets	**Pilot — Phase II**
USSD / SMS fallback	Yes	Works on any phone with no data at all. Cannot carry voice, but carries notifications,	**Build — Phase II**

OPTION	DOES IT WORK WITHOUT DATA?	ASSESSMENT	POSTURE
		balances, results, confirmations and simple menus — which is most of what an institution needs to reach a parent with	
Satellite (LEO)	It is data, from a different source	Could change rural institutional connectivity materially. Currently cost-prohibitive for our customers. Volume IX §7.2 posture: Watch, revisit on cost per Mbps	Watch
Mesh / peer-to-peer local networking	Yes, within range	Genuinely useful in a bounded campus or an emergency; range and routing make it unsuitable as a general calling solution. Honest scope: campus-local, not wide-area	Pilot — Phase III
Store-and-forward voice messages	Sends when data returns	Not a call, but solves much of what a call was needed for, at a fraction of the requirement — and matches how our markets already communicate	**Build — Phase I**

Table 170 — The options, assessed

8.3 The product position

SpaceTalk will state, in its own marketing: **“Works on the worst connection you have, and reaches people who have none.”** Both halves are deliverable — the first through codec optimisation, adaptive degradation and offline-first design; the second through PSTN bridging and USSD/SMS fallback.

It will not claim data-free calling, because that would be a lie, and in a product whose entire wedge is verified trust, a lie in the marketing is a contradiction of the product.

CHAPTER 9 — AI IN COMMUNICATION

Every user gets an assistant. Every institution can configure agents. Both operate under Volume IV Chapters 7–9 without modification.

CAPABILITY	VALUE	GOVERNANCE
Conversation summarisation	Catching up on 400 messages in 30 seconds — the single most requested capability in group messaging	Labelled AI; original always available
Thread & room search in natural language	“What did the bursar say about the deadline?”	Scoped to what the user may already see
Translation (message & live)	Decisive in multilingual institutions and diaspora communities	Original retained; limits disclosed
Draft replies	Suggested, never sent automatically	User sends; never auto-send
Proofreading & rewriting	Register-appropriate, in the user’s language	—
Meeting scheduling	Across calendars and time zones	—
Reminders & follow-ups	From the conversation itself	—
File retrieval	“Send me last term’s fee schedule”	Permission-checked, always
Workflow triggers	“Approve this” from inside the conversation	Approval chains per Volume III §7.1
Institutional Q&A	Grounded in the institution’s own records, with citations	Volume IV §7.2
Meeting summaries & action extraction	Attributed, editable, never authoritative on its own	Consent to record required first

Table 171 — Ai in Communication

The boundaries, non-negotiable:

- **The assistant never reads an end-to-end encrypted conversation unless the user explicitly invokes it in that conversation**, and when invoked, that is visible in the room.
- **No message content is used to train third-party models** (Volume IV §9).
- **No behavioural profiling for advertising.** There is no advertising (Volume I §14.3).
- **High-stakes content is routed, not answered** — a safeguarding disclosure in a message goes to a named human immediately, and the AI’s role is to *detect and escalate*, never to assess (Volume IV §8.1).
- **AI is labelled at the point of interaction**, in the user’s language.

CHAPTER 10 — WORKSPACE

The Slack/Teams/Discord replacement for institutions, distinguished by costing what our markets can pay and running on the devices they have.

Departments and org structure inherited from the StromeX record · roles and permissions from StromeX Identity · channels, threads and communities · shared documents and collaborative notes · calendars and scheduling · projects and tasks · **approval workflows and digital signatures inside the conversation** · meeting rooms and booking · knowledge base · directory with verified roles · guest access with explicit scope · retention, legal hold and export · admin console with audit · device management · an **AI secretary** per team.

The differentiator against Slack and Teams is not features — it is that the institution already has its people, roles, departments and records in StromeX. The workspace is not configured; it already exists. For an institution that has spent six weeks setting up a Slack org and still has stale membership, this is the whole argument.

CHAPTER 11 — EDUCATION, FINANCE, COMMERCE & SECTOR EDITIONS

11.1 Education edition

Class and cohort rooms created automatically from the timetable · parent–teacher channels with verified identity on both sides · announcement channels with acknowledgement tracking · assignment distribution and submission · attendance from within the room · live teaching with recording and transcription · AI tutors available in-conversation · digital library access · examination and result notification · fee reminders with payment inline · safeguarding disclosure channel with restricted, logged access and immediate human escalation.

The impersonation problem this solves is concrete and expensive: fraudulent “school” accounts on consumer messengers demanding fee payments are a live and growing fraud in our markets. A verified-identity channel is the answer, and it is Wedge 1 in its most saleable form.

11.2 Finance

Wallet · invoices and payment requests inside a conversation · subscriptions · donations, zakat and sadaqah with designation tracking · fee instalments · payroll notifications · transaction history · budgeting.

Built on StromeX Pay (Volume III Division 6) and constrained by it. Every financial capability is subject to the licensing and compliance regime of each jurisdiction; a feature ships in a market only when it is lawful there, assessed by counsel, not by analogy to another market. Escrow in particular is heavily regulated and is offered only where licensed.

11.3 Commerce

Digital storefront in a channel · bookings and appointments · digital and physical goods · subscriptions · creator monetisation · AI services sold in-conversation. Marketplace terms per Volume III §16.1 — 20% rev-share, deliberately below the 30% platform norm.

11.4 Sector editions

Government — sovereignty, in-country residency, federation, records retention, FOI handling, emergency broadcast, inter-agency federation, air-gapped deployment (Volume VII §8). **Healthcare** — restricted clinical channels, consent management, retention per health-data law, no clinical decision-making by AI (Volume VII §7). Available only in jurisdictions where the health-data regime has been mapped and implemented, never by assumption. **Faith institutions** — congregation channels, service scheduling, donation designation, memorisation groups, multilingual and Arabic-first (Volume VII §3). **Enterprise** — compliance, audit, governance, analytics, device management, retention, legal hold, admin console (Volume VII §14).

CHAPTER 12 — IDENTITY

One StromeX identity across every product (Volume III §3.3): passkeys and WebAuthn preferred, MFA available on every tier and mandatory for privileged roles, device management with a visible device list, account recovery verified out-of-band, enterprise SSO, and **verified institutional roles displayed on messages**.

The role display is Wedge 1 made visible. A message can carry *"Registrar · Sultan Hanafi Royal Schools · verified"*, checkable by anyone, backed by the same credential infrastructure that signs the institution's certificates (Volume IV §5). Revoking someone's role revokes their ability to speak as that role, immediately, everywhere.

Privacy commitments: phone number is optional, not the identity · a user may hold personal and institutional identities distinctly and see clearly which they are acting as · leaving an institution removes the institutional identity but not the personal account or its personal history · the institution never gains access to a user's personal conversations, in any configuration.

CHAPTER 13 — AUTOMATION & THE OPEN PLATFORM

No-code automation for every user: triggers, conditions, actions, a visual builder, and the Volume III §7.2 prebuilt packs surfaced inside conversations. A bursar should be able to build “when a fee is 14 days overdue, message the parent in their language, with a payment link, and notify me if unopened after 3 days” without writing anything.

The open platform: bots · mini-apps running inside rooms · plugins · custom AI agents · enterprise integrations · REST API · webhooks · SDKs · federation bridges to other protocols. Terms and quality gates per Volume V Chapter 10, including the standing commitment not to clone a successful marketplace extension.

CHAPTER 14 — TRUST, SAFETY & ABUSE PREVENTION

A communication platform without a safety plan is a liability, and a platform serving children without one is indefensible.

VECTOR	COUNTERMEASURE
Impersonation	Verified identity is the core defence (Wedge 1); unverified accounts are visibly marked as such
Spam & scams	Rate limiting, reputation, pattern detection, one-tap reporting, and **institutional channels that cannot be created by non-verified parties**
Fraud (fee scams)	Payment requests inside institutional channels are cryptographically bound to the institution; a payment request from an unverified party is blocked and flagged, not merely warned about
Child safety	Strictest class (Volume IV §4.3). Adults cannot initiate contact with minors outside institution-sanctioned channels; minors' rooms are structurally supervised; disclosure channels escalate to a named human immediately
Harassment	Block, report, mute, leave; institutional escalation to a named safeguarding role; evidence preserved for the institution's process
CSAM	Detection where lawful and technically possible; immediate reporting to the relevant authority; permanent ban. No exceptions and no jurisdictional discretion
Federated abuse	Server reputation, allow/deny lists, and the ability to defederate a server entirely (Chapter 5.3)
Misinformation in broadcast	Forwarding limits on unverified content; verified institutional sources visibly distinguished
Account takeover	MFA/passkeys, device list, anomalous-login detection, out-of-band recovery
Bulk data extraction	Rate limits, anomaly detection, and an audit trail visible to the institution

Table 172 — Trust, Safety & Abuse Prevention

The moderation position: institutions moderate their own communities, with our tools and our escalation paths; StromeX enforces the platform-level rules (illegal content, CSAM, coordinated fraud) everywhere and without negotiation. **We do not moderate political speech**, and we do not build tools for a government to monitor its citizens' private communication — Volume I §7.4, and it is a condition of the Government edition existing at all.

CHAPTER 15 — BUSINESS MODEL

Never advertising. Volume I §14.3. A platform carrying children's conversations will not be monetised through attention, ever.

STREAM	SHAPE	NOTES
Free tier	Personal use, small groups, an institution up to 50 members	Genuinely useful (Volume I §5.1)
Institution subscription	Per active member per month, banded	The primary line
Workspace	Per seat, per month	Priced against our markets, not against Slack
Storage & retention	Beyond generous included allowance	Real marginal cost
AI credits	Volume III Chapter 18, shared wallet	Consumption
Calling	Free on-net; PSTN and SMS at cost + margin	Pass-through disclosed
Commerce & payments	0.4% platform fee (Volume III Division 6)	—
Marketplace	20% rev-share	Below platform norm, deliberately
API & platform	Free to a threshold; commercial tier above	—
Enterprise & compliance	Retention, legal hold, admin, audit, device management	Where the enterprise value genuinely is
Government & sovereignty	On-premise, air-gapped, federation, residency	Highest ACV, Volume III §16.2 terms
White-label	Partners and institutions under their own brand	Volume III §16.2

Table 173 — Business Model

The strategic point: the free consumer tier is not where the money is and is not supposed to be. It exists so the network exists. Revenue comes from institutions, which is exactly the customer StromeX already has.

CHAPTER 16 — ROADMAP

Gated by Volume VIII. SpaceTalk is a Phase II product; nothing here may pull resources from Phase I foundations.

STAGE	WHEN	SCOPE
Protocol & core	Late Phase I (2029–30)	Event store, sync, identity integration, encryption model, offline-first client. **No public launch**
Education edition	Phase II (2031)	Launch inside the existing school customer base — the cold-start advantage, used deliberately
Workspace	Phase II (2031–32)	Institutional collaboration
Calling & media	Phase II (2032)	WebRTC, adaptive degradation, PSTN bridge, USSD/SMS fallback
Federation	Phase II (2032–33)	Matrix interoperability; the government wedge opens
Commerce & payments	Phase II (2033)	Per jurisdiction, as licensing permits
Open platform	Phase II (2033–34)	Bots, mini-apps, marketplace
Government & sovereign editions	Phase II–III (2034–36)	On-premise, air-gapped, residency
Sector editions	Phase III (2036+)	Health, finance, enterprise
Operator partnerships	Phase III	Zero-rating pilots (Chapter 8.2)

Table 174 — Roadmap

CHAPTER 17 — RISKS & WHAT WOULD KILL IT

#	RISK	LIKELIHOOD	IMPACT	POSITION
S1	**Network effects prove insurmountable even institutionally** — communities keep using WhatsApp alongside SpaceTalk	**High**	High	Accept and design for it: SpaceTalk must win the *institutional* conversation, not the social one. Coexistence is the expected state for years, and the product must not require exclusivity to be valuable
S2	Client quality falls behind on mid-range Android	Medium	Very high	The Skype failure mode. C2 is a permanent release gate, not a goal
S3	Federation complexity consumes the engineering budget	Medium	High	Federate at the server layer only; do not federate features that do not need it; ship the closed product first and federate second
S4	Safety incident involving a minor	Medium	**Catastrophic**	Chapter 14 is built before launch, not after. This risk alone justifies delaying launch
S5	Regulatory action on encryption in a market	Medium	High	The Chapter 4.3 position is defensible precisely because it is honest and disclosed; jurisdictions requiring hidden access are markets we decline
S6	Telecom/operator dependency for PSTN and zero-rating	Medium	Medium	Multi-operator; never make a core capability depend on one partner
S7	It distracts Phase I	**High**	High	Gated to Phase II by Volume VIII. **This is the most likely way**

#	RISK	LIKELIHOOD	IMPACT	POSITION
				SpaceTalk damages StromeX — not by failing, but by being started too early
S8	An incumbent adds verified institutional identity	Low–Medium	High	They would need an issuing authority. Our answer is to be further ahead on records, workflow and payments by then

Table 175 — Risks & What Would Kill It

The honest summary

SpaceTalk is the highest-ceiling and highest-risk product in the corpus. Its ceiling is genuinely the WeChat precedent — a communication surface that becomes infrastructure. Its risk is that messaging is the most brutally network-effect-dominated category in software, and that ambitions in this category have destroyed better-funded companies than ours.

It is worth building for one reason: StromeX will already be inside the institutions whose communities need to communicate. That is an entry no consumer competitor has and none can buy. If that advantage is used — institution by institution, with verified identity as the wedge and federation as the government route — SpaceTalk is a serious product. If it is instead launched as a general-purpose messenger hoping to win users one at a time, it will fail, expensively, and it will take Phase II's focus with it.

APPENDICES

Apparatus & Appendices

The appendices extract the operational instruments of the corpus — the gates, checklists and definitions a reader must be able to find without navigating a volume. They are summaries of, and subordinate to, the chapters they draw from. Where an appendix and its source chapter differ, the chapter governs.

Appendix A — Phase Gate Checklists

From Volume VIII, Chapter 2. A phase ends when its gate is passed, not when its years elapse. Running Phase I into 2032 because the gate was not met is a correct decision; entering Phase II with the gate unmet is not.

Gate I → II — assessed from 2030

#	CRITERION	THRESHOLD
1	Paying institutions	≥ 250
2	Net revenue retention	≥ 105%
3	Gross revenue retention	≥ 92%
4	Recurring share of revenue	≥ 55%
5	Blended gross margin	≥ 60%
6	CAC payback	≤ 15 months
7	Referral share of new customers	≥ 35%
8	Certified delivery partners	≥ 15
9	Reference customers willing to take calls	≥ 10
10	Executive layer operating without the founder for 60 days	Demonstrated by absence
11	Independent penetration test passed, no unresolved high findings	Yes
12	Free-tier monthly active institutions	≥ 5,000

Table 176 — Gate I → II — assessed from 2030

Gate II → III — assessed from 2035

#	CRITERION	THRESHOLD
1	Revenue	≥ growth-scenario floor
2	Recurring share of revenue	≥ 65%
3	Net revenue retention	≥ 115%
4	Countries with local leadership	≥ 3
5	Ecosystems beyond education at depth	≥ 2
6	Partner-delivered revenue	≥ 30%
7	Board with independent directors	Functioning
8	Accounts	Audited
9	Rule of 40	≥ 30

Table 177 — Gate II → III — assessed from 2035

Criterion 10 of Gate I is the one most likely to be quietly skipped and the one that matters most. It is assessed by the founder actually being absent for sixty days.

Appendix B — Market-Entry Checklist

From Volume VIII, Chapter 10. Every item must be complete and signed off before the first customer contract in a new country.

Legal & regulatory

#	ITEM	STATUS
B1	Entity incorporated	☐
B2	Tax registration	☐
B3	Employment compliance reviewed	☐
B4	Data protection regime mapped and implemented	☐
B5	Sector regulator requirements documented	☐
B6	Trademark filed	☐
B7	Contracts localised by local counsel	☐
B8	Sanctions and export-control screening	☐
B9	Anti-corruption exposure assessed	☐

Table 178 — Legal & regulatory

Commercial

#	ITEM	STATUS
B10	Market sized by the Volume II §2.2 method with dated inputs	☐
B11	Price band assigned	☐
B12	Competitors mapped	☐
B13	Three reference prospects identified	☐
B14	Partner or acquisition target assessed	☐
B15	Payment rails live	☐
B16	Local currency handling	☐
B17	Collection risk assessed	☐

Table 179 — Commercial

Product, operational & financial

#	ITEM	STATUS
B18	Language complete; curriculum and sector standards mapped	☐
B19	Local integrations built (payments, examination boards, regulators)	☐
B20	Data residency in place if required	☐
B21	Offline and low-bandwidth verified on local networks	☐

#	ITEM	STATUS
B22	Local leader hired, with authority	☐
B23	Support hours covered; delivery capacity confirmed in writing	☐
B24	Entry budget approved; break-even modelled with stated assumptions	☐
B25	FX exposure hedged or accepted explicitly	☐
B26	Exit criteria defined — what would make us withdraw, and by when	☐

Table 180 — Product, operational & financial

Item B26 is the one most often omitted and the most valuable. A market entry with no pre-agreed failure condition will be defended indefinitely by the people who championed it.

Appendix C — Metric Definitions

From Volume IX, Chapter 4. Every metric carries its denominator, its period and its trend. A metric definition may not be changed without restating history and saying that it was changed.

METRIC	DEFINITION	TARGET
Net revenue retention	Recurring revenue from the cohort of customers present twelve months ago, including expansion and contraction, divided by that cohort's recurring revenue then.	105–112% Phase I; ≥115% Phase II
Gross revenue retention	As above, excluding expansion. Isolates churn.	≥ 92%
CAC payback	Loaded acquisition cost per customer divided by monthly gross profit per customer.	< 12 months low-touch; < 18 consultative
Time to first value	Days from contract signature until the customer uses something real in production.	< 14 days
Adoption rate	Active users by role divided by licensed users, measured at 90 days.	Reported beside every bookings figure
Referral share	New customers originating from an existing customer or their network, divided by all new customers.	≥ 35% by end of Phase I
Rule of 40	Revenue growth percentage plus operating margin percentage.	≥ 30 at Gate II; ≥ 40 from 2034
Blended gross margin	Group gross profit divided by group revenue, across all eleven revenue engines.	60% (2030) → 76% (2040)
Unused-module count	Modules purchased and unused for 60 days. Triggers intervention or proactive removal.	Driven to zero
Cancellation friction	Clicks to cancel divided by clicks to subscribe.	< 1, measured and reported

Table 181 — Metric definitions and targets

Glossary

Terms are defined as this corpus uses them. Where a term also has a general industry meaning, the definition here governs internally and the cross-reference points to the chapter in which the term is established.

TERM	DEFINITION
ACV	Annual contract value. The recurring revenue a customer contributes in a year, excluding one-time implementation and build fees.
Band A / B / C	The three regional price bands defined in Volume I §6.4. All catalogue prices are quoted at Band C (list); Band A applies a 0.32× multiplier and Band B a 0.65×. The band follows the institution's operating jurisdiction, not its billing address.
CAC	Customer acquisition cost. Loaded sales and marketing cost divided by customers acquired in the period.
CAC payback	The number of months of gross profit from a customer required to recover that customer's acquisition cost.
Craft standard	The requirement in Volume VI §2.1 that creative work be indistinguishable from that of a good London or New York studio while remaining reproducible by a regional printer on the first attempt.
Credential	Any institution-issued, cryptographically signed record of an award, qualification, identity or transaction that a third party may need to verify. The subject of Volume IV, Chapter 5.
Degraded mode	The defined, designed and tested behaviour of a surface under failure — no network, slow network, provider outage. Required of every surface by Volume IV §10.2.
Ecosystem	A complete sector solution spanning all eight architectural layers, rather than a product sold into a sector. The unit of strategic expansion; see Volume I, Chapter 2.
Entrenched provision	A clause amendable only by board supermajority with published rationale and thirty days' notice. Enumerated in Volume IX §9.2.
Federation	Server-to-server interoperation across independently operated instances, allowing an institution to own its data while remaining reachable. The basis of the SpaceTalk government strategy; Volume X, Chapter 5.
Foundation tier	The permanently free tier. Not a trial and not a demo; governed by the Free Constitution in Volume I, Chapter 5.
Gate	A set of conditions that must be met before a roadmap phase may end. A phase ends when its gate is passed, not when its years elapse. Volume VIII, Chapter 2.
GRR	Gross revenue retention. Recurring revenue retained from existing customers excluding expansion; measures churn alone.
Guilloche	Fine mathematical line-work used in security printing, extremely difficult to reproduce by scan-and-print. The primary visual security element of a StromeX credential; Volume VI §6.2.
Human-in-the-loop	The requirement that a named human decides, and is recorded as deciding, wherever the stakes ladder classifies a decision as high. Volume IV §8.1.
LTV	Lifetime value. The discounted gross profit expected from a customer over the expected life of the relationship.

TERM	DEFINITION
No-fork rule	The prohibition on forking shared platform layers for a single customer without written CTO approval and a sunset date. Volume IV §3.2.
NRR	Net revenue retention. Recurring revenue from existing customers including expansion and contraction. Above 100% means the existing base grows without new customers.
Pricing Council	The standing body that owns Volume III, meets quarterly, and reviews every price against delivered margin. Volume I §8.4.
Reference Implementation	A delivered institutional deployment used as proof of method and as a permanent sales asset. Four are recorded in this edition.
Rule of 40	Revenue growth percentage plus operating margin percentage. A common composite measure of whether a software company balances growth against profitability.
Stakes ladder	The four-level classification — low, medium, high, prohibited — determining how much autonomy an AI system may exercise over a decision. Volume IV §8.1.
System of record	The authoritative store of an institution's people, money, documents, credentials and decisions. Identified in Volume II §9 as the group's only genuinely durable moat.
Value ledger	The running, honest account of what an institution has gained from StromeX, shown at every renewal including the periods where the numbers are unimpressive. Volume I §4.4.
Verification token	A prepaid unit purchased by an institution for verification-adjacent services carrying real marginal cost. Verification itself is always free to the verifier. Volume III §3.5.
Wave	A group of industry ecosystems opened in the same roadmap phase. Volume I §2.2.

Table 182 — Glossary of terms

References & Bibliography

Every external figure in this corpus is attributed at the point of use. This list consolidates those sources and records what each was used for. Sources were consulted in August 2026; where a source reports a market figure that moves, the figure is dated in the text.

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Internal sources

Claims in this corpus about what the group has built trace to these. Catalogue lines in Volume III are marked built, partial or specified-not-built on this basis.

StromeX MVP · apps/api, apps/web, infra

FastAPI backend with multi-provider AI routing, memory, spaced-repetition tutor and PDF export; Next.js frontend; Docker Compose infrastructure. Establishes the canonical stack in Volume IV §2.2.

Independent MVP Audit · docs/04-STROMEX-INDEPENDENT-AUDIT.md

Security, scalability, reliability and performance audit including a critical server-side request forgery vulnerability found, reproduced and fixed. Carried forward as a standing engineering rule in Volume IV §15.

Reference Implementation №1 · Sultan Hanafi Royal Schools

Four-language institutional build (English, Arabic RTL, Yorùbá, French) with approximately thirty-five page templates per locale; four verification surfaces; certificate generation with documented identifier architecture, security model and signing-key deployment; digital identity and card system; finance platform; eight role portals; personalisation centre; and approximately one hundred and twenty institutional system, governance and design documents.

Reference Implementation №2 · WorldWide English College

A seventy-three-table credential and commerce schema across fifteen migrations, covering enrolment integrity, competencies, evidence, awards, credential signing, issued documents, verifying institutions, payments, instalment plans, multi-currency routing, corporate seats and alumni chapters. The evidential basis for Volume IV §4.2.

Reference Implementation №3 · Al-Madeenah College

Bilingual static build system with offline service worker, verification surfaces, font subsetting pipeline and document generation to PDF and DOCX.

Reference Implementation №4 · Institutional portal system

Multi-role portal implementation covering student, staff, parent, applicant and administrative roles, admissions pipeline, attendance, timetable, assessment, finance and library.

Index

Generated by the build. Page references point to where each concept is established or substantively discussed; passing mentions are not indexed. Word index markers are also embedded throughout the corpus, so a live index can be built in Word if preferred.

A

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Colophon

This edition is composed for A4 at 210 × 297 mm. Display matter is set in a serif book face and running text in a companion serif, with a sans for tables, captions and navigation and a monospaced face for code and architecture figures — the four typographic roles defined in Volume I, Chapter 10. The StromeX brand faces, **Archivo** and **Fraunces**, ship with this edition in `docs/bible/publication/fonts/`; the build selects them automatically wherever they are installed in the rendering environment, so the corpus can be reissued in full brand typography without editing a line of it.

The rules, tables and figures follow the design laws in Volume I, Chapter 12: typography carries the interface, space is a component, colour carries meaning, and structural devices encode something true about the content rather than decorating it. Tables use horizontal rules only; the vertical grid is carried by alignment, as it is in every well-set financial document.

This DOCX is the authoritative master. The press-quality PDF edition is generated from it and is content-identical. Both are produced from the markdown corpus of record under version control at `docs/bible/`, by the build recorded at `docs/bible/publication/`, so that the publication can be regenerated exactly whenever the corpus is amended.

THE STROMEX EDITORIAL BIBLE · Edition II · VERSION 2.0

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